

20 marks

Construct the discrete frequency distribution from the following data.

10, 20, 15, 35, 30, 25, 40, 35, 30, 32, 45, 35, 33, 23, 17, 12

Marks	Frequency (f)
50	5
67	3
11	3
79	2
30	2

N = 15

Gamma Distribution

Construct age distribution of 20 persons having class width 10

10, 20, 15, 35, 30, 25, 40, 35, 30, 32, 45, 35, 33, 23, 17, 12

No. of classes (K) = $1 + \frac{3.322 \log_{10} N}{\log_{10} 10}$

$$= 1 + \frac{3.322 \log_{10} 17}{\log_{10} 10}$$

class width = 10

Age class	Frequency
0-10	10
10-20	10
20-30	10
30-40	10
40-50	10

N = 17

Birth weights (kilograms) of 20 children were recorded as follows

10	2.1	2.7	2.8	3.5	3.1	3.7
20	3.2	2.7	2.5	2.7	3.7	
30	3.5	4.1	3.9	2.8	2.2	
40	2.6	2.9	3.5	4.1	3.9	2.8

Construct frequency distribution using Sturge's formula.

51" Birth weight of 30 children

N = 30

Smallest value (s) = 2.0

Largest value (L) = 4.2

No. of classes (K) = $1 + \frac{3.322 \log_{10} N}{\log_{10} 10}$

$$= 1 + \frac{3.322 \log_{10} 30}{\log_{10} 10}$$

$$\text{class size} = \frac{L - s}{K} = \frac{4.2 - 2.0}{6} = 0.37$$

Class	Tally
2.0-2.4	
2.4-2.8	
2.8-3.2	
3.2-3.6	
3.6-4.0	
4.0-4.4	

Beta Distribution

The following are measurements of finger strength (in ounces) of sample of 54 kines therapists

32.5	13.2	35.4	21.3	29.4	31.6	20.3	34.5
21.2	28.3	27.1	25.0	20.7	22.5	30.2	27.7
37.3	33.7	20.4	21.9	23.3	17.3	20.0	18.8
20.6	20.5	21.8	27.5	32.5	20.6	28.8	30.3
35.4	34.1	27.5	29.6	22.2	21.3	22.2	37.0
36.9	24.6	28.3	24.8	27.1	25.4	34.5	23.6
						37.9	

Group these measurements into a distribution having the class 15.0-19.9, 20.0-24.9, ...

$$E = \frac{\sum f_i x_i}{N}$$

Q.1
1.2-3
- 2.4-3
- 2.9-3
- 3.4-3
- 3.9-3

Frequency Tally
||

Frequency (f)
2
13
24
11
4
N = 54 B

Convert the class interval into inclusive class interval when common interval is equal to 0.2
class interval

0-10
30-20
20-30
30-40
40-50

Frequency
3
10
24
27
5
Cf = lower limit of 2nd - upper limit of first
2
as 12 = lower limit of 2nd - upper limit of first
lower limit of 2nd = 10 + 0.2 = 10.2

inclusive class interval
0-10.2
10.2-20.2
20.2-30.2
30.2-40.2
40.2-50.2

Frequency
3
10
24
27
5
N = 69 B
0.2-10 0.1-9.5
10.2-20 10.1-19.5
20.2-30 20.1-29.5
30.2-40 30.1-39.5
40.2-50 40.1-49.5

Convert the class interval into exclusive class interval
class interval

5-8.9
9-12.9
13-16.9
17-20.9
21-24.9

Frequency
1
7
3
25
4

Q.2 Correction factor (Cf) = $\frac{\text{Lower limit of 2nd class} - \text{Upper limit of 1st class}}{2}$
 $= \frac{9-8.9}{2} = \frac{0.1}{2} = 0.05$

Frequency
class interval
4.95-8.95
8.95-12.95
12.95-16.95
16.95-20.95
20.95-24.95
N = 46 B

Q.3 Construct an ordinary frequency table from the following data

Expenditure in Rs	No. of person	Expenditure in Rs	No. of person
Above 0	100	Above 1000	35
Above 1000	96	Above 5000	23
Above 2000	76	Above 6000	15
Above 3000	49	Above 7000	3

Q.4 Ordinary frequency distribution

Class	No. of person (f)
0-1000	100-96 = 4
1000-2000	96-76 = 20
2000-3000	76-49 = 27
3000-4000	49-35 = 14
4000-5000	35-23 = 12
5000-6000	23-15 = 8
6000-7000	15-3 = 12
7000 & more	3
N = 100 B	

Construct an ordinary frequency table from following data

production in mill	No. Companies
Below 20	10
Below 30	34
Below 40	61
Below 50	65
Below 60	67
Below 70	69

ordinary frequency class	distribution No. of companies
below 20	10
20-30	$34-10=24$
30-40	$61-34=27$
40-50	$65-61=4$
50-60	$67-65=2$
60-70	$69-67=2$
	$N=69$ @

In a village of 55 houses, the number of male people are recorded as follows.

2	3	4	2	3	5	2	1
3	2	2	4	3	1	0	5
2	1	0	3	2	4	3	2
1	5	1	3	2	1		

Represent the above data into discrete frequency distribution.

Data (")	frequency
0	2
1	6
2	9
3	7
4	3
5	3
	$N=30$

the following data refers to the weights of large oil tins (in thousands of tons):

140	130	142	144	136	153	144	155
120	210	240	220	260	200	305	309
138	292	278	268	265	241	235	231
222	202	206	207	199	180	128	122
170	190	201	180	194	220	224	226
246	292	310	302	282	274	261	194
200	345						

Construct a frequency distribution of equal length

50. No. of Tins of Tins $N=50$

Smallest weight (S) = 130

Largest " (L) = 345

No. of classes (K) = $1 + 3.322 \log_{10} 50$

= 6.64 $\approx$ 7

Size of classes = $\frac{L-S}{K} = \frac{345-130}{7} = 27.8 \approx 28$

Class	Tally	Frequency
130 - 152		9
152 - 184		5
184 - 211		11
211 - 238		6
238 - 265		6
265 - 292		6
292 - 319		7
		$N=50$

Represents the following set of marks out of 50 obtained by 40 students

46	12	14	16	17	21	34	35
40	42	3	7	9	2	17	18
24	3	7	18	4	40	37	36
32	29	26	39	41	5	3	2
38	26	16	12	12	21	24	25

from a frequency distribution by taking a class interval

Class interval	Tally	Frequency (f)
0-5		4
5-10		4
10-15		2
15-20		5
20-25		4
25-30		4
30-35		2
35-40		4
40-45		4
45-50		1
		<u>No. = 40</u>

The min temperature (in °C) for Kathmandu for the month August, as reported by the meteorological department is given below.

Min	24	25	22	21.2	30	24.3	27
14	24	25	22	21.2	30	24.3	27
16	17	26	30	31	19.5	26.2	18.3
16	17.2	22.1	24	26.1	23	28	30
24	19	23.3	26	27	14	18	

No. of days of temp. (N) = 31

Lowest temp. (S) = 14

Highest " (L) = 31

No. of class (K) = $1 + 3.322 \log_{10} N = 1 + 3.322 \log_{10} 31 = 5.954 \approx 6$

Class interval = $\frac{L-S}{K} = \frac{31-14}{6} = 2.855 \approx 2.9$

Class	Tally	Frequency (f)
14-16.9		4
16.9-19.8		5
19.8-22.7		3
22.7-25.6		5
25.6-28.5		5
28.5-31.4		4
		<u>N = 31</u>

(13) The following are the monthly rents (in Rs.) of 40 shops. Tabulate data by grouping in interval of Rs. 8.

36	44	74	72	42	46	54	53
84	31	62	54	53	34	82	78
74	22	71	67	30	41	54	52
59	64	67	69	70	72	82	80
47	39	34	38	47	40	54	56

Class	Frequency	Tally
30-38	5	
38-46	6	
46-54	6	
54-62	1	
62-70	5	
70-78	7	
78-86	5	
		<u>N = 40</u>

(14) Prepare a frequency distribution taking a class interval of 10 by following figure relating to the wages in Rs. of 50 workers.

28	17	46	54	58	82	84	85
19	21	55	72	61	64	52	51
49	43	39	17	26	29	32	52
51	62	67	20	24	78	19	21
26	29	32	41	45	58	63	67
81	21	61	51	41	31	21	18
24	48						

Class	Frequency	Tally
10-20	5	
20-30	9	
30-40	4	
40-50	7	
50-60	9	
60-70	7	
70-80	5	
80-90	5	
		<u>N = 50</u>

Form a frequency distribution from the following data by 'Inclusive' method's taking 4 as the magnitude of class intervals:

3	7	9	18	16	26	8	40
13	36	42	4	17	26	20	31
14	17	30	28	35	27	11	14
20	24	5	8	36	24	28	34
36	38	30	32				

class	Frequency	Tally
0-4	2	
5-9	5	
10-14	5	
15-19	3	
20-24	4	
25-29	4	
30-34	5	
35-39	6	
40-44	2	
	$N=36$	

11) Change the following frequency distribution from inclusive method to exclusive method:

class	frequency
0-4	5
5-9	4
10-14	2
15-19	16
20-24	4
25-29	3
30-34	2

And construct the less than and more than frequency distribution.

Correction factor $(C_f) = \frac{\text{lower limit of last class} - \text{upper limit of 1st class}}{2}$

$$= \frac{5-4}{2} = \frac{1}{2} = 0.5$$

exclusive class	frequency
-0.5-4.5	5
4.5-9.5	4
9.5-14.5	2
14.5-19.5	16
19.5-24.5	4
24.5-29.5	3
29.5-34.5	2
	$N=36$

Less than frequency distribution

Less than	frequency
Less than 4.5	5
" 9.5	$5+4=9$
" 14.5	$9+2=11$
" 19.5	$11+16=27$
" 24.5	$27+4=31$
" 29.5	$31+3=34$
" 34.5	$34+2=36$

More than frequency distribution

More than	frequency
More than -0.5	36
" 4.5	$36-5=31$
" 9.5	$31-4=27$
" 14.5	$27-2=25$
" 19.5	$25-16=9$
" 24.5	$9-4=5$
" 29.5	$5-3=2$

12) Construct the ordinary frequency distribution table from following data.

109 - 121

Marks	No. of students
Above 0	30
" 10	77
" 20	72
" 30	71
" 40	64
" 50	62
" 60	42
" 70	36
" 80	28
" 90	14
" 100	0

Ordinary frequency distribution table

Class	Frequency
0 - 10	$30 - 77 = 3$
10 - 20	$77 - 72 = 5$
20 - 30	$72 - 71 = 1$
30 - 40	$71 - 64 = 7$
40 - 50	$64 - 62 = 2$
50 - 60	$62 - 42 = 20$
60 - 70	$42 - 36 = 6$
70 - 80	$36 - 28 = 8$
80 - 90	$28 - 14 = 14$
90 - 100	$14 - 0 = 14$
	$N = 80$

15) Construct an ordinary frequency table from the following data

Daily wages (in Rs)	No. of workers
Below 10	10
" 20	24
" 30	28
" 40	38

Below 50	44
" 60	47
" 70	49
" 80	58

16) Ordinary frequency distribution table

Class	Frequency
0 - 10	10
10 - 20	$24 - 10 = 14$
20 - 30	$28 - 24 = 4$
30 - 40	$38 - 28 = 10$
40 - 50	$44 - 38 = 6$
50 - 60	$47 - 44 = 3$
60 - 70	$49 - 47 = 2$
70 - 80	$58 - 49 = 9$

$N = 58$ (3)

19) The weights (in gram) of 40 oranges picked at random are follows

45, 55, 30, 110, 75, 100, 40, 60, 65, 40, 100, 75, 70, 65, 70, 40, 50, 60, 65, 55, 45, 90, 85, 75, 85, 75, 70, 110, 105, 80, 30, 55, 30, 70

Construct an ordinary frequency table

Solⁿ No. of weights (in gram) of orange (N) = 40

Lowest No. (L) = 30

Highest No. (H) = 110

No. of classes (K) = $1 + 3.322 \log_{10} 40 = 6.322 \approx 7$

Size of classes = $\frac{H-L}{K} = \frac{110-30}{7} = 11.428 \approx 12$

Class	Frequency	Tally
30 - 43	7	
43 - 56	7	
56 - 69	5	
69 - 82	11	
82 - 95	4	
95 - 108	4	
108 - 121	2	

2

The following table gives the weight (to the nearest kg) of 40 students at a college. Construct a frequency distribution with 7 classes.

40, 37, 42, 80, 53, 54, 77, 72, 96, 52, 57, 63, 32, 42, 46, 34, 56, 88, 70, 66, 60, 50, 59, 53, 73, 83, 93, 27, 22, 86, 54, 37, 42, 40, 50, 80, 90, 91, 34, 85.

Lowest weight out of 40 students (L) = 32

Highest " " " (H) = 96

(N) = 7

We have,

$$\text{Size of classes} = \frac{H-L}{K} = \frac{96-32}{7} = 9.28 \approx 10$$

Class	Frequency	Tally
32 - 42	8	
42 - 52	6	
52 - 62	3	
62 - 72	3	
72 - 82	4	
82 - 92	5	
92 - 102	5	
	<u>N = 40</u>	

DIAGRAMMATIC AND GRAPHICAL REPRESENTATION

When an observer or researcher collect the numerical data statistical enquiry he/she classifies and tabulate them. Then by diagram and graph, the presentation of data in the form of geometric figure is done like points, lines, bars, rectangles or circle etc. In this method, we get a clear picture of the variation in the value of variable.

→ Importance of diagrammatic & graphical representation.

- They give a bird's eye view of a set of numerical data & can present the data in to simple form.
- They give delight to the eye and leave an ever lasting impression on the mind.
- They give the required information in less time and without any mental strain.
- They are useful to compare the two or more set of numerical data.

→ Difference between diagrams & graph.

Diagrams	Graphs
① They are constructed on the plane paper.	① They are constructed on the graph paper.
② They are used for only for comparison.	② They are used to compare two or more data to study the mathematical relationship between two variables.
③ The numerical data are presented by bars, rectangle, circle, cubes, etc.	③ Numerical data are presented in terms of points and lines.
④ They will be of one, two, or three dimension.	④ They generally will be of two dimension.
⑤ Construction of diagram are not so easy.	⑤ Construction of graphs is easy compared with diagram.

Diagrammatic	Graphical
Bar diagram	Histogram
Simple bar diagram	Frequency polygon
Multiple " "	" Curve
Sub-divided " "	Ogive Curve
Percentage " "	
Two dimensional " "	
Pie diagram (Angular diagram)	

Simple bar diagram: It is a diagram for only one variable.
Subdivided bar diagram: This is used for two or more components.
Two dimensional diagram

1. Rectangular diagram
 2. Pie diagram

Circular diagram
 In this case, height & breadth is taken into consideration for the purpose of representation.
Pie diagram: is a diagram in the form of a circle whose area represent the total value.

$$\frac{\text{Total value}}{1} = \frac{360}{\text{Total}}$$

Histogram: presentation of frequency distribution (continuous) in the graphical form is called histogram which consist of set of adjacent vertical rectangles, classes are plotted on the x-axis and their frequency on the y-axis.

Frequency polygon

For a grouped frequency distribution, a frequency polygon is obtained by joining the mid point of the upper sides of the adjacent rectangles by means of a straight line. Similarly, frequency curve is obtained by joining the mid point of the upper sides of the adjacent rectangles by means of a curve line.

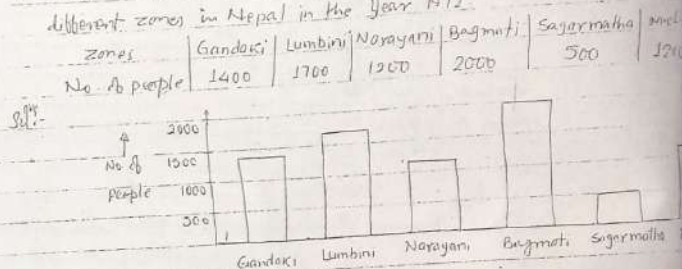
Histogram for unequal class-size

Rules:-

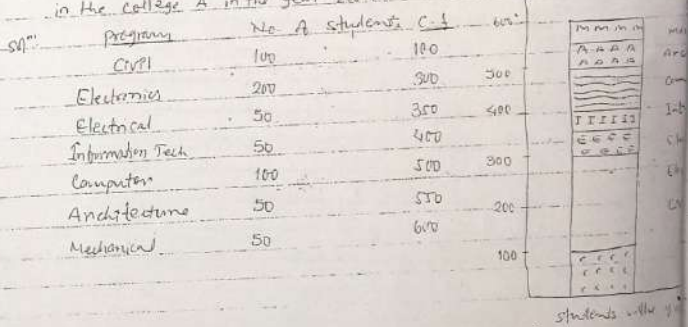
- (i) frequency of small class-size is same (not changed)
- (ii) if class-size of other class is n times than small class size, it is divided by n .

Exercise-2

1. Draw the simple bar diagram of the total number of people in different zones in Nepal in the year 1972.



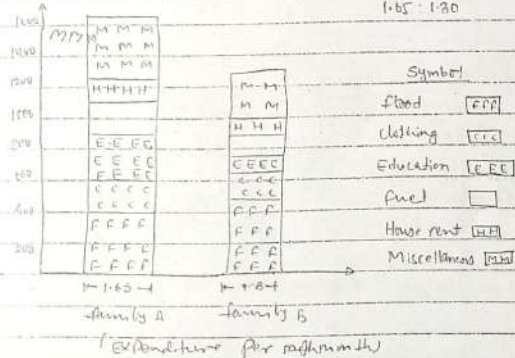
2. Draw the subdivided bar diagram of the total number of students in the college A in the year 2017.



Draw the subdivided bar diagram of total expenditure of two family A and B in the year 2019.

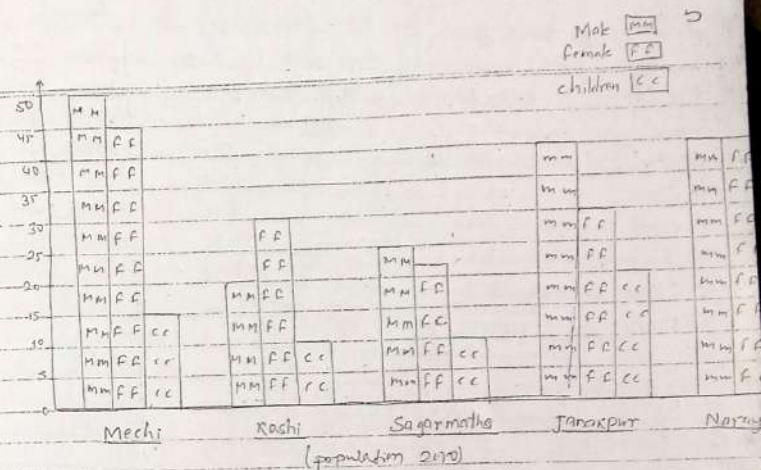
Item	family A	C.I	family B	C.I
Food	400	400	500	500
Clothing	200	600	150	600
Education	900	900	100	750
Fuel	300	300	150	900
House rent	150	1200	100	1000
Miscellaneous	400	1600	300	1800

Ratio of the family A & family B 1650 : 1300
1.65 : 1.30



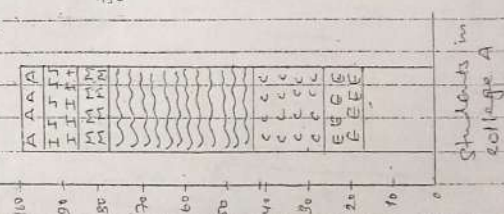
Draw the multiple bar diagram of the following data.

Zones	population (2010)		
	Male	Female	children
Mechi	50	45	15
Roshi	20	30	10
Sagarmatha	25	22	10
Jangrapur	40	30	20
Nayab	40	40	15



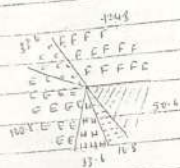
8) Represent the following data of the total number of students in college A by using percentage bar diagram.

SN	Program	No. of student	In %	C.I in %
1	Electronics	200	$\frac{200}{1150} \times 100 = 17.39$	17.39
2	Electrical	100	8.69	24.68
3	Civil	200	17.39	43.43
4	Computer	400	34.78	78.25
5	Mechanical	100	8.69	86.95
6	Information Tech	100	8.69	95.65
7	Architecture	50	4.35	100



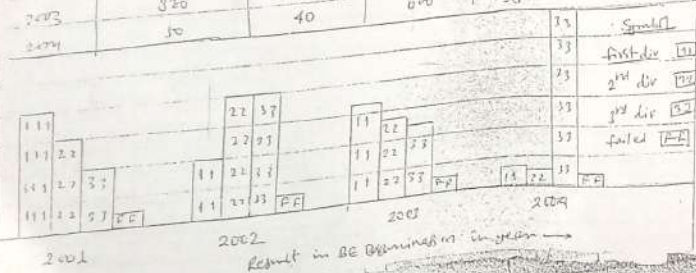
The following table gives the total expenditure of a family in a year 2007. represent this data by a pie-diagram.

Item	Expenditure	angle in degree	C. degree	Symbol
Food	2500	$\frac{2500 \times 360}{7500} = 120^\circ$	120.00	[FD]
clothing	700	$\frac{700 \times 360}{7500} = 33.6^\circ$	33.60	[CL]
Education	2100	$\frac{2100 \times 360}{7500} = 100.8^\circ$	100.80	[ED]
House rent	700	$\frac{700 \times 360}{7500} = 33.6^\circ$	33.60	[HR]
Fuel	350	$\frac{350 \times 360}{7500} = 16.8^\circ$	16.80	[F]
Other	1050	$\frac{1050 \times 360}{7500} = 50.4^\circ$	50.40	[O]
Total	7500			



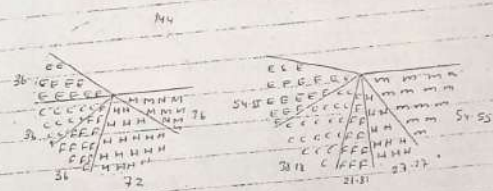
The following table give the result of the BE examination of university for past four years.

Years	No. of student passed in				Failed
	1st div	2nd div	3rd div	4th div	
2001	400	300	400	50	60
2002	400	400	240	60	60
2003	320	280	600	30	30
2004	50	40			



(15) The following data relate to the expenditure of family A and family B. represent these data by angular diagram.

Item	Family A	in degree	Family B	in degree
Food	200	$\frac{200 \times 360}{500} = 144^\circ$	200	$\frac{200 \times 360}{660} = 109.1^\circ$
Education	50	36	70	38.2
clothing	50	36	40	21.8
Fuel	50	36	50	27.3
house rent	100	72	100	54.5
Miscellaneous	50	36	60	32.7
Square root	22.36			
Ratio of circle	1.0			



(16) Represent the following frequency distribution by histogram.

Scores	Frequency
30-39	2
40-49	5
50-59	2
60-69	12
70-79	14
80-89	12
90-99	2

This is inclusive class size no.
 Correction factor $CF = \frac{40-35}{2} = 0.5$
 giv.

exclusive class	frequency
22.5-27.5	2
27.5-32.5	5
32.5-37.5	2
37.5-42.5	12
42.5-47.5	14
47.5-52.5	12
52.5-57.5	2

here class size is same i.e. 10. histogram of above problem is as follows

Draw a histogram representing the frequency distribution

Monthly wages (in Rs)	No. of workers
15	2
20	15
25	10
30	25
35	3
40	4
45	5

Histogram of above value is given in graph attached here.
 Monthly wages is mid value of class interval of size 5

Draw histogram and frequency curve of the following frequency distribution

Age (in yrs)	No. of Golfers (h)
182-183	1
183-184	1
184-185	4
185-186	24
186-187	12
187-188	5
188-189	3

23 Draw less than and more than ogive curve of the following frequency distribution

Marks	10-20	20-30	30-40	40-50	50-60	60-70
No. of students	4	6	10	20	18	2

And then find median value.
 Solⁿ first we prepare less than and more than frequency.

For less than ogive		for more than ogive	
Marks	less than cf	Marks	more than cf
less than 20	4	More than 10	60
" 30	4+6=10	" 20	60-4=56
" 40	10+10=20	" 30	56-6=50
" 50	20+20=40	" 40	50-10=40
" 60	40+18=58	" 50	40-20=20
" 70	58+2=60	" 60	20-18=2

the graph of less than and more than ogive is given in graph.
 from the fig. median is 45.

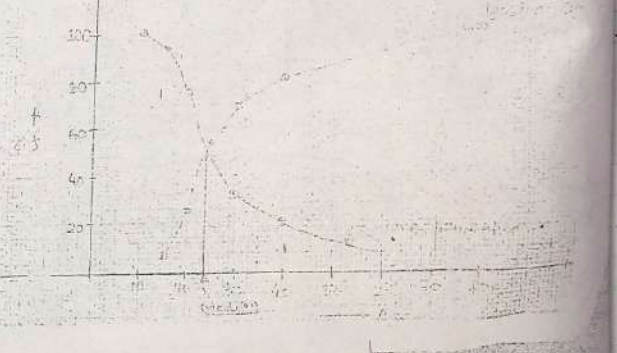
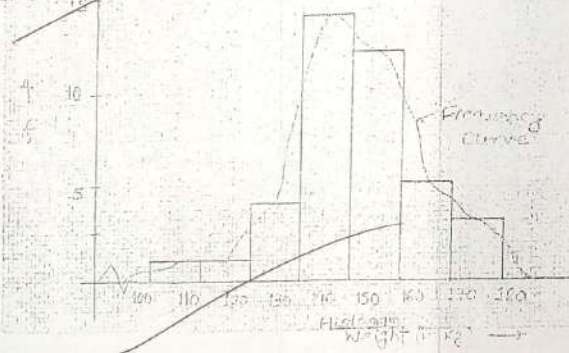
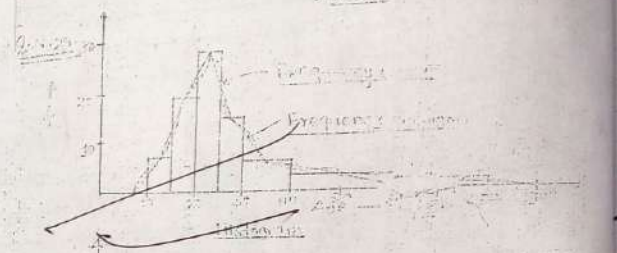
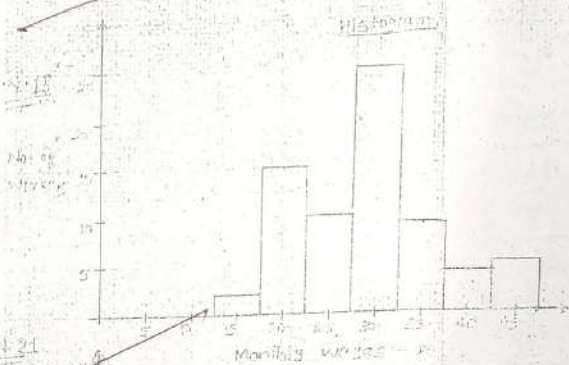
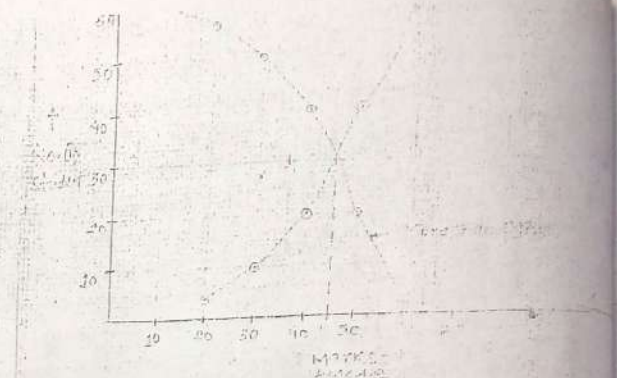
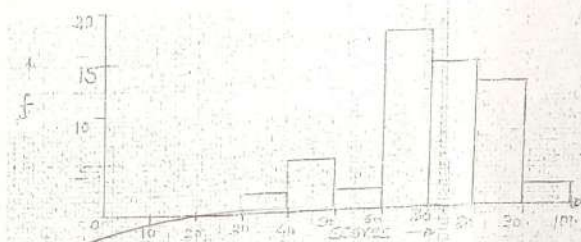
24 The following data represent age workers and no. of workers in a certain survey. Construct histogram, frequency polygon, E.Curve and ogive & median

Age	10-15	15-20	20-25	25-30	30-40	40-60	60-80
No. of workers	7	19	28	15	12	12	8

Solⁿ frequency of small class (i.e. 5) should be constant frequency
 class size 10 is adjusted by 2, 2, 40 on

Age	class size	changed frequency
10-15	5	14
15-20	5	38
20-25	5	56
25-30	5	75
30-40	10	30
40-60	20	15
60-80	20	8

histogram, frequency polygon & frequency curve are in graph



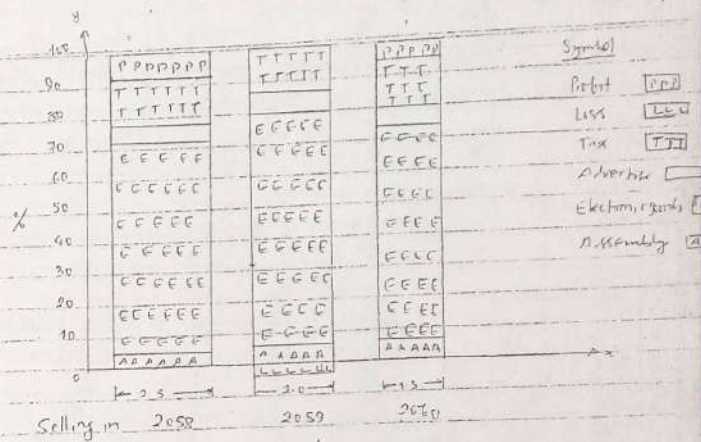
60-80 20 8/4-2

histogram, frequency polygon & frequency curve are in graph

for less than & more than ogive

less than ogive		More than ogive	
Age	c.f	Age	c.f
less than 15	7	More than 10	101
" 20	$7+10=26$	" 15	$101-7=94$
" 25	$26+8=34$	" 20	$94-10=84$
" 30	$34+15=49$	" 25	$84-25=59$
" 40	$49+12=61$	" 30	$59-15=44$
" 60	$61+12=73$	" 40	$44-12=32$
" 80	$73+8=81$	" 60	$32-12=20$

from the figure, median = 25 B



represent the following by a suitable diagram drawn on the percentage basis

Particular	year 2058			year 2059			year 2060		
	Cost	In %	Cf in %	Cost	In %	Cf in %	Cost	In %	Cf in %
1. Cost of Purchase									
(a) Assembled cost	2000	4	4	2000	5	5	2000	6.67	6.67
(b) Electronic goods cost	24000	60	72	30000	75	80	20000	66.67	73.33
(c) Advertised cost	3000	6	78	8000	75	87.5	2000	6.67	80
2. Tax cost	15% of 2500	15	93	15%	15%	102.50	15%	15	95
3. Selling Price per Comp	50,000			40000			50000		
Profit (+), Loss (-)	+3500	7	100	-1000	-2.5	100	+1000	2.5	100

Ratio in year of selling in diff year = $50000:40000:30000$
 $2.5:2:1.5$

4/ Measure of Central Tendency

① Arithmetic Mean

The arithmetic mean of n variate values $x_1, x_2, x_3, \dots, x_n$ is denoted by $\bar{x}$ & defined by

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} = \frac{\sum x}{n}$$

In terms of frequency

$$\bar{x} = \frac{\sum fx}{n}$$

② Median

Median divides the distribution into two equal parts. Median is thus, a potential average or positional average. It is denoted by M_d and

— first of all, keep all the distribution in ascending order

$$M_d = \left(\frac{N+1}{2} \right)^{th} \text{ term}$$

for continuous class

$$\text{Median} = l + \frac{N/2 - C.f.}{f} \times i$$

where, l = lower limit of the median class

i = (upper limit - lower limit) of median class

$C.f.$ = Cumulative frequency which is just previous value of $N/2$

f = frequency of median class

③ Mode — Mode is the value which occurs frequently in a set of observation and around which the other items of the set cluster densely.

for a continuous frequency distribution:-

Modal class = class having largest frequency

$$l \text{ Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i$$

⇒ Geometric Mean

It is used (i) to find rate of population growth & interest

(ii) In index number

Let $x_1, x_2, x_3, \dots, x_n$ be n variate values

Geometric mean is defined by

$$G = \sqrt[n]{x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n} = (x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n)^{1/n}$$

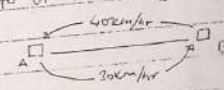
Taking log both side

$$\begin{aligned} \log G &= \log (x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n)^{1/n} \\ &= \frac{1}{n} \log (x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n) \\ &= \frac{1}{n} (\log x_1 + \log x_2 + \dots + \log x_n) \\ &= \frac{1}{n} \sum_{i=1}^n \log x_i = \frac{\sum \log x}{n} \end{aligned}$$

$$G = \text{Anti log} \left[\frac{\sum \log x}{n} \right]$$

⇒ Harmonic Mean

It is used to find average speed



→ Discrete

Let $x_1, x_2, \dots, x_n$ be n variate values

$$\text{defined by } HM = \frac{N}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} = \frac{N}{\sum \frac{1}{x_i}}$$

N = No. of items, $x_1, x_2, \dots, x_n$ are items

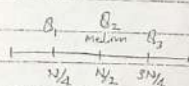
→ Interval of frequency

$$HM = \frac{N}{\sum f \cdot \frac{1}{x}}$$

partition value

1. Quartile
2. Decile
3. Percentile

1. Quartile



First Quartile Q_1

3rd " Q_3

2. Decile - One tenth of observation $N/10$

eg: 3rd decile = $\frac{3N}{10}$

7th decile = $\frac{7N}{10}$

3. Percentile $\rightarrow N/100$

eg 80th percentile = $\frac{80N}{100}$

1. Find Suitable average

- a. The following are the month monthly salaries of 10 persons
Salaries (Rs.): 275, 300, 250, 380, 420, 275, 126, 176, 210, 200

Solⁿ we have

$$\bar{x} = \frac{\sum x}{n} = \frac{2882}{10} = 288.2$$

- b. Rank of beauty contest is given below

7, 9, 5, 4, 2, 7, 8, 1, 1, 3, 9, 7, 8, 2, 15

S ⁿ	Item (x)	f	f ^c	f ^c
1	1	1	1	1
2	2	1	2	2
3	3	1	3	3
4	4	1	4	4
5	5	2	6	6
6	6	1	7	7
7	7	3	10	10
8	8	2	12	12
9	9	2	14	14
		N=15	15	15

$$\text{Median} = \frac{N+1}{2} = \frac{15+1}{2} = 8^{\text{th}} \text{ term}$$

$$\text{Median} = \text{Item whose } f \geq 8 \text{ i.e. } 8 \text{ is } 6.$$

- c. Size of shoes is given below

33 1/2, 22 1/2, 22, 35, 26, 36 1/2, 35 1/2, 36, 16, 15 1/2, 22 1/2, 35, 22 1/2

S ⁿ	Size of shoes	No. (f)	
1	15 1/2	1	here higher no. of frequency
2	16	1	here average size of shoes is 2
3	22 1/2	1	
4	22 1/2	3	
5	33 1/2	1	
6	35	2	
7	36	2	
8	36 1/2	1	

Q. Ratio of husbands and wives is given below

1.2, 0.96, 0.84, 1.23, 1.1, 1.02, 0.89, 1.13, 1.21, 1.05, 0.89

Sum we have, $\bar{x} = \frac{\sum x}{n} = \frac{1.2 + 0.96 + 0.84 + 1.23 + 1.1 + 1.02 + 0.89 + 1.13 + 1.21 + 1.05 + 0.89}{11}$

$= 1.0573$

Q. A student runs from A to C through B and back to B.

Speed (A to B), 10 mph
Speed (B to C), 9 mph
Speed (C to B), 3 mph
Speed (B to A), 7 mph

Solⁿ



we have,

$H.M. = \frac{N}{\sum \frac{1}{x}} = \frac{4}{(\frac{1}{10} + \frac{1}{9} + \frac{1}{3} + \frac{1}{7})} = \frac{4}{0.465} = 8.60$

-16-

Q. Calculate an arithmetic mean and median for the following distribution

x	1	2	3	4-6	7-9	10-12	13-20	21-24	25-36
f	10	5	4	9	6	2	1	10	15
$\sum f = 61$									
$\sum fx = 910$									
$\bar{x} = \frac{\sum fx}{N} = \frac{910}{61} = 14.89$									

for median

x	f	$C.f$
1	10	10
2	5	15
3	4	19
4-6	9	28
7-9	6	34
10-12	2	36
13-20	1	37
21-24	10	47
25-36	15	62

Now, $N/2 = 61/2 = 30.5$. Then we get $C.f$ just greater than 30.5

So median class is 25-36

$L = 6.5, C.f = 27, i = (36 - 27) = 9, f = 15$

$M_d = L + \frac{(N/2 - C.f)}{f} \times i = 6.5 + \frac{30.5 - 27}{15} \times 9 = 8.5$

Q. Calculate the arithmetic mean for following data:-

Height (in cm)	219	216	213	210	207	204	201	198
No. of persons	2	4	6	10	11	7	5	4
$\sum f = 50$								
$\sum fx = 10377$								
$\bar{x} = \frac{\sum fx}{N} = \frac{10377}{50} = 207.54$								

Find the average marks of students from the following data:

Marks	No. of students	mean	f	fx
0-10	3	5	3	15
10-20	5	15	5	75
20-30	7	25	7	175
30-40	10	35	10	350
40-50	12	45	12	540
50-60	15	55	15	825
60-70	18	65	18	1170
70-80	10	75	10	750
80-90	8	85	8	680
90-100	5	95	5	475
Total	80			4150

$N = 80$
 $\bar{x} = \frac{\sum fx}{N} = \frac{4150}{80} = 51.875$
 Median class = 50-60
 $N/2 = 40$
 $L = 50, i = 10, c.f = 37$
 $N/2 = 40, i = 15$
 $\text{Median} = L + \frac{N/2 - c.f}{f} \times i = 50 + \frac{40 - 37}{15} \times 10 = 52$

Two hundred persons were interviewed by a public opinion polling agency. The frequency distribution gives the age of the persons interviewed.

Age Group	Frequency (f)	Mean (x)	fx
9.5-19.5	32	14.5	464
19.5-29.5	42	24.5	1029
29.5-39.5	40	34.5	1380
39.5-49.5	56	44.5	2492
49.5-59.5	20	54.5	1090
59.5-69.5	6	64.5	387
69.5-79.5	2	74.5	149
79.5-89.5	2	84.5	169
Total	200		7160

$N = 200$
 $\bar{x} = \frac{\sum fx}{N} = \frac{7160}{200} = 35.8$

Calculate the arithmetic mean from the following data

Class Interval	Frequency	Midpoint	fx
0-1	8	0.5	4
1-2	8	1.5	12

Class Interval	Frequency	Midpoint	fx
3-5	10	4	40
5-10	12	7.5	90
10-15	18	12.5	225
15-25	11	20	220
25-35	10	30	300
35-45	9	40	360
45-60	8	52.5	420
Total	78		1735

$\bar{x} = \frac{\sum fx}{N} = \frac{1735}{78} = 22.24$

Find the class intervals, if the arithmetic mean of the following distribution is 38 and assumed mean is 35.

Step Deviation (d)	Frequency (f)
-3	5
-2	10
-1	25
0	50
1	20
2	10

Mean = 38, A = 35

d	f	fd
-3	5	-15
-2	10	-20
-1	25	-25
0	50	0
1	20	20
2	10	20

Now, from the formula of step deviation

$$\bar{x} = A + \frac{\sum fd}{N} \times h$$

$$38 = 35 + \frac{-20}{100} \times h$$

$$3 = -\frac{20}{100} \times h$$

$$h = 150$$

Now, $A = 35$, $h = 150$

Class Interval	Frequency
0-15	5
15-30	10
30-45	25
45-60	50
60-75	20
75-90	10

- (9) The average height of a group of 25 children was calculated to be 78.4 cm. It was later discovered that one value was misread as 69 cm instead of the correct value 96 cm. Calculate correct mean.
 Sol: Incorrect average ht ($\bar{x}$) = 78.4 cm

$$\text{No. of children (N)} = 25$$

$$\text{Total height of the children} = 78.4 \times 25 = 1960$$

Now,

$$\text{Correct Total height} = 1960 - 69 + 96 = 1987$$

$$\text{Correct average ht } (\bar{x}) = \frac{1987}{25} = 79.48 \text{ cm}$$

- (10) A candidate obtains the following percentages in an examination: Math 60, Statistics 75, Basic Electrical 63, Physics 59 and Computer Programming 55. Find the weighted mean if the weights 2, 1, 5, 5, 3 are allocated to the subjects respectively.

Sol:

$$\bar{w} = \frac{w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + w_5x_5}{w_1 + w_2 + w_3 + w_4 + w_5} = \frac{60 \times 2 + 75 \times 1 + 63 \times 5 + 59 \times 5 + 55 \times 3}{2 + 1 + 5 + 5 + 3} = \frac{606.5}{10} = 60.65\% = 60.65\%$$

- (11) From the following frequency distribution, calculate the missing frequency

No. of Tablets	No. of persons	Mid-value	f ₂
4-8	11	6	66
8-12	13	10	130
12-16	16	14	224
16-20	14	18	252
20-24	7 (p)	22	22p
24-28	9	26	234
28-32	12	30	310
32-36	6	34	204
36-40	4	38	152

$$N = 70 + p$$

$$1772 + 22p$$

The average no. of tablets to cure a person is 20

we have

$$\bar{x} = \frac{\sum fx}{N}$$

$$20 = \frac{1772 + 22p}{70 + p}$$

$$1800 + 20p = 1772 + 22p$$

$$2p = 28$$

$$p = 14$$

- (12) The mean marks obtained by 300 students in the subject of Maths are 45. The mean of the top 100 students was found to be 70 and the mean of the last 100 was found to be 20. What is the mean of the remaining 100 students?

$$\text{Sol: The mean marks of 300 students} = 45$$

$$\text{Total marks} = 45 \times 300 = 13500$$

$$\text{Top 100 mean} = 70$$

$$\text{Total marks of top 100} = 70 \times 100 = 7000$$

$$\text{Last 100 mean} = 20$$

$$\text{Total marks of last 100} = 100 \times 20 = 2000$$

$$\text{Remaining 100 students mean} = \frac{13500 - 7000 - 2000}{100} = 45$$

- (13) In a certain examination, average grade of all students in class A is 68.4 and that of all students in class B is 71.2. If the average grade of all students in class A to class B is 70, find the ratio of the number of students in class A to the number in class B.

$$\text{Sol: Let the no. of students in class A be } x$$

$$\text{Let the no. of students in class B be } y$$

$$\text{Total grade in A} = 68.4x$$

$$\text{Total grade in B} = 71.2y$$

$$\text{Average grade (A+B)} = 70$$

$$\text{Average grade} = \frac{\text{Total grade of A} + \text{Total grade of B}}{\text{No. of students in A} + \text{No. of students in B}}$$

$$70 = \frac{68.4x + 71.2y}{x + y}$$

$$70x + 70y = 6842 + 7174$$

$$70x + 70y = 7194 - 70y$$

$$167 = 123$$

$$\frac{x}{y} = \frac{12}{16} = \frac{3}{4} = 3:4$$

The mean of marks obtained in an examination by a group of 100 students was found to be 49.96 the mean of the marks obtained in the same examination by another group of 200 students was 52.32 find the mean of the marks obtained by both the groups of students taken together.

$$\text{The mean marks obtained by 100 students} = 49.96$$

$$\text{Total marks obtained by 100 students} = 49.96 \times 100 = 4996$$

$$\text{The mean marks obtained by 200 students} = 52.32$$

$$\text{Total marks obtained by 200 students} = 52.32 \times 200 = 10464$$

$$\text{Mean marks obtained by both the groups} = \frac{4996 + 10464}{100 + 200} = 51.55$$

A distribution consists of three components with frequencies 300, 200 & 100 having their means 16, 8 and 4 respectively. find the mean of the combined distribution.

$$\text{Mean of the combined distribution} = \frac{(300 \times 16 + 200 \times 8 + 100 \times 4)}{(300 + 200 + 100)} = 8$$

Mean weight of 150 student in a certain class is 70 kg. The mean weight of boys in the class is 70 kg and that of the girls is 55 kg. find the number of boys and number of girls in the class.

$$\text{Let no. of boys} = x$$

$$\text{Girls} = 150 - x$$

$$\text{Total weight of boys} = 70x$$

$$\text{Girls} = 55(150 - x)$$

$$\text{Total weight of boys \& girls} = 150 \times 60 = 9000$$

$$70x + (150 - x)55 = 9000$$

$$\text{or, } 70x + 8250 - 55x = 9000$$

$$15x = 9000 - 8250 = 750$$

$$x = 50$$

$$\text{No. of boys} = 50$$

$$\text{No. of girls} = (150 - x) = 100$$

(12) Find the median from the following table:

x:	5	7	9	11	13	15	17	19
f:	1	2	3	9	11	8	5	4
cf:	1	3	6	15	26	34	39	43

Here, $N = 43$, $N/2 = 21.5$.
Then cumulative frequency just greater than $N/2$ is 26.
∴ Median class value is 13.

(10) Calculate the mean and median from the following table:

Class Interval	Frequency (f)	Mid class (x)	f.x	cf
6.5-7.5	5	7	35	5
7.5-8.5	12	8	96	17
8.5-9.5	25	9	225	42
9.5-10.5	48	10	480	90
10.5-11.5	32	11	352	122
11.5-12.5	6	12	72	128
12.5-13.5	1	13	13	129
Total	129		1273	

$$\text{Mean}(\bar{x}) = \frac{\sum fx}{N} = \frac{1273}{129} = 9.87$$

$$N/2 = 129/2 = 64.5$$

Cumulative frequency just greater than $N/2$ is 90

Median class: 9.5-10.5

here, $l = 9.5$, $f = 10.5 - 9.5 = 1$, $C.F = 42$, $f = 48$

$$\begin{aligned} \text{Median } (M_d) &= l + \frac{N/2 - C.F}{f} \times h \\ &= 9.5 + \frac{45 - 42}{48} \times 1 \\ &= 9.57 \\ &= 9.57 \end{aligned}$$

(20) Compute the Median from the following data.

less than	Mid value	frequency	C.F	here
120-130	125	6	6	$N/2 = 300/2 = 150$
130-134	132	25	31	Cumulative frequency just greater than $N/2$
134-140	137	48	79	is 267
140-150	145	72	151	Median value - this class = 150-140
150-158	154	116	267	$l = 150$, $h = 10$, $C.F = 151$, $f = 116$
158-160	159	60	327	
160-170	165	32	359	Median $(M_d) = 150 + \frac{150 - 151}{116} \times 10$
170-180	175	22	381	$= 150 + \frac{150 - 151}{116} \times 10$
180-200	190	3	384	$M_d = 155.203$
		$N = 300$		

(21) Find the Mode of the following frequency distribution

Size	frequency	Size	frequency
1	8	9	20
2	8	10	45
3	15	11	29
4	23	12	6
5	35		
6	40		
7	52		
8	28		

here, higher no. of frequency is 45
Mode = 10

(22) Find the mode and median from the following table

Age	No. of students	C.F
0-10	2	2
10-20	18	20
20-30	30	50
30-40	45	95
40-50	35	130
50-60	20	150
60-70	6	156
70-80	3	159

for mode
Maximum No. of frequency
Modal class = 30-40
 $l = 30$, $h = 10$, $f_1 = 45$, $f_2 = 35$
 $\Delta_1 = f_1 - f_0 = 45 - 30 = 15$, $\Delta_2 = f_2 - f_1 = 35 - 45 = -10$
Mode = $l + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times h = 30 + \frac{15}{15 - 10} \times 10 = 30 + 30 = 60$

(23) Calculate the mode of the following distribution monthly wages

Monthly wages (Rs.)	No. of workers
50-70	4
70-90	44
90-110	38
110-130	28
130-150	6
150-170	8
170-190	12
190-210	2
210-230	2

for median
 $N/2 = 159/2 = 79.5$
Cumulative frequency is greater
Median class = 90-110
 $M_d = l + \frac{N/2 - C.F}{f} \times h$
 $= 90 + \frac{79.5 - 50}{38} \times 20 = 90 + 15.5 = 105.5$

Maximum No. of workers is 44
Modal class = 70-90
 $l = 70$, $h = 20$, $f_1 = 44$, $f_0 = 4$, $f_2 = 38$
 $\Delta_1 = f_1 - f_0 = 44 - 4 = 40$
 $\Delta_2 = f_2 - f_1 = 38 - 44 = -6$
Mode = $l + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times h = 70 + \frac{40}{40 - 6} \times 20 = 70 + 17.3 = 87.3$

An incomplete distribution of families according to their expenditure per week is given below. The median & mode of the distribution are Rs. 25 and Rs. 24 respectively. Calculate the missing frequencies:-

Expenditure	No. of families	Cf
0-10	14	14
10-20	9 (x) let	14+x
20-30	27	41+x
30-40	9 (y) let	41+y
40-50	15	56+x+y

for Median

$$N/2 = \frac{56+x+y}{2}$$

Cumulative frequency is greater

$$\text{Then } N/2 \text{ is } 41+x$$

$$\text{Median class} = 20-30$$

$$L = 20, M = 25, C_1 = 14+x, f = 27$$

$$M = L + \frac{N/2 - C_1}{f} \times h$$

$$25 = \frac{56+x+y}{2} - \frac{(14+x)}{27} \times 10$$

$$27 = \frac{56+x+y-28-2x}{27} \times 10$$

$$\text{or } 27-x+y = 27$$

$$x-y = 0 \quad \text{--- (1)}$$

solving eqⁿ (1) & (11)

$$27 - \frac{x}{2} = 27$$

$$(27 - \frac{x}{2} = 27) \times 2$$

$$x = 25$$

$$y = x-1 = 25-1 = 24$$

for Mode

$$\text{Maximum frequency} = 27$$

$$\therefore \text{Modal class} = 20-30$$

$$L = 20, f_1 = 9, f_2 = 27, f_3 = 9, P_0 = x, P_4 = y$$

$$C_1 = f_1 - f_0 = 27-x$$

$$C_2 = f_3 - f_2 = 27-y$$

we have,

$$\text{Mode} = L + \frac{C_1}{C_1 + C_2} \times h$$

$$\text{or, } 24 = 20 + \frac{27-x}{27-x+27-y} \times 10$$

$$\text{or, } 4 = \frac{270-10x}{54-x-y}$$

$$\text{or, } 216-4x-4y = 270-10x$$

$$\text{or } 6x-4y = 54$$

$$3x-2y = 27 \quad \text{--- (11)}$$

$$\therefore x = 25$$

$$y = 24$$

(28) Compute the geometric mean of the following data:

x:	10	15	18	20	25
f:	2	3	5	6	4
Σf	x	f	$\log x$	$f \log x$	$G = \text{Antilog} \left[\frac{\Sigma f \log x}{N} \right]$
10	2	1		2	$= \text{Antilog} \frac{2.5907}{20}$
15	3	1.176	3.528		$= 18.2$
18	5	1.255	6.275		
20	6	1.301	7.806		
25	4	1.398	5.592		
	$N = 20$		$\Sigma f \log x = 25.293$		

(29) Find the G.M. and Harmonic Mean of the following data

Variable	Frequency (f)	Mid (x)	$f \log x$	f/x
75-125	10	100	20	$10 \times \frac{1}{100} = 0.10$
125-175	20	150	43.52	0.133
175-225	30	200	69.03	0.15
225-275	6	250	14.39	0.024
275-325	4	300	9.91	0.0133
	$N = 90$		$\Sigma f \log x = 156.85$	$\Sigma f/x = 0.42$

$$\text{Geometric Mean (G)} = \text{Antilog} \left[\frac{\Sigma f \log x}{N} \right]$$

$$= \text{Antilog} \left[\frac{156.85}{90} \right]$$

$$= 174.06 // 2$$

$$\text{Harmonic Mean (H.M)} = \frac{N}{\Sigma f/x} = \frac{90}{0.42}$$

$$= 214.2857 //$$

(30) Calculate A.M, G.M and H.M of the following data

	$\log x$	$1/x$	
32	1.505	0.03125	$A.M = \frac{32+35+36+37+39+41}{6} = 36.67$
35	1.544	0.02857	$G.M = \text{Antilog} \left[\frac{\Sigma f \log x}{N} \right] = \text{Antilog} \left[\frac{9.77}{6} \right] = 36.67$
36	1.556	0.02777	
37	1.568	0.02703	$H.M = \frac{N}{\Sigma 1/x} = \frac{6}{0.1644} = 36.44$
39	1.591	0.02564	
41	1.613	0.02439	
	$\Sigma \log x = 9.37$	$\Sigma 1/x = 0.1644$	Now, $A.M > G.M > H.M$ proved

(21) The marks obtained by 25 students in a test are given below

Roll No.	Marks (x)	No. of students (f)	$\frac{1}{f}$	$f \cdot \frac{1}{x}$
11	11	3	0.0909	0.2727
12	12	7	0.0833	0.5831
13	13	8	0.0769	0.6152
14	14	3	0.0754	0.2142
15	15	2	0.0666	0.1332
H.M = 9	$N = 23$			$\Sigma f \cdot \frac{1}{x} = 1.8267$

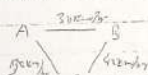
$$\text{Harmonic Mean (H.M.)} = \frac{N}{\Sigma f \cdot \frac{1}{x}} = \frac{23}{1.8267} = 12.65$$

(22) Compute the harmonic mean of the following data

Class	Frequency (f)	Mid (x)	$F \cdot \frac{1}{x}$
0-10	4	5	0.8
10-20	6	15	0.4
20-30	10	25	0.4
30-40	7	35	0.2
40-50	5	45	0.066
	$\Sigma f = 30$		$\Sigma F \cdot \frac{1}{x} = 1.867$

$$\text{H.M.} = \frac{N}{\Sigma F \cdot \frac{1}{x}} = \frac{30}{1.867} = 16.071$$

(23) Three cities A, B, C are equidistant from each other. A infinitesimal from A to B at 30 km/hr, from B to C at 40 km/hr and from C to A at 50 km/hr. Determine the average speed.



$$\text{Average Speed} = \frac{\text{Total distance travelled}}{\text{Time taken}} = \frac{3}{\frac{1}{30} + \frac{1}{40} + \frac{1}{50}} = 38.597 \text{ km/hr}$$

(24) Find the median, quartiles, 7th deciles and 85th percentile for the following data.

Monthly Rent	No. of families (f)	C.F.	Cumulative
20-40	6	6	
40-60	9	15	
60-80	11	26	
80-100	14	40	
100-120	20	60	
120-140	15	75	
140-160	10	85	
160-180	8	93	
180-200	7	100	

for median

$$N/2 = 100/2 = 50$$

C.F. is greater than $N/2$ is 60

Median class = 100-120

$$M_d = 100 + \frac{50 - 40}{20} \times 20 = 110$$

for 7th deciles

$$\frac{7N}{10} = \frac{7 \times 100}{10} = 70$$

C.F. is greater than $7N/10 = 75$

class = 120-140

$$D_7 = 120 + \frac{70 - 60}{20} \times 20$$

$$= 120 + \frac{10}{20} \times 20$$

$$= 133.33$$

for quartiles

$$\text{1st Quartile} = N/4 = 100/4 = 25$$

First quartile class is 40-60

class = 60-80

$$Q_1 = 60 + \frac{25 - 15}{20} \times 20$$

$$= 60 + \frac{10}{20} \times 20 = 70$$

$$\text{for } Q_3, \frac{3N}{4} = \frac{3 \times 100}{4} = 75$$

C.F. is greater than $3N/4$ is 85

class = 120-140

$$Q_3 = 120 + \frac{75 - 60}{20} \times 20$$

$$= 140$$

for 85th percentile

$$\frac{85N}{100} = \frac{85 \times 100}{100} = 85$$

C.F. is greater than $85N/100$ is 93

class = 160-180

$$P_{85} = 160 + \frac{85 - 75}{20} \times 20$$

$$= 170$$

$$M_d = 110, Q_1 = 70, Q_3 = 140, P_{85} = 170$$

Q. An incomplete frequency distribution is given as follows.

Variable	Frequency	C.F	hence,
10-20	12	12	$N/2 = 229/2 = 114.5$
20-30	30	42	C.f is greater than $N/2$ is 114.5
30-40	7 (x)	42+x	class = 60-70
40-50	65	107+x	$M_1 = 1 + N/2 - C.f \times i$
50-60	9 (y)	107+x+y	$A_1 = 40 + \frac{114.5 - (42+x)}{(107+x+y)} \times 10$
60-70	25	132+x+y	
70-80	18	229	

And $x+y = (229 - 12 - 30 - 65 - 18 - 9) = 95$

$34 + y = 79$

$y = 79 - 34 = 45$

$\therefore x = 34$
 $y = 45$

Q. Find the median, lower and upper quartiles for following distribution

Classes	No. of students	Ordering marks	L	C.F
below 10	15	0-10	15	15
" 20	25	10-20	35-15=20	35
" 30	60	20-30	60-35=25	60
" 40	84	30-40	84-60=24	84
" 50	94	40-50	94-84=10	94
" 60	127	50-60	127-94=33	127
" 70	198	60-70	198-127=71	198
" 80	249	70-80	249-198=51	249

hence, $N/2 = 249/2 = 124.5$

C.f is greater than $N/2$ is 127

Median class = 50-60

$M_2 = 50 + \frac{124.5 - 94}{84} \times 10$

for lower quartile

$N/4 = 249/4 = 62.25$

C.f is greater than $N/4$ is 84

class = 30-40

$Q_1 = 30 + \frac{62.25 - 25}{84} \times 10$

for upper quartile

$3N/4 = 3 \times 249/4 = 186.75$

C.f greater than $3N/4$ is 198

class = 60-70

$Q_3 = 60 + \frac{186.75 - 127}{71} \times 10$
 $Q_3 = 68.415$

Q. The median and mode of the following wage distribution are known to be Rs. 33.5 and Rs. 34 resp. Three frequency values from the table are however, missing. You are required to find out these values.

Wages in Rs.	No. of persons	C.f
0-10	4	4
10-20	16	20
20-30	7 (x)	20+x
30-40	7 (y)	20+x+y
40-50	7 (z)	20+x+y+z = 220
50-60	6	226
60-70	4	230

hence,

$N/2 = \frac{230}{2} = 115$

C.f is greater than $N/2$ is $20+x+y$

class = 30-40

Median = $30 + \frac{115 - (20+x)}{y} \times 10$

$33.5 - 30 = \frac{95 - 20 - x}{y} \times 10$
 $3.5 \times y = 75 - x$
 $7 + 0.35y = 7.5 - x$

$33.5 - 50 = \frac{115 - 20 - x}{y} \times 10$

$3.5 \times y = 95 - x$

$7 + 0.35y = 9.5 - x$

$x = 60$
 $y = 70$
 $z = 40$

hence,
 $M_0 = 1 + \frac{0.1}{0.1+0.2} \times 4$

$Q_1 = 30 + \frac{4-x}{7-x+y+z-20} \times 10$

$1.24 - 30 = \frac{4-x}{7-x+y+z-20} \times 10$

$1.24 - 30 = \frac{4-x}{y} \times 10$

$7 + 0.24y = 30$

$7 + 0.24y = 30$

$7 + 0.24y = 30$

$7 + 0.24y = 30$

$7 + 0.24y = 30$

$7 + 0.24y = 30$

Q5 Find the mode of the following frequency distribution

Size	frequency (f)
1	3
2	8
3	15
4	23
5	35
6	40
7	32
8	28
9	20
10	15
11	14
12	6

Max^m frequency = 40
 $\therefore$ mode = 6

Q6 Given below is the distribution of 140 candidates obtaining marks 8 or higher in certain examination. All marks are given in whole numbers.

Mark	Class	f
8	10-20	140-125 = 15
10	20-30	137-118 = 19
12	30-40	118-100 = 18
14	40-50	100-75 = 25
16	50-60	75-45 = 30
18	60-70	45-25 = 20
20	70-80	25-9 = 16
22	80-90	9-2 = 7
24	90-100	2
26		0

CHAPTER-5

Measure of Dispersion

The measure of dispersion is the scatterness of the from the central value.

- Range
- Quartile deviation
- Mean deviation
- Standard deviation

(i) Range

Range is defined as

$$\text{Range} = \text{Largest Item} - \text{Smallest Item} \\ = L - S$$

$$\text{Coefficient of range} = \frac{L - S}{L + S} \quad (0 < x < 1)$$

(ii) Quartile deviation (Q.D.) is defined as

$$Q.D. = \frac{Q_3 - Q_1}{2} \quad \text{where } Q_1, Q_2, Q_3 \text{ are first, 2nd, 3rd, 4th}$$

It is also called semi-Interquartile range.

(iii) Mean Deviation

a) Mean deviation from mean

Let $x_1, x_2, \dots, x_n$ be n variate value with mean

Median (M_d) & Mode (M_o)

then mean deviation from mean

$$= \frac{\sum |x - \bar{x}|}{n} \quad \text{or} \quad \frac{\sum |x - \bar{x}|}{n}$$

b) Mean deviation from median

$$= \frac{\sum |x - M_d|}{n} \quad \text{or} \quad \frac{\sum |x - M_d|}{n}$$

c) Mean deviation from mode

$$= \frac{\sum |x - M_o|}{n} \quad \text{or} \quad \frac{\sum |x - M_o|}{n}$$

Standard deviation

Let $x_1, x_2, \dots, x_n$ be n variate values, then population standard deviation is denoted by σ and defined by

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} \text{ or } \sqrt{\frac{\sum x^2}{n} - (\bar{x})^2} \text{ or } \sqrt{\frac{\sum f(x - \bar{x})^2}{n}}$$

Variance

It is defined as square of standard deviation in variance.

$$\sigma^2 = \frac{\sum (x - \bar{x})^2}{n} \text{ or } \frac{\sum x^2}{n} - (\bar{x})^2 \text{ or } \frac{\sum f(x - \bar{x})^2}{n}$$

Coefficient of Variance

It is denoted by C.V. & defined by

$$C.V. = \frac{\sigma}{\bar{x}} \times 100\%$$

It is always expressed in percentage

Note: - The set of data is more consistent or uniform where C.V. is less than that of other set of data.

Combined mean and combined standard deviation

1st set of data 2nd set of data n^{th} set of data

Size = n_1 Size = n_2 Size = n_n

Mean = $\bar{x}_1$ Mean = $\bar{x}_2$ Mean = $\bar{x}_n$

S.D = σ_1 S.D = σ_2 S.D = σ_n

When they are combined,

$$\text{Combined Size (n)} = n_1 + n_2 + \dots + n_n$$

$$\text{Combined mean } (\bar{x}) = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + \dots + n_n\bar{x}_n}{n_1 + n_2 + \dots + n_n}$$

$$\text{Combined standard deviation } (\sigma) = \sqrt{\frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2) + \dots + n_n(\sigma_n^2 + d_n^2)}{n_1 + n_2 + \dots + n_n}}$$

$$\text{where, } d_1 = \bar{x}_1 - \bar{x}$$

$$d_2 = \bar{x}_2 - \bar{x}$$

$$d_n = \bar{x}_n - \bar{x}$$

① Calculate the quartile deviation of 63 students in commercial method.

Given below:

Marks	No. of students	Cf	for B_1
0-10	5	5	$N/4 = 63/4 = 15.75$
10-20	7	12	Cf is greater than $N/4$ is 22
20-30	10	22	Class = 20-30
30-40	16	38	$B_1 = 20 + \frac{15.75 - 12}{10} \times 10 = 23.75$
40-50	11	49	for B_3
50-60	7	56	$3N/4 = 3 \times 63/4 = 47.25$
60-70	3	59	Cf is greater than $3N/4$ is 49
70-80	2	61	Class = 40-50
80-90	2	63	$B_3 = 40 + \frac{47.25 - 49}{11} \times 10 = 48.41$
90-100	0	63	Now, Quartile deviation (Q.D.) = $B_3 - B_1$

② Find the mean deviation from mean of the following data:

Class	Frequency	Marks (x)	Σfx	$ x - \bar{x} $	$\Sigma f x - \bar{x} $
0-4	8	2	16	11.05	88.4
6-12	10	9	90	5.05	50.5
12-18	12	15	180	0.95	11.4
18-24	9	21	189	6.95	62.55
24-30	5	27	135	12.95	64.75
	N=44		$\Sigma fx = 610$	$\Sigma f x - \bar{x} = 222.6$	

$$\text{Mean } (\bar{x}) = \frac{\Sigma fx}{N} = \frac{610}{44} = 13.86$$

$$\text{Mean deviation from mean} = \frac{\Sigma f|x - \bar{x}|}{N} = \frac{222.6}{44} = 5.06$$

③ Following figures gives the production of non-fatty dry milk for twelve months of 2007. Calculate (i) Range (ii) Mean deviation from (iii) Variance (iv) Coefficient of variation.

Month	Production (million lbs)	$f(x - \bar{x})$
January	82.5	1.975
February	81.4	3.845
March	95.8	10.975
April	111.5	26.075
May	121.4	45.275
June	124.5	41.075
July	98.2	13.275
August	74.2	9.275
September	53.2	32.275
October	50.3	35.175
November	49.3	36.175
December	67.1	18.375

① Range = $L - S = 124.5 - 49.3 = 75.2$

(ii) Mean deviation about mean

$$\frac{\sum (x - \bar{x})}{n} = \frac{293.55}{12} = 24.46$$

(iii) Variance

$$\sigma^2 = \frac{\sum (x - \bar{x})^2}{n} = \frac{8781.10}{12} = 731.758$$

(iv) Coefficient of variance

$$C.V. = \frac{\sigma}{\bar{x}} \times 100\% = \frac{27.05}{31.66\%}$$

Mean ($\bar{x}$) = $\frac{\sum x}{n} = \frac{1025.10}{12} = 85.425$

④ The following table gives the number of branches and number of plants in a patch of field. Calculate the mean deviation about median and find variance of the number of branches.

No. of branches (x)	No. of plants (f)	$f(x - M_d)$	$f \times f(x - \bar{x})^2$
2	16	32	3.3856 x 16
3	13	39	0.7521 x 13
4	19	76	0.0081 x 19
5	12	60	1.3456 x 12
6	5	30	4.6656 x 5
7	4	28	9.0025 x 4

Now, Maximum frequency among all values is 19

∴ Median $N/2 = 69/2 = 34.5$

C.f. is greater than $N/2$, 64.8

∴ Median $M_d = 4$

Mean deviation from median = $\frac{\sum f(x - M_d)}{n} = \frac{79}{69} = 1.144$

⑤ Age distribution of hundred life insurance policy holder is as follows. Calculate the mean deviation from median age.

Age	Exclusive age	Frequency	$f(x - M_d)$
17-19.5	18.5	9	180
20-25.5	22.5	16	360
26-35.5	30.5	12	360
36-40.5	38.5	26	1020
41-50.5	45.5	14	630
51-55.5	53.5	12	640
56-60.5	58.5	6	350
61-70.5	65.5	5	325

Now, $N = 100$
 $N/2 = 50$
 C.f. is greater than $N/2$, 63

Median class = 35.5 - 40.5

$M_d = 35.5 + \frac{50 - 63}{26} \times 5 = 38.25$

Mean deviation from median =

⑥ Calculate mean & standard deviation of the following distribution

Size (x)	f	Mutually	$f(x - \bar{x})^2$
2.5-7.5	12	5	7503.0
7.5-12.5	25	10	11205.0
12.5-17.5	65	15	14634.75
17.5-22.5	121	20	12112.0
22.5-27.5	175	25	4383.75
27.5-32.5	198	30	0
32.5-37.5	174	35	4391.00
37.5-42.5	120	40	11988.00
42.5-47.5	46	45	14840.10
47.5-52.5	27	50	10754.0
52.5-57.5	9	55	5622.75
57.5-62.5	3	60	2699.10

$\bar{x} = \frac{\sum fx}{N} = \frac{80005}{1000} = 80.005$

$\sigma^2 = \frac{\sum f(x - \bar{x})^2}{n} = \frac{100194.85}{1000}$

- ① A cellular manufacture is considering the production of a new type of collar to attract young men. The following statistics of neck circumference are available based upon the measurement of typical group of college students: Compute mean, standard deviation and variance.

Mid value (x)	No. of students (f)	f ₁	f(x - $\bar{x}$) ²
12.5	4	50	11.92
13.5	12	242	28.745
13.5	30	405	15.387
14.5	65	852	3.33
14.5	66	987	4.811
15.5	29	435	17.104
15.5	18	272	20.03
16.5	1	16	3.13
16.5	1	165	5.15

$$N = 231$$

$$\Sigma f x = 3287.50$$

$$\Sigma f(x - \bar{x})^2 = 119.342$$

$$\bar{x} = \frac{\Sigma f x}{N} = \frac{3287.50}{231} = 14.23$$

$$\text{Standard deviation } (\sigma) = \sqrt{\frac{\Sigma f(x - \bar{x})^2}{N}} = \sqrt{\frac{119.342}{231}} = 0.516$$

$$\sigma = 0.718$$

- ② Two persons participated in fire shooting competition and were able to hit the target correctly out of 10 shots as given below. Find which of the competitors is more consistent in shooting performance.

Competitor A (x _i)	(x _i - $\bar{x}_1$) ²	Competitor B (y _i)	(y _i - $\bar{y}_2$) ²
6	11.56	12	9
12	6.76	15	36
12	6.76	7	4
10	0.36	7	4
7	5.76	4	25
	$\Sigma (x - \bar{x}_1)^2 = 31.2$		$\Sigma (y - \bar{y}_2)^2 = 78$

$$\text{Mean A } (\bar{x}_1) = \frac{6+12+12+10+7}{5} = 9.4$$

$$\text{Mean A B } (\bar{y}_2) = \frac{12+15+7+7+4}{5} = 11$$

$$\sigma_1 = \sqrt{\frac{\Sigma (x - \bar{x}_1)^2}{n}} = \sqrt{\frac{31.2}{5}} = 2.5$$

$$\sigma_2 = \sqrt{\frac{\Sigma (y - \bar{y}_2)^2}{n}} = \sqrt{\frac{78}{5}} = 3.95$$

Coefficient of variance of A

Coefficient of variance of B

$$C.V. = \frac{\sigma_1}{\bar{x}_1} \times 100 = \frac{2.5}{9.4} \times 100 = 26.57\%$$

$$C.V. = \frac{\sigma_2}{\bar{y}_2} \times 100 = \frac{3.95}{11} \times 100 = 35.88\%$$

The competitor A is more consistent than B because coefficient of variance of A is less than coefficient of variance of B.

- ③ Scores of two golfers for 24 rounds were as follows:

Score of Golfer A	f ₁	f ₁	f(x - $\bar{x}_1$) ²
70	1	70	36
71	1	71	25
72	3	216	48
73	1	73	9
74	3	222	72
75	2	150	2
76	1	76	0
77	2	154	2
78	4	312	16
79	2	158	38
80	2	160	32
81	2	162	50
	$N = 24$	$\Sigma f x = 1824$	$\Sigma f(x - \bar{x}_1)^2 = 280$

$$\text{Mean } (\bar{x}_1) = \frac{\Sigma f x}{N} = \frac{1824}{24} = 76$$

-28

Golfer B plays is more consistent than A because coefficient
 of variance of B is less than coefficient of variance of A.
 (10) Lives of two models of refrigerators turned in for new models
 in a recent survey are: what is the average life of each model
 of these refrigerators? which model has the variability?

$$\begin{aligned} \bar{x}_1 &= \frac{\sum x_1}{N} = \frac{256}{50} = 5.12 \text{ years} \\ \bar{x}_2 &= \frac{\sum x_2}{N} = \frac{308}{50} = 6.16 \text{ years} \end{aligned} \quad \left. \vphantom{\begin{aligned} \bar{x}_1 &= \frac{\sum x_1}{N} = \frac{256}{50} = 5.12 \text{ years} \\ \bar{x}_2 &= \frac{\sum x_2}{N} = \frac{308}{50} = 6.16 \text{ years} \end{aligned}} \right\} e.$$

Find the missing information from the following data.

	Group I	Group II	Group III	Combined
Number	50	?	90	200
S.D.	6	7	?	7.745
Mean	113	?	115	116

Combined size (n) = $n_1 + n_2 + n_3$
 $200 = 50 + n_2 + 90$
 $n_2 = 60$

Combined mean ($\bar{x}$) = $\frac{n_1\bar{x}_1 + n_2\bar{x}_2 + n_3\bar{x}_3}{n_1 + n_2 + n_3}$
 $116 = \frac{50 \times 113 + 60\bar{x}_2 + 90 \times 115}{50 + 60 + 90}$

Combined standard deviation (σ)

$$= \sqrt{\frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2) + n_3(\sigma_3^2 + d_3^2)}{n_1 + n_2 + n_3}}$$

$d_1 = \bar{x}_1 - \bar{x} = 113 - 116 = -3$
 $d_2 = \bar{x}_2 - \bar{x} = 120 - 116 = 4$
 $d_3 = \bar{x}_3 - \bar{x} = 115 - 116 = -1$

$$7.745^2 = \frac{50\{6^2 + (-3)^2\} + 60\{7^2 + 4^2\} + 90\{9^2 + (-1)^2\}}{200}$$

$$1997.05 = \frac{50 \times 45 + 60 \times 65 + 90 \times 82}{200}$$

$$63^2 = 63.066$$

$$\sigma_2 = 7.99848$$

A student obtained the mean and standard deviation of 100 observations as 40 and 5 resp. It was later discovered that he had wrongly copied down an observation as 50 instead of 40. Calculate the correct mean and standard deviation.

Mean of observation (n) = 100
 Incorrect mean ($\bar{x}$) = 40
 " deviation (σ) = 5

Total no. of observation = $n \cdot \bar{x} = 40 \times 100 = 4000$

Correct observation = $4000 - 50 + 40 = 3990$

Correct mean ($\bar{x}$) = $\frac{3990}{100} = 39.9$

for standard deviation

$$\sigma = \sqrt{\frac{\sum x^2}{n} - (\bar{x})^2}$$

$$5 = \sqrt{\frac{\sum x^2}{100} - 40^2}$$

$$(25 + 160) \times 100 = \sum x^2$$

$$\sum x^2 = 162500 \text{ (Incorrect)}$$

Corrected $\sum x^2 = 162500 - 80^2 + 40^2$
 $= 162500 - 2500 + 1600$
 $= 161600$

Corrected $\sigma = \sqrt{\frac{\sum x^2}{n} - (\bar{x})^2}$
 $= \sqrt{\frac{161600}{100} - 39.9^2}$
 $= 4.898$

The scores of two golfers for 10 rounds each are which may be regarded as the more consistent player?

Player A	$(x - \bar{x}_1)^2$	Player B	$(x - \bar{x}_2)^2$
58	9	84	182.25
59	4	56	210.25
60	1	92	462.25
54	49	65	30.25
65	16	86	240.25
66	25	78	56.25
57	121	44	702.25
75	196	54	272.25
60	64	28	56.25
52	181	68	6.25

$\sum x = 610$
 $\bar{x}_1 = \frac{\sum x}{n} = \frac{610}{10} = 61$

$$\sigma_1 = \sqrt{\frac{\sum (x - \bar{x}_1)^2}{n}}$$

$$= \sqrt{\frac{776}{10}}$$

$$= 8.82$$

$CV_A = \frac{\sigma_1}{\bar{x}_1} \times 100 = \frac{8.82}{61} \times 100 = 14.47\%$

player A is more consistent

$\sum x = 705$
 $\bar{x}_2 = \frac{\sum x}{n} = \frac{705}{10} = 70.5$

$$\sigma_2 = \sqrt{\frac{\sum (x - \bar{x}_2)^2}{n}}$$

$$= \sqrt{\frac{2218.5}{10}}$$

$$= 47.12$$

$CV_B = \frac{\sigma_2}{\bar{x}_2} \times 100$
 $= \frac{47.12}{70.5} \times 100$
 $= 66.84\%$

(A) The heights and weights of 10 army men are given below

In which characteristics are they more variable?

Sn ⁿ Height (in cm)	$(x - \bar{x})^2$	weights (in kg)	$(y - \bar{y})^2$
170	16	75	0
172	4	74	1
168	36	75	0
177	9	76	1
173	25	77	4
179	25	73	4
171	9	76	1
173	1	75	0
173	16	74	1
173	1	75	0

$$\Sigma x = 1740$$

$$\Sigma (x - \bar{x})^2 = 142$$

$$\Sigma y = 750$$

$$\Sigma 12$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{1740}{10} = 174$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{750}{10} = 75$$

$$s_2 = \sqrt{\frac{\Sigma (y - \bar{y})^2}{n}} = \sqrt{\frac{12}{10}} = 1.09$$

$$\text{variance } (s_2^2) = 1.2$$

$$s_1 = \sqrt{\frac{\Sigma (x - \bar{x})^2}{n}} = \sqrt{\frac{142}{10}} = 3.77$$

$$\text{Variance } (s_1^2) = 14.2$$

Height is more variable i.e. $s_1^2 = 14.2$

The following are scores of two batsmen A and B in a series of 10 innings. Who is the better run getter and who is more consistent?

Score of Batsman A	$(x - \bar{x})^2$	Score of Batsman B	$(y - \bar{y})^2$
12	1444	97	289
115	4225	12	324
6	1936	16	196
73	529	42	144
7	1849	4	676
19	961	51	441
119	4761	37	49
36	296	48	324
84	1156	43	169
29	441	0	900
$\Sigma x = 500$	$\Sigma (x - \bar{x})^2 = 17498$	$\Sigma y = 310$	$\Sigma (y - \bar{y})^2 = 312$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{500}{10} = 50$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{310}{10} = 31$$

$$s_1 = \sqrt{\frac{\Sigma (x - \bar{x})^2}{n}} = \sqrt{\frac{17498}{10}} = 41.83$$

$$C.V. = \frac{s_1}{\bar{x}} \times 100 = \frac{41.83}{50} \times 100 = 83.66\%$$

$$s_2 = \sqrt{\frac{\Sigma (y - \bar{y})^2}{n}} = \sqrt{\frac{312}{10}} = 17.79$$

$$C.V. = \frac{s_2}{\bar{y}} \times 100 = \frac{17.79}{31} \times 100 = 57.4\%$$

The better run getter player is A because his average is 50 & B player B is more consistent than A because C.V. of B is 57.4%.

(D) The average and standard deviation of the heights of two groups of students, numbering 100 and 200 respectively, are given below. Find the average and standard deviation of the heights of all the 300 students taken together.

	Group I	Group II
Average (cm)	150	165
S.D. (cm)	5	10

Snⁿ we have

$$\text{Combined mean } (\bar{x}) = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} = \frac{100 \times 150 + 200 \times 165}{300} = 160$$

$$\text{Combined Standard deviation } (s) = \sqrt{\frac{n_1 (s_1^2 + d_1^2) + n_2 (s_2^2 + d_2^2)}{n_1 + n_2}}$$

$$d_1 = \bar{x} - \bar{x}_1 = 160 - 150 = 10$$

$$d_2 = \bar{x} - \bar{x}_2 = 160 - 165 = -5$$

$$s = \sqrt{\frac{100(5^2 + 10^2) + 200(10^2 + (-5)^2)}{300}} = 11.05$$

Q. The first of the two groups has 100 items with mean 45 and variance 45. If the combined group has 250 items with mean 51 and variance 130, find the mean & standard deviation of second group.

1st group	2nd group	Combined
$n_1 = 100$	$n_2 = ?$	$n = 250$
$\bar{x}_1 = 45$	$\bar{x}_2 = ?$	$\bar{x} = 51$
$\sigma_1^2 = 45$	$\sigma_2^2 = ?$	$\sigma^2 = 130$
$\sigma_1 = 7$	$\sigma_2 = ?$	$\sigma = 11.40$
$n_2 = n - n_1 = 250 - 100 = 150$		

Combined mean ($\bar{x}$) = $\frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$

$$51 = \frac{100 \times 45 + 150 \times \bar{x}_2}{250}$$

$$\bar{x}_2 = \frac{250 \times 51 - 100 \times 45}{150}$$

$$\bar{x}_2 = 57$$

Combined standard deviation

$$\sigma = \sqrt{\frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}}$$

$$d_1 = \bar{x} - \bar{x}_1 = 51 - 45 = 6$$

$$d_2 = \bar{x} - \bar{x}_2 = 51 - 57 = -6$$

$$(11.4)^2 = \frac{100(45 + 36) + 150(\sigma_2^2 + 36)}{250}$$

$$\sigma_2^2 = \frac{250 \times 130 - 8500}{150} = 16$$

$$\sigma_2 = 4$$

Q. The means of two samples of size 50 and 100 resp 53.1 and 50.3 and the standard deviations are 8.7. Obtain the mean & standard deviation of the sample of size 150 obtained by combining the two samples.

Sample 1	Sample 2	Combined
$n_1 = 50$	$n_2 = 100$	$n = 150$
$\bar{x}_1 = 54.1$	$\bar{x}_2 = 52.8$	$\bar{x} = ?$
$\sigma_1 = 8$	$\sigma_2 = 7$	$\sigma = ?$

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} = \frac{50 \times 54.1 + 100 \times 52.8}{150} = 53.567$$

$$\sigma = \sqrt{\frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}}$$

$$d_1 = \bar{x} - \bar{x}_1 = 53.567 - 54.1 = -0.533$$

$$d_2 = \bar{x} - \bar{x}_2 = 53.567 - 52.8 = 0.767$$

$$\sigma = \sqrt{\frac{50(8^2 + 0.284) + 100(7^2 + 0.588)}{150}} = 7.56$$

Q. A distribution consists of three components with frequencies 200, 250 and 100 having means 25, 30 and 15, standard deviations 3, 4 and 5. Show that the mean of the combined group is 26 and its standard deviation is 2 approximately.

Comp 1	Comp 2	Comp 3	Combined Comp
$n_1 = 200$	$n_2 = 250$	$n_3 = 100$	$n = 550$
$\bar{x}_1 = 25$	$\bar{x}_2 = 30$	$\bar{x}_3 = 15$	$\bar{x} = ?$
$\sigma_1 = 3$	$\sigma_2 = 4$	$\sigma_3 = 5$	$\sigma = ?$

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + n_3\bar{x}_3}{n_1 + n_2 + n_3} = \frac{200 \times 25 + 250 \times 30 + 100 \times 15}{550} = 26.36$$

$$\sigma = \sqrt{\frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2) + n_3(\sigma_3^2 + d_3^2)}{n_1 + n_2 + n_3}}$$

$$d_1 = \bar{x} - \bar{x}_1 = 26.36 - 25 = 1.36$$

$$d_2 = \bar{x} - \bar{x}_2 = 26.36 - 30 = -3.64$$

$$d_3 = \bar{x} - \bar{x}_3 = 26.36 - 15 = 11.36$$

$$\sigma = \sqrt{\frac{200(3^2 + 1.85) + 250(4^2 + 13.25) + 100(5^2 + 129)}{550}} = 7.80$$

Q. Find

Marks	Mid	Req	f	$f(x - \bar{x})^2$
29.5 - 34.5	32	5	160	731.25
34.5 - 39.5	37	10	370	562.50
39.5 - 44.5	42	15	630	922.50
44.5 - 49.5	47	30	1410	1571.25
49.5 - 54.5	52	5	260	281.25
54.5 - 59.5	57	5	285	751.25
Total 70			3115	$\Sigma f(x - \bar{x})^2 = 2857.50$

$$\bar{x} = \frac{\sum fx}{N} = \frac{3115}{70} = 44.50$$

Mean of the passed boys = 44.50

No. " " = 70

Overall mean = 38

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} = \frac{70 \times 44.5 + 30 \times \bar{x}_2}{100}$$

$$\text{or } 38 = \frac{70 \times 44.5 + 30 \bar{x}_2}{100}$$

$$\bar{x}_2 = \frac{3800 - 70 \times 44.5}{30} = 29.83$$

$$\sigma_B = \sqrt{\frac{\sum (1x - \bar{x}_1)^2}{N}}$$

$$= \sqrt{\frac{8687.50}{70}} = 6.196$$

Combined S.D = 10

-30-
32

$$\sigma^2 = \frac{n_B (\sigma_B^2 + d_1^2) + n_F (\sigma_F^2 + d_2^2)}{n_1 + n_2}$$

$$d_1 = \bar{x} - \bar{x}_B$$

$$100 \times 100 = 70 (6.196^2 + (-6.5)^2) + 30 (\sigma_F^2 + 15.12^2)$$

$$= (38 - 44.5)$$

$$\text{or, } (\sigma_F^2 + 15.12^2) = \frac{10000 - 5644.83}{30}$$

$$d_2 = 38 - 29.83$$

30

80 - mark 9

700626.224

Theory of probability:-

Random experiment

An experiment is called random experiment if its outcome (result) cannot be predicted with certainty.

Example:- Tossing a coin, Rolling a dice (1 to 6) are random experiment.

Sample space:- A set of all possible outcomes for a random experiment is called sample space and is denoted by S . For example

(i) In a coin tossing

$$S = \{H, T\}$$

(ii) In a dice rolling

$$S = \{1, 2, 3, 4, 5, 6\}$$

(iii) If two coins are tossed

$$S = \{HH, HT, TH, TT\} \quad \textcircled{HT} \quad \textcircled{TH}$$

Equally Likely event

Events are said to be equally likely if chances of obtaining each event is equal. Example In a fair coin

$$\begin{aligned} P(\text{obtaining head}) &= \frac{1}{2} \\ P(\text{ " Tail }) &= \frac{1}{2} \end{aligned}$$

Mutually exclusive event

Two or more events are called mutually exclusive if the happening of any one of them excludes the happening of all other in the same experiment.

For example:- In a coin tossing:

Event A = obtaining head

" B = " Tail

- In a dice rolling

Event A = obtaining even number (2, 4, 6)

" B = " odd " (1, 3, 5)

Various approaches to probability

Mathematical or classical or prior approach

If there are n exhaustive, mutually exclusive and equally likely outcomes of an experiment and m of them are favourable to an event A , then probability of happening event A is defined by

$$P(A) = \frac{\text{favourable no. of cases}}{\text{Total no. of cases}} = \frac{m}{n}$$

This definition was given by James Bernoulli.

Example :- (i) In a coin tossing

$$P(\text{obtaining head}) = \frac{\text{fav. no. of cases}}{\text{Total no. of cases}} = \frac{1}{2}$$

(ii) In rolling a dice $S = \{1, 2, 3, 4, 5, 6\}$
 $P(\text{obtaining 3}) = \frac{\text{fav. no. of cases}}{\text{Total no. of cases}} = \frac{1}{6}$

(iii) A box containing 7 red, 6 black & 5 green balls
 A ball is drawn
 $P(\text{obtaining red}) = \frac{\text{fav. no. of cases}}{\text{Total no. of cases}} = \frac{7}{18}$

7 R
6 B
5 G

Statistical approaches / empirical

Suppose that an event A occurs ' m ' times in ' n ' repetitions of a random experiment where n is sufficiently large. Then probability of an event A is

$$P(A) = \lim_{n \rightarrow \infty} \frac{m}{n}$$

Modern or Axiomatic Approach

Let A be any event of sample space S . Then probability of an event A i.e. $P(A)$ satisfies the following axioms:

Axiom I: $P(A)$ is defined & non-negative

" II: $P(S) = 1$

" III: If $A_1, A_2, A_3, \dots, A_n$ are mutually exclusive events then

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$$

Permutation

$$P(n, r) = \frac{n!}{(n-r)!}$$

- digit
 - position
 - chair
 A, B, C
 AB, BA
 BC, CB
 AC, CA

Combination

$$C(n, r) = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$\frac{n!}{(n-r)!r!}$
 A, B, C
 $r=2$
 AB/BA
 BC/CB
 AC/CA

Theorem-1
 Probability of impossible event is zero i.e. $P(\phi) = 0$

Proof

we have,

$$S \cup \phi = S$$

Taking Probability both side

$$P(S \cup \phi) = P(S)$$

$$P(S) + P(\phi) = P(S)$$

Since S & ϕ are mutually exclusive

$$P(\phi) = 0 \quad \text{proved}$$

Theorem-2

If $\bar{A}$ is complementary event of A , then
 $P(\bar{A}) = 1 - P(A)$

Proof

we have,

$$A \cup \bar{A} = S$$

Taking Prob both side

$$P(A \cup \bar{A}) = P(S)$$

$$P(A) + P(\bar{A}) = P(S)$$

(Since A & $\bar{A}$ are mutually exclusive)

$$P(A) + P(\bar{A}) = 1$$

[$\because P(S) = 1$]

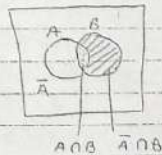
$$P(\bar{A}) = 1 - P(A)$$

proved



for any events A & B defined in sample space S

$$P(\bar{A} \cap B) = P(B) - P(A \cap B)$$



Proof:

$$B = (A \cap B) \cup (\bar{A} \cap B)$$

Adding prob. both side

$$P(B) = P[(A \cap B) \cup (\bar{A} \cap B)]$$

Since $(A \cap B)$ & $(\bar{A} \cap B)$ are mutually exclusive

$$P(B) = P(A \cap B) + P(\bar{A} \cap B)$$

$$P(\bar{A} \cap B) = P(B) - P(A \cap B) \text{ proved}$$

Similarly,

$$P(A \cap \bar{B}) = P(A) - P(A \cap B)$$

~~addition theorem of probability~~

A and B are not mutually exclusive

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof:

From Figure

$$A \cup B = A \cup (\bar{A} \cap B)$$

Adding prob. both side

$$P(A \cup B) = P[A \cup (\bar{A} \cap B)]$$

$$P(A) + P(B) = P(A) + P(\bar{A} \cap B) \text{ [Since } A \text{ and } \bar{A} \cap B \text{ are mutually exclusive]}$$

--- (i)

Also

$$B = (A \cap B) \cup (\bar{A} \cap B)$$

Adding prob. both side

$$P(B) = P[(A \cap B) \cup (\bar{A} \cap B)]$$

Since $A \cap B$ and $\bar{A} \cap B$ are mutually exclusive

$$P(B) = P[A \cap B] + P[\bar{A} \cap B]$$

$$P(\bar{A} \cap B) = P(B) - P(A \cap B) \text{ --- (ii)}$$

Adding eqn (i) & (ii), (i) becomes

$$P(A) + P(B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) \text{ proved}$$

If A, B, C are not mutually exclusive

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

Proof:

$$P(A \cup B \cup C)$$

Suppose

$$B \cap C = \emptyset$$

Then,

$$P(A \cup B \cup C) = P(A \cup B)$$

$$= P(A) + P(B) - P(A \cap B) \text{ [using addition theorem]}$$

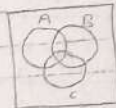
$$= P(A) + P(B \cup C) - P(A \cap (B \cup C))$$

$$= P(A) + P(B) + P(C) - P(B \cap C) - [P(A \cap B) + P(A \cap C)]$$

$$= P(A) + P(B) + P(C) - P(B \cap C) - P(A \cap B) - P(A \cap C)$$

$$= P(A) + P(B) + P(C) - P(B \cap C) - P(A \cap B) - P(A \cap C) + P(A \cap B \cap C)$$

proved



#

Conditional probability

Conditional probability of an event A given that an event B is already occurred is denoted by $P(A/B)$ & defined by

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

e.g.

Name of student	English (E)	Maths (M)	Name of student	English
A	P	P ✓	H	P
B	P	F	I	P
C	F	P	J	P
D	P	P ✓	K	P
E	F	F	L	P
F	P	F		
G	F	P		

passed in English = 8 passed in Math = 6
 failed " = 4 failed " = 6
 pass in both = 4

$$P(\text{passed in English}) = 8/12 \quad P(\text{passed in Math}) = 6/12$$

$$P(E \cap M) = P(E \cap M) = 4/12$$

$$P(E/M) = 4/6 \quad P(M/E) = 4/8$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B/A) = \frac{P(B \cap A)}{P(A)}$$

$$\text{then } P(A \cap B) = \frac{P(A) P(B/A)}{P(B)} \quad [\because A \cap B = B \cap A]$$

$$P(B/A) = \frac{P(B) P(A/B)}{P(A)}$$

VS Bayes's theorem on conditional probability

Let $E_1, E_2, \dots, E_n$ be mutually exclusive & exhaustive event of sample space S with $P(E_i) \neq 0, i=1, 2, \dots, n$, then for any arbitrary event A of S ,

$$P(E_r/A) = \frac{P(E_r) P(A/E_r)}{\sum_{i=1}^n P(E_i) P(A/E_i)} \quad r=1, 2, \dots, n$$

Proof

Let us draw a diagram from the diagram

$$A = (E_1 \cap A) \cup (E_2 \cap A) \cup \dots \cup (E_n \cap A)$$

Using prob. with add

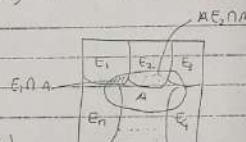
$$P(A) = P[(E_1 \cap A) \cup (E_2 \cap A) \cup \dots \cup (E_n \cap A)]$$

Since $E_1 \cap A, E_2 \cap A, \dots$ and $E_n \cap A$ are mutually exclusive

$$P(A) = P(E_1 \cap A) + P(E_2 \cap A) + \dots + P(E_n \cap A)$$

$$= P(E_1) P(A/E_1) + P(E_2) P(A/E_2) + \dots + P(E_n) P(A/E_n)$$

$$P(A) = \sum_{i=1}^n P(E_i) P(A/E_i) \quad \dots (i)$$



$$L.H.S = P(E_r/A) = \frac{P(E_r \cap A)}{P(A)}$$

$$P(E_r/A) = \frac{P(E_r) P(A/E_r)}{\sum_{i=1}^n P(E_i) P(A/E_i)} \quad \text{for } r=1, 2, \dots, n$$

why? (i)

⇒ Multiplication Law of probability

If the events A and B are independent, then

$$P(A \text{ and } B) = P(A \cap B) = P(A) \cdot P(B)$$

If the events A and B are dependent

$$P(A \cap B) = P(A) \cdot P(B/A)$$

⑥ What is the probability of drawing a queen from a pack of cards?
 52 favorable no. of cases (m) = no. of Queens = 4
 Total no. of cases = 52
 $P(\text{drawing Queen}) = \frac{\text{fav. no. of case}}{\text{Total}} = \frac{4}{52} = \frac{1}{13}$ Q

⑦ There are 5 white and 3 black balls in an urn. What is the probability of getting a black ball in drawing a single ball from the bag?

52 fav. no. of case (m) = no. of black ball = 3
 Total no. of ball = 8
 $P(\text{drawing a black ball}) = \frac{\text{fav. no. of case}}{\text{Total}} = \frac{3}{8} = 0.375$ Q

⑧ Three unbiased coins are tossed simultaneously. Find the probability of getting exactly all tails.

Sample space of 3 unbiased coins = {HHH, HHT, HTH, TTH, THT, HTT, TTT, TTT}
 $P(\text{exactly all tails}) = \frac{1}{8}$
 fav. no. of case = 1
 Total = 8
 $P(\text{exactly all tails}) = \frac{1}{8} = 0.125$

⑨ At least one tail

$P(\text{at least one tail}) = 1 - P(T=1)$
 $= P(\text{one tail}) + P(\text{two tail}) + P(\text{three tail})$
 $= \frac{3}{8} + \frac{3}{8} + \frac{1}{8}$
 $= \frac{7}{8} = 0.875$ Q

⑩ only one tail

$P(\text{only one tail}) = 1 - P(T=1)$
 $= P(\text{one tail})$
 $= \frac{3}{8} = 0.375$

⑪ at most one tail

$P(\text{at most one tail}) = 1 - P(T \leq 1)$
 $= P(\text{one tail}) + P(\text{two tail})$
 $= \frac{3}{8} + \frac{3}{8} = \frac{6}{8} = 0.75$

⑫ at least one head

$P(\text{at least one head}) = 1 - P(H=1)$
 $= P(\text{one head}) + P(\text{two head}) + P(\text{three head})$
 $= \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = \frac{7}{8} = 0.875$ Q

⑬ at most one tail/head

$P(\text{at most one tail/head}) = 1 - P(H \leq 1)$
 $= P(\text{one head}) + P(\text{zero head})$
 $= \frac{3}{8} + \frac{1}{8} = 0.5$

⑭ In a single throw of two dice, find the probability of getting a sum of 7 or 11 or 4 or less than 5.

⑭ⁱ $P(\text{sum } 7) = \frac{6}{36} = \frac{1}{6}$
 fav. no. of case = 6 { (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) }
 Total no. of case = 6x6 = 36 ways
 $P(\text{sum } 7) = \frac{6}{36} = \frac{1}{6}$
 $P(\text{sum } 11) = \frac{2}{36} = \frac{1}{18}$
 fav. no. of case = 2 { (5,6), (6,5) }
 $P(\text{sum } 11) = \frac{2}{36} = \frac{1}{18}$

⑭ⁱⁱ $P(7 \text{ or } 11) = P(\text{sum } 7) + P(\text{sum } 11)$
 $= \left(\frac{1}{6} + \frac{1}{18} \right) = 0.222$ Q

⑭ⁱⁱⁱ $P(\text{exactly } 4) = \frac{3}{36} = \frac{1}{12}$

fav. no. of case = 3 { (1,3), (2,2), (3,1) }
 $P(\text{sum } 4) = \frac{3}{36} = \frac{1}{12} = 0.083$ Q

⑩ $P(\text{less than 5}) = ?$

fav. no. of case = 6 $\{(1,1), (1,2), (1,3), (1,4), (2,2), (3,1)\}$
 $P(\text{less than 5}) = \frac{6}{36} = \frac{1}{6} = 0.166 \quad \text{Q}$

⑪ An urn contains four white, eight black, six red and two green balls. If three balls are drawn random. Determine the probability that

(a) All white balls

Sol: $P(\text{all white balls}) = ?$

fav. no. of case (w) = $4C_3$

total no. of case = $20C_3$

$\therefore P(\text{all white}) = \frac{4C_3}{20C_3} = 0.005 \quad \text{Q}$

4 W
8 B
6 R
2 G

total = 20

(b) all red ball

fav. no. of case (w) = $6C_3$

$P(\text{all red ball}) = \frac{6C_3}{20C_3} = 0.015$

(c) 1 white, 1 black & 1 green.

$P(1W, 1B, 1G) = \frac{4C_1 \times 8C_1 \times 2C_1}{20C_3}$
 $= 0.0561 \quad \text{Q}$

⑫ Two balls are drawn in succession from a bag containing 6 white and 2 black balls. what is the probability that both balls are of same colour.

Sol: $P(\text{same colour}) = ?$

fav. no. of case = $(6C_2 + 2C_2)$

total no. of case = $8C_2$

$P(\text{same colour}) = \frac{(6C_2 + 2C_2)}{8C_2} = 0.525 \quad \text{Q}$

6 W
2 B

total = 8

⑬ A bag contains 5 red, 4 black and 2 white balls. what is the probability of drawing a red and white ball if two balls are drawn one after other with replacement?

Sol: $P(\text{drawing red and white ball}) = ?$

$P(R, W \text{ or } W, R) = ?$

$= P(R_1) \cdot P(W_2) + P(W_1) \cdot P(R_2)$

$= \left(\frac{5}{11} \times \frac{2}{11} + \frac{2}{11} \times \frac{5}{11}\right) = \frac{10}{121} = 0.083 \quad \text{Q}$

⑭ Four cards are drawn from a well shuffled pack of 52 cards. find the probability that

(a) all are diamond

fav. no. of case = $13C_4$

total no. of case = $52C_4$

$P(\text{all are diamond}) = \frac{13C_4}{52C_4} = 0.0026$

(b) There is one card of each suit

fav. no. of case = $13C_1 \times 13C_1 \times 13C_1 \times 13C_1$

total no. of case = $52C_4$

$P(\text{one card of each suit}) = \frac{13C_1 \times 13C_1 \times 13C_1 \times 13C_1}{52C_4} = 0.1055 \quad \text{Q}$

(c) There are two spade & two hearts

fav. no. of case = $13C_2 \times 13C_2$

total no. of case = $52C_4$

$P(\text{Two spade & Two heart}) = \frac{13C_2 \times 13C_2}{52C_4} = 0.0271 \quad \text{Q}$

⑮ The following is the frequency distribution of annual commission from a survey of 300 media sale persons

Annual Commission (Rs)	Frequency
0 - 4999	15
5000 - 9999	25
10000 - 14999	35
15000 - 19999	45
20000 - 24999	70
25000 or above	30

Based on this information, what is the probability that a media persons make a commission

between 9999 and 10000 (exclusive)

$$\text{fav no of case} = 25$$

$$\text{total no of case} = 300$$

$$P(\text{bet}^n 9999 \text{ \& } 10000) = \frac{25}{300} = 0.083$$

less than 15000

$$\text{fav no of case} = (35 + 25 + 15) = 75$$

$$\text{total no of case} = 300$$

$$P(\text{less than } 15000) = \frac{\text{fav}}{\text{total}} = \frac{75}{300} = 0.25$$

20000 or more

$$\text{fav no of case} = (70 + 30) = 100$$

$$\text{total no of case} = 300$$

$$P(20000 \text{ or more}) = \frac{\text{fav}}{\text{total}} = \frac{100}{300} = 0.333$$

betⁿ 14999 and 20000 (exclusive)

$$\text{fav no of case} = 125$$

$$\text{total no of case} = 300$$

$$P(\text{bet}^n 14999 \text{ \& } 20000) = \frac{\text{fav}}{\text{total}} = \frac{125}{300} = 0.4166$$

Let A and B be two possible outcomes of an experiment and suppose

$$P(A) = 0.4, P(A \cup B) = 0.7 \text{ and } P(B) = 2$$

for what choice of x, A and B are mutually exclusive

$$P(A) = 0.4$$

$$P(B) = 2$$

$$P(A \cup B) = 0.7$$

Since A and B are mutually ~~exclusive~~ exclusive

$$P(A \cup B) = P(A) + P(B) = 0$$

$$0.7 = 0.4 + 2$$

$$x = 0.3$$

for what choice of x, A and B are independent.

Since A and B are independent

$$P(A \cap B) = P(A) \cdot P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$A \cup B = P(A) + P(B) - P(A) \cdot P(B)$$

$$0.7 = 0.4 + 2 - 0.4x$$

$$0.3 = 0.6x$$

$$x = 0.5/0.6 = 0.833$$

(16) A market research organization claims that among 100 house wives interviewed 62 regularly buy product X, 53 regularly buy product Y, 18 regularly buy both A these product and 13 don't buy these products at all. Should the result of this survey be questioned?

100 of house wives = 100

$$P(\text{buying product X}) = 62/100 = 0.62$$

$$P(\text{ " " Y}) = 53/100 = 0.53$$

$$P(\text{buying both X \& Y}) = P(X \cap Y) = 18/100 = 0.18$$

$$P(\text{buying at least one}) = P(X \cup Y) = P(X) + P(Y) - P(X \cap Y) = 0.62 + 0.53 - 0.18 = 0.97$$

$$P(\text{Not buying X \& Y}) = 13/100 = 0.13$$

$$\text{Total probability} = 0.97 + 0.13 = 1.10 \text{ which is false}$$

(17) A construction company is bidding for two contracts 'A' and 'B'. probability that the company will get contract A is $3/5$, the probability that the company will get contract B is $1/3$ and the probability that the company will get contract B is with the is $1/8$. What is the probability that the company will get contract A or B?

$$P(\text{Contract A}) = 3/5$$

$$P(\text{Contract B}) = 1/3$$

$$P(\text{both A \& B}) = P(A \cap B) = 1/8$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{3}{5} + \frac{1}{3} - \frac{1}{8} = 0.808$$

(18) There are three radio stations A, B and C which can receive in a city of 2000 families. The following information is available on the basis of a survey.

1200	families	listen to radio A
1100	"	" " " B
800	"	" " " C
865	families	listen to radio station A and B
450	"	" " " A and C
400	"	" " " B and C
100	"	" " " A, B, C

Find the probability who listen one radio station.

Solⁿ $P(A) = \frac{1200}{2000} = 0.60$

$P(B) = \frac{1100}{2000} = 0.55$

$P(C) = \frac{800}{2000} = 0.40$

$P(A \cap B) = \frac{865}{2000} = 0.4325$

$P(A \cap C) = \frac{450}{2000} = 0.225$

$P(B \cap C) = \frac{400}{2000} = 0.20$

$P(A \cap B \cap C) = \frac{100}{2000} = 0.05$

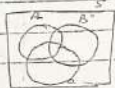
$P(\text{listen at least one radio station}) = P(A \cup B \cup C) = ?$

we have,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

$$= 0.60 + 0.55 + 0.40 - 0.4325 - 0.225 - 0.20 + 0.05$$

$= 0.7425 = B$



(19) The odds against a certain event S to 2 and odds in favour of another independent event are 6 to 5, find the chance that at least one of the events will happen.

Solⁿ:- $P(A) = \frac{2}{2+5} = \frac{2}{7}$ [5:2]

$P(B) = \frac{6}{6+5} = \frac{6}{11}$ [6:5]

$P(A \cap B) = P(A) \cdot P(B)$ Since A & B are independent

$= \frac{2}{7} \times \frac{6}{11} = \frac{12}{77}$

$P(\text{at least one event}) = P(A \cup B) = ?$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= \frac{2}{7} + \frac{6}{11} - \frac{12}{77}$

$= 0.635 = B$

(20) A box contains six red, four white and blue balls. A person draws four balls from the box at random. Find the probability that among the balls drawn there is at least one ball of each color.

Solⁿ $P(\text{at least one ball of each color}) = ?$

$P(\text{at least one ball of each color}) = \frac{6C1 \times 4C1 \times 5C2 + 6C2 \times 4C1 \times 5C1 + 6C3 \times 4C1 \times 5C0}{15C4}$

$= \frac{6C1 \times 4C1 \times 5C2 + 6C2 \times 4C1 \times 5C1 + 6C3 \times 4C1 \times 5C0}{15C4}$

$= \frac{6 \times 4 \times 10 + 6 \times 6 \times 5 + 15 \times 4 \times 1}{1365} = 0.527$

(21) The chances of winning at two race horses are $\frac{1}{3}$ and $\frac{1}{6}$ are the probability that at least one will win when the runner (A) in different races and (B) in the same race.

Solⁿ (A) $P(A) = \frac{1}{3}$

$P(B) = \frac{1}{6}$

$P(A \cup B) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Since A & B are independent

$P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$

$= \frac{1}{3} + \frac{1}{6} - \frac{1}{3} \times \frac{1}{6}$

$= 0.416 = B$

(B) $P(\text{in same race}) = P(A \cap B)$ Since A & B are independent

$= P(A) \cdot P(B)$

$= \frac{1}{3} \times \frac{1}{6}$

$= \frac{1}{18}$

$= \frac{1}{18} = 0.055 = B$

A card is drawn from a bag containing 10 cards numbered 1 through 10. Let A denotes the events that the number on card drawn is less than 7. B is the event that the number on the card drawn is a multiplier of 2. Find the probability of the events.

Ans: (S) = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

A = {1, 2, 3, 4, 5, 6}

B = {2, 4, 6, 8, 10}

(a) $P(A) = \frac{\text{fav. no.}}{\text{Total}} = \frac{6}{10} = 0.6$

(b) $P(B) = \frac{5}{10} = 0.5$

(c) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= 0.6 + 0.5 - 0.3$
 $= 0.8$

(d) $P(A \cap B) = ?$

$\bar{A} = \{7, 8, 9, 10\}$

$\bar{B} = \{1, 3, 5, 7, 9\}$

$\bar{A} \cap \bar{B} = \{7, 9\}$

$P(\bar{A} \cap \bar{B}) = \frac{2}{10} = 0.2$

Three newspapers A, B and C are published in a certain city. It is estimated that from a family that 20% read A, 16% read B, 14% read C, 5% read A & B, 5% read A & C, and 4% read B & C and 2% read all the three newspapers. What are the probabilities that a normally chosen person (a) doesn't read any paper (b) doesn't read C (c) read A but not B (d) read only one of these papers.

Ans: (a) doesn't read any paper (b) doesn't read C (c) read A but not B (d) read only one of these papers.

$P(A) = 20\% = 0.2$

$P(B) = 16\% = 0.16$

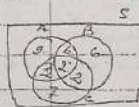
$P(C) = 14\% = 0.14$

$P(A \cap B) = 5\% = 0.05$

$P(A \cap C) = 5\% = 0.05$

$P(B \cap C) = 4\% = 0.04$

$P(A \cap B \cap C) = 2\% = 0.02$



$P(\text{at least read one newspaper}) = P(A \cup B \cup C)$

$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$
 $= 0.2 + 0.16 + 0.14 - 0.05 - 0.05 - 0.04 + 0.02$
 $= 0.35$

(a) doesn't read any paper = $1 - P(A \cup B \cup C)$
 $= 1 - 0.35$
 $= 0.65$

(b) doesn't read C = $1 - P(C)$
 $= 1 - 0.14$
 $= 0.86$

(c) read A but not B
 $P(A) - P(A \cap B)$
 $0.2 - 0.05 = 0.15$

(d) only one of these papers =
 $P(A) - P(A \cap B) - P(A \cap C) + P(A \cap B \cap C)$
 $= 0.2 - 0.05 - 0.05 + 0.02$
 $= 0.12$

$P(\text{exactly one}) = P(A \cup B \cup C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + 2P(A \cap B \cap C)$
 $= 0.35 - 0.05 - 0.05 - 0.04 + 2 \times 0.02$
 $= 0.22$

An urn contains 12 balls of which 4 are white, five balls are drawn at random and laid on side, their colour recorded. Another ball is drawn, what is the prob. that it is white.

Ans: fav. no. of case = 4C1

Total no. of case = 12C1

$P(\text{white}) = \frac{\text{fav. no. of case}}{\text{Total no. of case}} = \frac{4C1}{12C1} = \frac{4}{12} = \frac{1}{3}$



(25) Suppose a wildcatter assigns a prior probability of 0.6 for gas. A seismic survey might be used that augurs favourable in 85% of applications where gas was known to be present and unfavourably in 90% of sites where no gas was found. Find the posterior probability of gas.

- (a) Given a favourable seismic result.
(b) Given an unfavourable seismic result.

(27) The contents of urns I, II and III are as follows:

- 1 white, 2 black and 3 red balls.
2 white, 1 black and 1 red ball.
4 white, 5 black and 3 red balls.

One urn is chosen at random and two balls drawn. They are found to be white and red. What is the probability that they came from urn II?

Soln: Let Event E_1 = Urn I is chosen $\Rightarrow P(E_1) = \frac{1}{3}$

" E_2 = Urn II " $\Rightarrow P(E_2) = \frac{1}{3}$

" E_3 = Urn III " $\Rightarrow P(E_3) = \frac{1}{3}$

Let event A = obtaining one white & one red

$$P(A/E_1) = \frac{1C_1 \times 2C_1}{6C_2} = 0.20$$

$$P(A/E_2) = \frac{2C_1 \times 1C_1}{4C_2} = 0.333$$

$$P(A/E_3) = \frac{4C_1 \times 3C_1}{12C_2} = 0.1818$$

Now,

$$P(E_2/A) = \frac{P(E_2) P(A/E_2)}{P(E_1) P(A/E_1) + P(E_2) P(A/E_2) + P(E_3) P(A/E_3)}$$

$$= \frac{\frac{1}{3} \times 0.333}{0.333 \times 0.2 + 0.333 \times 0.333 + 0.333 \times 0.1818}$$

$$P(E_2/A) = 0.466 \text{ } \odot$$

(28) In Pokhara city, 25% of all cars emit excessive amount of pollutant. It is the probability is 0.92 that a car emitting excessive amount will fail the city vehicular test and the probability is that a car not emitting excessive amount of pollutants will pass the test. What is the probability that a car which fails the test emits excessive amount of pollutant?

The probabilities of the three persons A, B & C becoming VC are $\frac{1}{3}$, $\frac{1}{3}$ and $\frac{1}{3}$ respectively. The probabilities that the grade increment system will be introduced if A, B and C become VC are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ resp.

What is the probability that the grade increment system will be introduced if the grade increment system has been introduced. What is the probability that the VC appointed is Mr. B?

$$P(A) = \frac{1}{3}$$

$$P(B) = \frac{1}{3}$$

$$P(C) = \frac{1}{3}$$

Let event E = grade increment system will be introduced

$$P(B/A) = \frac{1}{2}$$

$$P(E/B) = \frac{1}{3}$$

$$P(E/C) = \frac{1}{4}$$

$$P(E) = P(A)P(E/A) + P(B)P(E/B) + P(C)P(E/C)$$

$$= \frac{1}{3} \times \frac{1}{2} + \frac{1}{3} \times \frac{1}{3} + \frac{1}{3} \times \frac{1}{4}$$

$$= \frac{1}{6} + \frac{1}{9} + \frac{1}{12} = \frac{3+4+2}{18} = \frac{9}{18} = \frac{1}{2}$$

$$P(B/E) = \frac{P(B)P(E/B)}{P(E)} = \frac{\frac{1}{3} \times \frac{1}{3}}{\frac{1}{2}} = \frac{\frac{1}{9}}{\frac{1}{2}} = \frac{2}{9} = 0.22$$

① what is the probability of drawing two aces from a shuffled deck 52 playing cards?

So: fav. no. of case (m) = 4C2

Total no. of case (n) = 52C2

$$P(\text{drawing two aces}) = \frac{m}{n} = \frac{4C2}{52C2} = 0.00452$$

② 3 of 20 types in storage are defective and 4 of them are random chosen from inspection. what is the probability that only one of defective types will be included?

fav. no. of case (m) = 3C1 x 17C3

Total no. of case (n) = 20C4

$$P(\text{obtaining only one of defective}) = \frac{m}{n} = \frac{3C1 \times 17C3}{20C4} = 0.421$$

3 defects
12 non-defects
T = 20

③ A card is drawn from a well-shuffled deck of 52 playing cards randomly. Find the probability that it will be king of diamond.

fav. no. of case (m) = 1C1

Total no. of case (n) = 52C1

$$P(\text{obtaining king of diamond}) = \frac{m}{n} = \frac{1C1}{52C1} = 0.01923$$

④ A committee of 4 members is selected at random from 4 men and 3 women. Find the probability that 3 men and 1 woman are selected.

fav. no. of case (m) = 4C3 x 3C1

Total no. of case (n) = 7C4

$$P(\text{committee of 3 men & 1 woman}) = \frac{m}{n} = \frac{4C3 \times 3C1}{7C4} = 0.34285$$

⑤ An experiment has three possible and mutually exclusive outcomes A, B and C. Check in each case whether the assignment of probabilities is permissible.

Since A, B & C are mutually exclusive outcomes

Then for

$$P(A) + P(B) + P(C) = \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \frac{20+15+12}{60} = \frac{47}{60} \neq 1$$

for B

$$P(A) + P(B) + P(C) = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$$

for C

$$P(A) + P(B) + P(C) = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$$

for D

$$P(A) + P(B) + P(C) = 0.57 + 0.2 + 0.3 = 1.07$$

from above value, permissible probability has C i.e. 1 & 2

6) If a card is drawn from well shuffled deck of 52 playing cards, what is the probability of drawing a red ace or a black queen?

$$\text{Sol}^n \text{ fav. no. of cases (n)} = 2C_1 = 2$$

$$\text{Total no. of cases (n)} = 52C_1 = 52$$

$$P(\text{drawing a red ace or a black queen}) = \frac{2}{52} = 0.03846$$

7) When we roll a pair of balanced dice, what are the probability of getting

a) 7 or 11 (b) 3 (c) 2 or 12 (d) 2, 3, or 12

$$\text{Total no. of cases} = 6 \times 6 = 36$$

$$\text{fav. no. of cases} = 6 \quad \{(3,4), (4,3), (2,5), (5,2), (1,6), (6,1)\}$$

$$P(\text{sum 7}) = \frac{6}{36} = \frac{1}{6}$$

$$\text{8) fav. no. of cases getting 11} = 2 \quad \{(6,5), (5,6)\}$$

$$P(\text{getting 11}) = \frac{2}{36} = \frac{1}{18}$$

$$P(\text{getting 7 or 11}) = \frac{2}{36} + \frac{2}{36} = \frac{4}{36} = \frac{1}{9}$$

$$\text{9) fav. no. of cases getting 3} = 2 \quad \{(1,2), (2,1)\}$$

$$P(\text{getting 3}) = \frac{2}{36} = \frac{1}{18}$$

$$\text{10) fav. no. of cases getting 2} = 1 \quad \{(1,1)\}$$

$$P(\text{getting 2}) = \frac{1}{36}$$

$$\text{fav. no. of cases getting 12} = 1 \quad \{(6,6)\}$$

$$P(\text{getting 12}) = \frac{1}{36}$$

$$P(\text{getting 2 or 12}) = \frac{1}{36} + \frac{1}{36} = \frac{1}{18}$$

$$\text{11) getting (2, 3 or 12)} = \left(\frac{1}{36} + \frac{1}{36} + \frac{1}{36} \right) = \frac{1+1+1}{36} = \frac{3}{36} = \frac{1}{12}$$

12) A Lottery sells tickets numbered from 00001 through 50000. the probability of drawing a number that is divisible by 250?

$$\text{Sol}^n \text{ fav. no. of cases (n)} = 250 \text{ divisible by } 250 \text{ i.e. } \{250, 500, \dots, 50000\}$$

$$\text{Total no. of cases (n)} = 50000$$

$$P(\text{drawing divisible by 250}) = \frac{\text{fav. no. of cases}}{\text{Total no. of cases}} = \frac{250}{50000} = 0.005$$

13) A car rental agency has 12 compact cars and 12 intermediate cars. If 4 of the cars are randomly selected for a safety test, what is the probability of getting two of each kind?

$$\text{Sol}^n \text{ fav. no. of cases (n)} = 12C_2 \times 12C_2$$

$$\text{Total no. of cases (n)} = 30C_4$$

$$P(\text{obtaining two of each kind}) = \frac{\text{fav. no. of cases}}{\text{Total no. of cases}} = \frac{12C_2 \times 12C_2}{30C_4} = 0.007$$

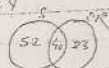
14) In a group of 160 graduates engineering students, 92 are enrolled in advanced courses in statistics, 63 are enrolled in a course in research, and 49 are enrolled in both. How many of these are not enrolled in either course?

$$\text{Sol}^n \text{ } P(A) = 92, P(B) = 63$$

$$P(A \cap B) = 49$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 92 + 63 - 49 = 106$$

$$P(\text{not enrolled}) = 1 - P(A \cup B) = 1 - \frac{106}{160} = \frac{54}{160} = \frac{27}{80}$$



92
63
49

15) If A and B are mutually exclusive events, $P(A) = 0.29$ and $P(B) = 0.37$

$$\text{16) } P(A \cup B) = P(A) + P(B) = 0.29 + 0.37 = 0.66$$

$$\text{17) } P(A \cap B) = P(A) + P(B) = 0.29 + 0.37 = 0.66$$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$= 0.27 + 0.03 - 0.72$$

$$= 0$$

$$P(A' \cap B') = P(A') + P(B') - P(A \cup B)$$

$$= 0.73 + 0.97 - [1 - P(A \cup B)]$$

$$= 0.73 + 0.97 - 1 + 0.72 = 0.72$$

Given $P(A) = 0.25$, $P(B) = 0.32$ and $P(A \cap B) = 0.14$

Q.1. (a) $P(A \cup B)$ (b) $P(A' \cap B)$ (c) $P(A \cap B')$ (d) $P(A' \cap B')$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 0.25 + 0.32 - 0.14$$

$$= 0.43$$

$$P(A' \cap B) = P(B) - P(A \cap B)$$

$$= 0.32 - 0.14$$

$$= 0.18$$

$$P(A \cap B') = P(A) - P(A \cap B)$$

$$= 0.25 - 0.14$$

$$= 0.11$$

$$P(A' \cap B') = 1 - P(A \cup B) = 1 - 0.43 = 0.57$$

If the probability that a Communication System will have high fidelity is 0.82 and the probability that it will have high selectivity is 0.18, what is the probability that a System will have high fidelity given high selectivity?

probability A high fidelity $P(A) = 0.82$

high fidelity high selectivity $P(A \cap B) = 0.18$

probability B high selectivity $P(B) = 0.18$

when A & B are independent

then $P(A \cap B) = P(A) \cdot P(B)$

$$P(B) = \frac{P(A \cap B)}{P(A)} = \frac{0.18}{0.82} = \frac{9}{41}$$

Q.14 If a card drawn from a pack of cards is black, what is the probability that next drawn from a pack of card is also black?

So, $P(\text{first drawn is a black card}) = \frac{26}{52}$

$$P(\text{next drawn is a black card}) = \frac{26}{52} \times \frac{25}{51}$$

$$= 0.245$$

Q.15 What is the probability of getting 2 head or blip of a balance

So, $S = 8$ {HHH, HHT, HTH, TTH, THT, HTT, TTT}

fav. no. of case = 1

$P(\text{getting 2 head}) = \frac{1}{8}$

Q.16 Two cards are drawn at random from an ordinary deck of 52 playing cards. what is the probability of getting two aces if

(a) the first card is replaced before the second card is drawn

(b) the first card is not replaced before the second card is drawn

So, (a) fav. no. of case = $4C_1 \times 4C_1$

total no. of case = 52C1

$$P(\text{getting two aces}) = \frac{4C_1 \times 4C_1}{52C1} = \frac{16}{52}$$

(b) fav. no. of case = $4C_1 \times 3C_1$

total no. of case = 52C2

$$P(\text{getting two aces}) = \frac{4C_1 \times 3C_1}{52C2} = \frac{12}{52}$$

Q.17 If $P(A) = 0.65$, $P(B) = 0.4$ and $P(A \cap B) = 0.24$ are the event A & B

So, In this question, for event A & B, independent, there must have value of A & B but here A & B have some values jointly. the event A & B are not independent

3) What is the probability of not rolling any 6's in 5 rolls of a balanced die

3a) $S = \{1, 2, 3, 4, 5, 6\}$

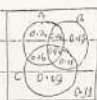
$$P(\text{obtaining 6's in one roll}) = \frac{1}{6}$$

$$P(\text{not " " " " " "}) = 1 - \frac{1}{6} = \frac{5}{6}$$

$$P(\text{not " " " " " " in 5 rolls}) = \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} = 0.4018 \quad \text{Q}$$

4) Given in probability features find

$$\begin{aligned} \text{C) } P(A/B) &= \frac{P(A \cap B)}{P(B)} \\ &= \frac{(0.06 + 0.04)}{(0.06 + 0.14 + 0.11 + 0.19)} = \frac{0.10}{0.40} \\ &= 0.25 \end{aligned}$$



$$\begin{aligned} \text{D) } P(B/C) &= \frac{P(B \cap C)}{P(C)} \quad P(C) = 0.4 \\ &= \frac{(0.06 + 0.19)}{0.6} = \frac{0.25}{0.6} = 0.416 \end{aligned}$$

$$P(C) = 0.4$$

$$P(C') = 1 - 0.4 = 0.6$$

$$\text{C) } P(A \cap B | C) = \frac{P(A \cap B \cap C)}{P(C)} = \frac{0.04}{0.4} = 0.10$$

$$\text{D) } P(A \cap B | C) = \frac{P(A \cap B \cap C)}{P(C)} = \frac{0.04}{0.4} = 0.10$$

5) Given $P(A) = 0.5$, $P(B) = 0.30$ and $P(A \cap B) = 0.15$ verify that

$$\text{C) } P(A/B) = P(A/B')$$

we have

$$\begin{aligned} \text{S- } P(A/B) &= \frac{P(A \cap B)}{P(B)} \\ &= \frac{0.15}{0.3} \\ &= 0.50 \end{aligned}$$



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now

$$P(B') = 1 - P(B) = 1 - 0.3 = 0.7$$

$$P(A \cap B') = P(A) - P(A \cap B) = 0.5 - 0.15 = 0.35$$

$$\text{R.H.S. } P(A/B') = \frac{P(A \cap B')}{P(B')} = \frac{0.35}{0.7} = 0.5$$

L.H.S. = R.H.S. proved

$$\text{(ii) } P(B/A) = P(B/A')$$

$$\text{L.H.S. } P(B/A) = \frac{P(B \cap A)}{P(A)} = \frac{0.15}{0.5} = 0.3$$

$$P(B \cap A') = P(B) - P(A \cap B) = 0.3 - 0.15 = 0.15$$

$$P(A') = 1 - P(A) = 1 - 0.5 = 0.5$$

$$\text{R.H.S. } P(B/A') = \frac{P(B \cap A')}{P(A')} = \frac{0.15}{0.5} = 0.3$$

L.H.S. = R.H.S. proved

② If the odds are 5 to 3 that event M will not occur, 2 to 1 that event N will occur, and 4 to 1 that they will not both occur are the two events M and N independent

$$\text{So } P(M) = \frac{3}{3+5} = \frac{3}{8}$$

$$P(N) = \frac{2}{1+2} = \frac{2}{3}$$

$$P(M \cap N) = \frac{1}{4+1} = \frac{1}{5}$$

$$M \text{ and } N \text{ independent when } P(M \cap N) = P(M) \cdot P(N)$$

$$= \frac{3}{8} \times \frac{2}{3} = \frac{1}{4} = 0.25$$

$$= 0.25 \neq P(M \cap N)$$

M and N are not independent i.e. (dependent)

(23) From a lottery of 50 tickets marked 1, 2, 3, ..., 50, 4 tickets are drawn. What is the chance that these marked 1 and 2 are among them?

Solⁿ - for no. of case (m) = $20_2 \times 48_2$

Total no. of case = 50_4

$P(\text{obtaining 1 and 2 marked}) = \frac{\text{for no. of case}}{\text{Total no. of case}} = \frac{20_2 \times 48_2}{50_4} = 0.008$

(24) A bag contains 10 balls, two of which are red, three blue and five black. Three balls are drawn at random from the bag, that is every ball has equal chance of being included in the three. What is the probability that

(i) three balls are of different colour

(ii) only two balls are of the same colour and

(iii) the balls are all of the same colour?

$P(\text{obtaining different colour}) = ?$

for no. of case (m) = $2C_1 \times 3C_1 \times 5C_1$

Total no. of case = $10C_3$

$P(\text{obtaining diff. colour}) = \frac{2C_1 \times 3C_1 \times 5C_1}{10C_3} = 0.24$

(ii) $P(\text{obtaining two balls are same colour}) = ?$

for no. of case = $\{(2C_2 \times 3C_1) + (2C_1 \times 5C_2) + (3C_2 \times 2C_1) + (3C_1 \times 5C_2) + (5C_2 \times 2C_1) + (5C_1 \times 2C_2)\}$

Total no. of case = $10C_3$

$P(\text{obtaining two balls are same colour}) = \frac{\text{for no. of case}}{\text{Total no. of case}}$

$= \frac{72}{10C_3} = 0.483$

(iii) $P(\text{obtaining all are same colour}) = ?$

for no. of case = $(3C_3 + 5C_3)$

Total no. of case = $10C_3$

$P(\text{obtaining all are same colour}) = \frac{(3C_3 + 5C_3)}{10C_3} = 0.0266$

(24) Two-third of the students in a class is boys and rest are girls. It is known that the probability of a girl getting a first is 0.25 and that of boy getting a first class is 0.20. Find the prob. that a student chosen at random will get first class marks in Subject.

Solⁿ - first class marks to Boys = $\frac{2}{3} \times 0.20 = 0.1333$

... girls = $\frac{1}{3} \times 0.25 = 0.0833$

$P(\text{obtaining a student first class marks}) = 0.1333 + 0.0833 = 0.2166$

(25) A bag contains 3 red and 5 black balls and a second bag contains 4 red and 6 black balls. A ball is drawn from each bag. Find the probability that

(i) both the balls are red and

(ii) one is red & other black.

Solⁿ - (i) $P(\text{both the balls are red}) = ?$

$P(\text{both balls are red}) = \left(\frac{3C_1}{8C_1} \times \frac{4C_1}{10C_1} \right) = 0.225$

(ii) $P(\text{one red & other black}) = \left(\frac{3C_1 \times 5C_1}{8C_2} \times \frac{6C_1 \times 4C_1}{10C_2} \right) = 0.2875$

(26) 6 bullets, of which three are blanks, are randomly inserted in a gun. What is the probability that the first three bullets fired are blanks?

Solⁿ $P(B_1 \cap B_2 \cap B_3) = P(B_1) \cdot P(B_2/B_1) \cdot P(B_3/B_1 \cap B_2)$

$= \frac{3}{6} \times \frac{2}{5} \times \frac{1}{4} = \frac{1}{20} = 0.05$

(27) The probability that a man will be alive 25 years hence is 0. The probability that his wife will be alive 25 years hence is 0.

Find the probability that 25 years hence

(i) both will be alive

(ii) only the man will be alive

(iii) " " " " " "

(iv) at least one of them will be alive.

94

$$(i) P(A) = 0.30, P(B) = 0.40$$

(ii) probability both will be alive

$$P(A \cap B) = ?$$

$$P(A \cap B) = P(A) \cdot P(B) \quad [\text{since } A \text{ \& } B \text{ are independent}]$$

$$= 0.30 \times 0.40$$

$$= 0.12$$

$$(iii) P(A \cap \bar{B}) = ?$$

$$= P(A) \cdot P(\bar{B})$$

$$= P(A) \cdot \{1 - P(B)\}$$

$$= P(A) \cdot (1 - 0.4)$$

$$= 0.3 \times 0.60$$

$$= 0.18$$

$$(iv) P(A \cup B) = ?$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 0.30 + 0.4 - 0.12$$

$$= \underline{0.58}$$

$$(v) P(\bar{A} \cap B) = ?$$

$$= P(\bar{A}) \cdot P(B)$$

$$= \{1 - P(A)\} \cdot P(B)$$

$$= (1 - 0.3) \cdot 0.40$$

$$= 0.28$$

-48-

(28)

An urn contains 12 balls of which 4 are white; five balls are drawn at random and laid aside their colour unnoted. Another ball is drawn, what is the probability that it is white.

bag contains 4 white, 8 black, 6 red and 2 green balls find the probability of getting either a white or a black or a green ball in a single draw

$$= \frac{4C_1 + 8C_1 + 2C_1}{20C_1}$$

$$= \frac{1+8+2}{20} = \frac{11}{10} = 0.70$$

4W
8B
6R
2G

$$T=20$$

from a group of 8 children, 5 boys and 3 girls, 3 children are selected at random. calculate the probabilities that the selected group contains:

no girl

$$\frac{5C_3 \times 3C_0}{8C_3} = \frac{10}{56} = \frac{5}{28}$$

only one girl

$$\frac{5C_2 \times 3C_1}{8C_3} = \frac{10 \times 3}{56} = \frac{15}{28}$$

only one particular girl

$$\frac{1C_1 \times 7C_2}{8C_3}$$

at least one girl

$$\frac{5C_3 \times 3C_0 + 5C_2 \times 3C_1 + 5C_1 \times 3C_2}{8C_3}$$

$$= \frac{10 + 15 + 1}{56} = \frac{26}{56} = \frac{13}{28}$$

more girls than boys

$$\frac{5C_2 \times 3C_1 + 5C_1 \times 3C_2}{8C_3}$$

$$= \frac{15 + 1}{56} = \frac{16}{56} = \frac{2}{7}$$

Q1 A university has to select an examiner from a list of 50 persons. 20 of them are women and 30 men, 10 of them know English and 15 of them are teachers and remaining 35 are not. what is the probability of the university selecting a English knowing woman teacher?

$$S^{th} \quad P(\text{women}) = \frac{20}{50} = \frac{2}{5} = P(E_1)$$

$$P(\text{knowing English}) = \frac{10}{50} = \frac{1}{5} = P(E_2)$$

$$P(\text{being teachers}) = \frac{15}{50} = \frac{3}{10} = P(E_3)$$

$$P(E_1 \cup E_2 \cup E_3) = ?$$

for independent

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) \times P(E_2) \times P(E_3)$$

$$= \frac{2}{5} \times \frac{1}{5} \times \frac{3}{10} = \frac{3}{125}$$

Q2 Ramesh and Suresh are asked to solve a problem. The probability of Ramesh solving it is $\frac{2}{3}$ and that of Suresh solving it is $\frac{3}{4}$. The probability that:

$$S^{th} \quad P(R) = \frac{2}{3}, \quad P(S) = \frac{3}{4}$$

(i) both can solve the problem

$$P(\text{both can solve}) = P(R \cap S) = P(R) \cdot P(S) = \frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$$

(ii) Ramesh solves it but Suresh cannot

$$P(R \cap \bar{S}) = P(R) \cdot P(\bar{S}) = P(R) \cdot [1 - P(S)] = \frac{2}{3} \cdot [1 - \frac{3}{4}] = \frac{2}{3} \times \frac{1}{4} = \frac{1}{6}$$

(iii) Suresh solves it but Ramesh cannot

$$P(\bar{R} \cap S) = P(\bar{R}) \cdot P(S) = [1 - P(R)] \cdot P(S) = (1 - \frac{2}{3}) \times \frac{3}{4} = \frac{1}{3} \times \frac{3}{4} = \frac{1}{4}$$

None of them can solve

$$\begin{aligned}
 P(\bar{R} \cap \bar{S}) &= P(\bar{R}) \cdot P(\bar{S}) \\
 &= [1 - P(R)] \cdot [1 - P(S)] \\
 &= (1 - \frac{2}{3}) \cdot (1 - \frac{3}{4}) \\
 &= \frac{1}{3} \times \frac{1}{4} \\
 &= \frac{1}{12} \quad \underline{\underline{Q}}
 \end{aligned}$$

at least one of them will solve it.

$$\begin{aligned}
 P(R \cup S) &= P(R) + P(S) - P(R \cap S) \\
 &= \frac{2}{3} + \frac{3}{4} - \frac{1}{12} \\
 &= \frac{8+9-1}{12} \\
 &= \frac{16}{12} \quad \underline{\underline{Q}}
 \end{aligned}$$

only one of them can solve.

$$\begin{aligned}
 &= P(A \cap \bar{B}) \text{ or } P(\bar{A} \cap B) \\
 &= 1 \cdot P(\bar{B}) + P(\bar{A}) \cdot P(B) \\
 &= P(A) [1 - P(B)] + [1 - P(A)] P(B) \\
 &= \frac{1}{2} \times \frac{1}{4} + \frac{1}{4} \times \frac{1}{2} = \frac{2 \times 1 + 1 \times 1}{4 \times 2} = \frac{3}{8} \quad \underline{\underline{Q}}
 \end{aligned}$$

A problem of statistics is given to three students A, B and C whose chance of solving it are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. Find the probability that

$$\begin{aligned}
 P(A) &= \frac{1}{2} & P(\bar{A}) &= 1 - P(A) = 1 - \frac{1}{2} = \frac{1}{2} \\
 P(B) &= \frac{1}{3} & P(\bar{B}) &= 1 - P(B) = 1 - \frac{1}{3} = \frac{2}{3} \\
 P(C) &= \frac{1}{4} & P(\bar{C}) &= 1 - P(C) = 1 - \frac{1}{4} = \frac{3}{4}
 \end{aligned}$$

(i) P(all of them can solve the problem) = ?

$$\begin{aligned}
 P(A \cap B \cap C) &= P(A) \cdot P(B) \cdot P(C) \\
 &= \frac{1}{2} \times \frac{1}{3} \times \frac{1}{4} = \frac{1}{24} \quad \underline{\underline{Q}}
 \end{aligned}$$

(ii) P(one of them can solve) = ?

$$\begin{aligned}
 P(\bar{A} \cap \bar{B} \cap \bar{C}) &= P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C}) \\
 &= \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} = \frac{1}{4} \quad \underline{\underline{Q}}
 \end{aligned}$$

(iii) P(problem will be solved) = ?

$$\begin{aligned}
 P(A \cup B \cup C) &= P(\text{at least one solve the problem}) \\
 P(A \cup B \cup C) &= 1 - P(\text{non-solved}) \\
 &= 1 - \frac{1}{4} \\
 &= \frac{3}{4}
 \end{aligned}$$

(iv) P(exactly two solve the problem)

$$\begin{aligned}
 &P(A \cap B \cap \bar{C}) \text{ or } A \cap \bar{B} \cap C \text{ or } \bar{A} \cap B \cap C \\
 &= P(A \cap B \cap \bar{C}) + P(A \cap \bar{B} \cap C) + P(\bar{A} \cap B \cap C) \\
 &= P(A) \cdot P(B) \cdot P(\bar{C}) + P(A) \cdot P(\bar{B}) \cdot P(C) + P(\bar{A}) \cdot P(B) \cdot P(C) \\
 &= \frac{1}{2} \times \frac{1}{3} \times \frac{2}{3} + \frac{1}{2} \times \frac{2}{3} \times \frac{1}{3} + \frac{1}{2} \times \frac{1}{3} \times \frac{1}{3} \\
 &= \frac{1}{3} + \frac{1}{3} + \frac{1}{9} \\
 &= \frac{3+3+1}{9} = \frac{7}{9} \quad \underline{\underline{Q}}
 \end{aligned}$$

(v) A can hit a target 4 times with 5 shots. B can hit 3 with 5. Can hit 3 with 5 shots. Each fire a volley, what is the probability

(i) two shot hit the target?

(ii) one shot hit the target?

$$\begin{aligned}
 \text{Sol}^n \text{ probability of A is } P(A) &= \frac{4}{5} & P(\bar{A}) &= 1 - \frac{4}{5} = \frac{1}{5} \\
 \text{" B is } P(B) &= \frac{3}{5} & P(\bar{B}) &= 1 - \frac{3}{5} = \frac{2}{5} \\
 \text{" C is } P(C) &= \frac{3}{5} & P(\bar{C}) &= 1 - \frac{3}{5} = \frac{2}{5}
 \end{aligned}$$

$$\begin{aligned}
 (i) &P(A \cap B \cap \bar{C}) \cup P(A \cap \bar{B} \cap C) \cup P(\bar{A} \cap B \cap C) \\
 &= P(A \cap B \cap \bar{C}) + P(A \cap \bar{B} \cap C) + P(\bar{A} \cap B \cap C) \\
 &= P(A) \cdot P(B) \cdot P(\bar{C}) + P(A) \cdot P(\bar{B}) \cdot P(C) + P(\bar{A}) \cdot P(B) \cdot P(C) \\
 &= \frac{4}{5} \times \frac{3}{5} \times \frac{2}{5} + \frac{4}{5} \times \frac{2}{5} \times \frac{3}{5} + \frac{1}{5} \times \frac{3}{5} \times \frac{3}{5} \\
 &= \frac{24}{125} + \frac{24}{125} + \frac{9}{125} \\
 &= \frac{57}{125} \quad \underline{\underline{Q}}
 \end{aligned}$$

$$\begin{aligned}
 (ii) &P(A \cap \bar{B} \cap \bar{C}) \cup P(\bar{A} \cap B \cap C) \cup P(\bar{A} \cap \bar{B} \cap C) \\
 &= P(A \cap \bar{B} \cap \bar{C}) + P(\bar{A} \cap B \cap C) + P(\bar{A} \cap \bar{B} \cap C) = P(A) \cdot P(\bar{B}) \cdot P(\bar{C}) + P(\bar{A}) \cdot P(B) \cdot P(C) + P(\bar{A}) \cdot P(\bar{B}) \cdot P(C) \\
 &= \left(\frac{4}{5} \times \frac{2}{5} \times \frac{2}{5} \right) + \left(\frac{1}{5} \times \frac{3}{5} \times \frac{3}{5} \right) + \left(\frac{1}{5} \times \frac{2}{5} \times \frac{2}{5} \right) = \frac{28}{125} \quad \underline{\underline{Q}}
 \end{aligned}$$

In certain college, 25% of the students failed in Mathematics, 10% of the students failed in Chemistry and 10% failed in both Mathematics and Chemistry. A student is selected at random. What is the prob. that he has failed in Mathematics given that he has failed in chemistry?

$$P(M) = 0.25$$

$$P(C) = 0.15$$

$$P(M \cap C) = 0.10$$

$$P(M/C) = ?$$

$$P(C/M) = ?$$

$$P(M/C) = \frac{P(M \cap C)}{P(C)} = \frac{0.10}{0.15} = \frac{2}{3}$$

$$P(C/M) = \frac{P(M \cap C)}{P(M)} = \frac{0.10}{0.25} = \frac{2}{5}$$

A box contains 3 red and 5 blue balls. Two balls are drawn at a time at random without replacement find the prob. of getting 1 red and 1 blue ball.

$$\begin{matrix} 3R \\ 5B \end{matrix}$$

$$T=8$$

$$P(\text{one red \& 1 blue}) = ?$$

$$P(\text{one red \& one blue}) = \frac{{}^3C_1 \times {}^5C_1}{{}^8C_2} = \frac{15}{28}$$

From a vessel containing 3 white and 5 black balls, 4 balls transferred into an empty vessel from this vessel 2 balls drawn and it found to be white. what is the probability that 2 balls transferred 3 are white and 1 black?

$$P(W) = \frac{{}^3C_2 \times {}^5C_2}{{}^8C_4} = \frac{5}{70} = \frac{1}{14}, P(B) = \frac{{}^3C_1 \times {}^5C_3}{{}^8C_4} = \frac{30}{70} = \frac{3}{7}, P(WB) = \frac{{}^3C_1 \times {}^5C_2}{{}^8C_4} = \frac{15}{70} = \frac{3}{14}$$

$$P(W_1) = \frac{{}^3C_1 \times {}^5C_1}{{}^8C_2} = \frac{5}{28} = \frac{1}{14}, P(B_1) = \frac{{}^3C_2}{{}^8C_2} = \frac{3}{14}$$

$$P(A/E_0) = 0, P(A/E_1) = 1/4, P(A/E_2) = 2/4, P(A/E_3) = 3/4, P(A/E_4) = 1$$

$$P(E_2/A) = ?$$

$$P(E_2/A) = \frac{P(E_2) \cdot P(A/E_2)}{\sum_{i=1}^4 P(E_i) \cdot P(A/E_i)}$$

$$= \frac{1/4 \times 2/4}{P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2) + P(E_3) \cdot P(A/E_3) + P(E_4) \cdot P(A/E_4)}$$

$$= \frac{1/4 \times 2/4}{1/4 \times 1/4 + 1/4 \times 2/4 + 1/4 \times 3/4 + 1/4 \times 1}$$

$$= \frac{1/8}{1/16 + 2/16 + 3/16 + 4/16} = \frac{1/8}{10/16} = \frac{1/8 \times 16}{10} = \frac{2}{10} = \frac{1}{5}$$

$$= \frac{3/56}{\frac{1}{2} \times \frac{1}{4} + \frac{1}{4} \times \frac{2}{4} + \frac{1}{4} \times \frac{3}{4} + 0 \times 1} = \frac{3/56}{(3/8 + 1/4 + 3/16)} = \frac{3/56}{(6/16 + 4/16 + 3/16)} = \frac{3/56}{13/16} = \frac{3}{13} \times \frac{16}{56} = \frac{3}{13} \times \frac{2}{7} = \frac{6}{91}$$

$$= \frac{3/56}{(3/8 + 1/4 + 3/16)} = \frac{3/56}{(6/16 + 4/16 + 3/16)} = \frac{3/56}{13/16} = \frac{3}{13} \times \frac{16}{56} = \frac{3}{13} \times \frac{2}{7} = \frac{6}{91}$$

$$= \frac{3}{13} \times \frac{16}{56} = \frac{3}{13} \times \frac{2}{7} = \frac{6}{91}$$

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$$= \frac{3}{13} \times \frac{16}{56} = \frac{3}{13} \times \frac{2}{7} = \frac{6}{91}$$

$$= \frac{3}{13} \times \frac{16}{56} = \frac{3}{13} \times \frac{2}{7} = \frac{6}{91}$$

Urn A contains 2 white, 1 black and 3 red balls, urn B contains 1 black and 4 red balls and urn C contains 4 white, 1 black and 1 red ball. One urn is chosen at random and 2 balls are drawn. The balls happen to be red & black. what is the probability that both balls from urn B?

511

$$\begin{matrix} 2W \\ 1B \\ 3R \end{matrix} \quad \begin{matrix} 1B \\ 4R \end{matrix} \quad \begin{matrix} 4W \\ 1B \\ 1R \end{matrix}$$

$$\text{let Event } E_1 = \text{Urn A is chosen} \Rightarrow P(E_1) = 1/3$$

$$E_2 = \text{Urn B} \Rightarrow P(E_2) = 1/3$$

$$E_3 = \text{Urn C} \Rightarrow P(E_3) = 1/3$$

$$\text{let event } M = \text{obtaining one red \& one black}$$

$$P(M/E_1) = \frac{{}^3C_1 \times {}^1C_1}{{}^4C_2} = \frac{3}{6} = \frac{1}{2}$$

$$P(M/E_2) = \frac{{}^4C_1 \times {}^1C_1}{{}^5C_2} = \frac{4}{10} = \frac{2}{5}$$

$$P(M/E_3) = \frac{{}^1C_1 \times {}^4C_1}{{}^5C_2} = \frac{4}{10} = \frac{2}{5}$$

$$P(E_2/M) = \frac{P(E_2) \cdot P(M/E_2)}{P(E_1) \cdot P(M/E_1) + P(E_2) \cdot P(M/E_2) + P(E_3) \cdot P(M/E_3)}$$

$$= \frac{1/3 \times 2/5}{1/3 \times 1/2 + 1/3 \times 2/5 + 1/3 \times 2/5} = \frac{2/15}{1/3 \times 1/2 + 2/15 + 2/15} = \frac{2/15}{1/6 + 4/15} = \frac{2/15}{5/10 + 4/10} = \frac{2/15}{9/10} = \frac{2}{15} \times \frac{10}{9} = \frac{4}{27}$$

There are 3 bags, each containing 5 white balls and 3 black balls. Also there are 2 bags, each containing 2 white balls and 4 black balls. A white ball is drawn at random. Find the probability that this white ball is from a bag of the first group.

First event = $E_1 \Rightarrow P(E_1) = \frac{3}{5}$
 " " = $E_2 \Rightarrow P(E_2) = \frac{2}{5}$

Let event A = obtaining one white ball.

$$P(A/E_1) = \frac{5 \times 3}{84} = \frac{5}{8}$$

$$P(A/E_2) = \frac{2 \times 4}{64} = \frac{2}{8} = \frac{1}{4}$$

$$P(E_1/A) = \frac{P(E_1) P(A/E_1)}{P(E_1) P(A/E_1) + P(E_2) P(A/E_2)}$$

$$= \frac{\frac{3}{5} \times \frac{5}{8}}{\frac{3}{5} \times \frac{5}{8} + \frac{2}{5} \times \frac{1}{4}} = \frac{\frac{3}{8}}{\frac{3}{8} + \frac{2}{20}} = \frac{\frac{3}{8}}{\frac{15}{40} + \frac{4}{20}} = \frac{\frac{3}{8}}{\frac{15+8}{20}} = \frac{\frac{3}{8}}{\frac{23}{20}} = \frac{3}{8} \times \frac{20}{23} = \frac{15}{23}$$

There are three machines A, B and C producing 1000, 2000 and 3000 articles per hour respectively. These machines are known to be producing 1%, 2% and 3% defective respectively. One article is selected at random from an hour production of the three machines and found to be defective. What is the probability that the article is produced from (i) Machine A, (ii) Machine B, (iii) Machine C?

Event E_1 = production of machine A
 " E_2 = " " B
 " E_3 = " " C

Then $P(E_1) = \frac{1000}{6000} = \frac{1}{6}$ and $P(E_3) = \frac{3000}{6000} = \frac{1}{2}$
 $P(E_2) = \frac{2000}{6000} = \frac{1}{3}$

Event A = total defective item.

$$P(A/E_1) = 1\% = 0.01$$

$$P(A/E_2) = 2\% = 0.02$$

$$P(A/E_3) = 3\% = 0.03$$

$$P(A) = P(E_1) P(A/E_1) + P(E_2) P(A/E_2) + P(E_3) P(A/E_3)$$

$$= \frac{1}{6} \times \frac{1}{100} + \frac{1}{3} \times \frac{2}{100} + \frac{1}{2} \times \frac{3}{100}$$

$$= \frac{1+4+9}{600} = \frac{14}{600}$$

$$P(E_1/A) = \frac{P(E_1/A)}{P(A)} = \frac{P(E_1) P(A/E_1)}{P(A)} = \frac{\frac{1}{6} \times \frac{1}{100}}{\frac{14}{600}} = \frac{1}{14}$$

$$P(E_2/A) = \frac{P(E_2/A)}{P(A)} = \frac{P(E_2) P(A/E_2)}{P(A)} = \frac{\frac{1}{3} \times \frac{2}{100}}{\frac{14}{600}} = \frac{2}{14}$$

$$P(E_3/A) = \frac{P(E_3/A)}{P(A)} = \frac{P(E_3) P(A/E_3)}{P(A)} = \frac{\frac{1}{2} \times \frac{3}{100}}{\frac{14}{600}} = \frac{9}{14}$$

CHAPTER 8

Random Variable and Probability Distribution

Random variable

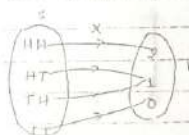
A random variable is a real valued function defined on sample space S (domain) and range $(-\infty, \infty)$ in a random expt.

Example

96. Two coins are tossed

Let x be real valued function of getting head.

$S = \{HH, HT, TH, TT\}$



Types

Discrete random variable

Continuous " "

Discrete Random variable

A real valued function defined on a discrete sample space is called discrete random variable. e.g. $x = \{H, T\}$

Continuous Random variable

A random variable x is said to be continuous if it can take all possible values between certain limits.

Example:-

$x = \{\text{height of people in Nepal lies bet } 5 \text{ to } 6 \text{ feet}\}$

$x = \{x: 5 \leq x \leq 6\}$

Probability mass function (p.m.f)

Let x be a discrete random variable with values $x_1, x_2, \dots, x_n$ and their probabilities $P(x_1), P(x_2), \dots, P(x_n)$. Then probability $P(x_i)$, $i = 1, 2, \dots, n$ is called probability mass function. (p.m.f) if it satisfies

$$(i) P(x_i) \geq 0 \text{ for all } i$$

$$(ii) \sum_{i=1}^n P(x_i) = 1 \text{ i.e. } P(x_1) + P(x_2) + \dots + P(x_n) = 1$$

Example,

In coin tossing

$x = \{H, T\}$

$$P(H) = \frac{1}{2}$$

$$P(T) = \frac{1}{2}$$

$$\text{here, } P(H) + P(T) = \frac{1}{2} + \frac{1}{2} = 1.$$

Def. Probability Distribution function (Cumulative prob. distribution)

The probability distribution function of a discrete random variable $x = \{x_1, x_2, x_3, \dots, x_n\}$ is denoted by $F(x_j)$ and defined

$$F(x_j) = P(x_1) + P(x_2) + \dots + P(x_j)$$

Example 96. In a dice $\{1, 2, 3, 4, 5, 6\}$

$$P(1) = \frac{1}{6}$$

$$F(1) = \frac{1}{6}$$

$$P(2) = \frac{1}{6}$$

$$F(2) = P(1) + P(2) = \frac{2}{6}$$

$$P(3) = \frac{1}{6}$$

$$F(3) = P(1) + P(2) + P(3) = \frac{3}{6}$$

$$P(4) = \frac{1}{6}$$

$$F(4) = P(1) + P(2) + P(3) + P(4) = \frac{4}{6}$$

$$P(5) = \frac{1}{6}$$

$$F(5) = P(1) + P(2) + P(3) + P(4) + P(5) = \frac{5}{6}$$

$$P(6) = \frac{1}{6}$$

$$F(6) = P(1) + P(2) + P(3) + P(4) + P(5) + P(6) = 1$$

$$\sum P(x) = 1$$

Probability density function (pdf)

probability density function is simply denoted by $f(x)$ the following properties

$$(i) f(x) \geq 0 \text{ for all } x$$

$$(ii) \int_{-\infty}^{\infty} f(x) dx = 1$$

Exercise-7

5. A random variable X has following probability function.

value of X, x :	-2	-1	0	1	2	3
$P(x)$:	0.1	K (0.1)	0.2	$2K$ (0.2)	0.3	K (0.1)

(i) find the value of K

we have,

$$\sum P(x) = 1$$

$$0.1 + K + 0.2 + 2K + 0.3 + K = 1$$

$$4K = 0.4$$

$$K = 0.1$$

(ii) find $P(X < 2)$, $P(-2 < X < 2)$, $P(0 \leq X < 2)$, $P(X \geq -2)$

$$P(X < 2) = P(-2) + P(-1) + P(0) + P(1)$$

$$= 0.1 + 0.1 + 0.2 + 0.2$$

$$= 0.6$$

$$P(-2 < X < 2) = P(-1) + P(0) + P(1)$$

$$= 0.1 + 0.2 + 0.2$$

$$= 0.5$$

$$P(0 \leq X < 2) = P(0) + P(1)$$

$$= 0.2 + 0.2$$

$$= 0.4$$

$$P(X \geq -2) = P(-2) + P(-1) + P(0) + P(1) + P(2) + P(3)$$

$$= 0.1 + 0.1 + 0.2 + 0.2 + 0.3 + 0.1$$

$$= 1$$

(iii) find $F(-2)$, $F(0)$, $F(3)$

$$F(-2) = P(-2) = 0.1$$

$$F(0) = P(-2) + P(-1) + P(0) = 0.1 + 0.1 + 0.2 = 0.4$$

$$F(3) = P(-2) + P(-1) + P(0) + P(1) + P(2) + P(3) = 1$$

(iv) Construct the cdf $F(x)$

$X: x$	-2	-1	0	1	2	3
$F(x)$	0.1	0.2	0.4	0.6	0.9	1

6. From a lot of 10 items containing 3 defective, a sample of 4 items is drawn at random. Let the random variable X denote the number of defective items in the sample. Answer the following when the sample is drawn without replacement.

(i) find the prob. distribution of X

$$X = \{ \text{No. of defective item} \}$$

$$= \{ 0, 1, 2, 3 \}$$

$X: x$	$P(x)$	$F(x)$
0	$\frac{{}^3C_0 \times {}^7C_4}{{}^{10}C_4} = 0.166$	0.166
1	$\frac{{}^3C_1 \times {}^7C_3}{{}^{10}C_4} = 0.5$	0.666
2	$\frac{{}^3C_2 \times {}^7C_2}{{}^{10}C_4} = 0.3$	0.966
3	$\frac{{}^3C_3 \times {}^7C_1}{{}^{10}C_4} = 0.033$	1

(ii) find $P(X \leq 1)$, $P(X < 1)$ and $P(0 \leq X < 2)$

$$P(X \leq 1) = P(0) + P(1) = 0.166 + 0.5 = 0.666$$

$$P(X < 1) = P(0) = 0.166$$

$$P(0 \leq X < 2) = P(1) = 0.5$$

7. A random variable X has the following probability function

value of $X: x$	0	1	2	3	4	5	6	7
$P(x)$	0	K	$2K$	$3K$	$4K$	$5K$	$6K$	$7K^2 + K$

(i) find K

we have,

$$\sum P(x) = 1$$

$$K + 2K + 3K + 4K + 5K + 6K + 7K^2 + K = 1$$

$$\text{or, } 10K^2 + 9K - 1 = 0$$

$$K = \frac{-9 \pm \sqrt{9^2 + 4 \times 10 \times 1}}{2 \times 10} = \frac{-9 \pm \sqrt{121}}{20} = \frac{-9 \pm 11}{20} = \frac{2}{20} = \frac{1}{10}$$

ii) Estimate $P(X < 6)$, $P(X \geq 6)$ and $P(0 < X < 5)$

$$\begin{aligned} P(X < 6) &= P(0) + P(1) + P(2) + P(3) + P(4) + P(5) + P(6) \\ &= K + 2K + 3K + 3K + K^2 + 0 \\ &= \frac{1}{10} + 2 \times \frac{1}{10} + 3 \times \frac{1}{10} + 3 \times \frac{1}{10} + \left(\frac{1}{10}\right)^2 \\ &= \frac{1}{10} + \frac{2}{10} + \frac{3}{10} + \frac{3}{10} + \frac{1}{100} \\ &= \frac{10 + 20 + 30 + 30 + 1}{100} = \frac{81}{100} \end{aligned}$$

$$\begin{aligned} P(X \geq 6) &= P(6) + P(7) \\ &= 2K^2 + 7K^2 + K \\ &= 8K^2 + K \\ &= 8 \times \frac{1}{100} + \frac{1}{10} = \frac{8 + 10}{100} = \frac{18}{100} \end{aligned}$$

$$\begin{aligned} P(0 < X < 5) &= P(1) + P(2) + P(3) + P(4) \\ &= 5 + 2K + 3K + 3K \\ &= 8K = \frac{8}{10} \end{aligned}$$

iii) Find the distribution function of X

x	0	1	2	3	4	5	6	7
$P(X \leq x)$	0	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{5}{10}$	$\frac{8}{10}$	$\frac{81}{100}$	$\frac{87}{100}$	1

Let $P(x)$ be the probability function of a discrete random variable which assumes the values x_1, x_2, x_3, x_4 such that

$$2P(x_1) = 3P(x_2) = P(x_3) = 5P(x_4)$$

find the prob. distribution & prob. distribution function.

$$\text{Sol } X = \{x_1, x_2, x_3, x_4\}$$

$$2P(x_1) = 3P(x_2) = P(x_3) = 5P(x_4) = K$$

$$P(x_1) = \frac{K}{2}, P(x_2) = \frac{K}{3}, P(x_3) = K \text{ \& } P(x_4) = \frac{K}{5}$$

we have

$$\sum P(x_i) = 0.1$$

$$P(x_1) + P(x_2) + P(x_3) + P(x_4) = 1$$

$$\frac{K}{2} + \frac{K}{3} + K + \frac{K}{5} = 1$$

$$15K + 10K + 30K + 6K = 1$$

$$61K = 1$$

$$61K = 20$$

$$K = \frac{20}{61}$$

x	x_1	x_2	x_3	x_4
$P(x)$	$15/61$	$10/61$	$20/61$	$6/61$
$F(x)$	$15/61$	$25/61$	$45/61$	1

⑤ Let $P(x) = \frac{x}{15}$, $x = 1, 2, 3, 4, 5$
elsewhere

(i) Find $P\{X = 1 \text{ or } 2\}$

$$\begin{aligned} X: x & 1 & 2 & 3 & 4 & 5 \\ P(x) & \frac{1}{15} & \frac{2}{15} & \frac{3}{15} & \frac{4}{15} & \frac{5}{15} \end{aligned}$$

$$P\{X = 1 \text{ or } 2\} = P(X = 1) + P(X = 2)$$

$$= \frac{1}{15} + \frac{2}{15} = \frac{3}{15} = \frac{1}{5}$$

(ii) $P\{1 < X < 5/2 \mid X > 1\}$

$$P(2) / \{P(2) + P(3) + P(4) + P(5)\}$$

$$= \frac{2/15}{2/15 + 3/15 + 4/15 + 5/15} = \frac{2}{14} = \frac{1}{7}$$

⑩ A random variable X has the density

$$f(x) = a + bx^2, \quad 0 < x < 1$$

Determine a and b such that $a + b = 2$ --- ①

we have

$$\text{Total probability} = \int_0^1 f(x) dx = 1$$

$$= \int_0^1 (a + bx^2) dx = 1$$

$$= \left[ax + \frac{bx^3}{3} \right]_0^1 = 1$$

$$= a + \frac{b}{3} = 1$$

$$\therefore 3a + b = 3 \quad \text{--- ②}$$

Solving eqⁿ: (i) & (ii)

$$a+b=2$$

$$3a+b=3$$

$$+2a = 1$$

$$a = \frac{1}{2}, b = \frac{3}{2}$$

(ii) Let x be a random variable with pdf

$$f(x) = ax \quad 0 \leq x \leq 1$$

$$= a \quad 1 \leq x \leq 2$$

$$= -ax+3a \quad 2 \leq x \leq 3$$

$$= 0 \quad \text{otherwise}$$

Compute

(i) the constant a

$$\text{Total probability} \int_0^3 f(x) dx = 1$$

$$\text{or, } \int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^3 f(x) dx = 1$$

$$\text{or, } \int_0^1 ax dx + \int_1^2 a dx + \int_2^3 (-ax+3a) dx = 1$$

$$2 \left[\frac{ax^2}{2} \right]_0^1 + [ax]_1^2 + \left[-\frac{ax^2}{2} + 3ax \right]_2^3 = 1$$

$$2 \left[\frac{a}{2} + (2a - a) - \frac{a}{2} + 6a - 6a \right] = 1$$

$$2 \left[\frac{a}{2} + a - \frac{a}{2} + 6a - 6a \right] = 1$$

$$2 \left[a + 6a - 6a \right] = 1$$

$$2 \left[a \right] = 1$$

$$a = \frac{1}{2}$$

(ii) $P(x \leq 1.5)$

$$\int_0^{1.5} f(x) dx$$

$$= \int_0^1 f(x) dx + \int_1^{1.5} f(x) dx$$

$$= \int_0^1 ax dx + \int_1^{1.5} a dx$$

$$= \left[\frac{ax^2}{2} \right]_0^1 + [ax]_1^{1.5}$$

$$= \frac{a}{2} + 1.5a - a$$

$$= \frac{a}{2} = \frac{1}{4}$$

(ii) Given the pdf

$$f(x) = \begin{cases} Kx(1-x), & \text{for } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{Total probability} \int_0^1 f(x) dx = 1$$

$$\int_0^1 Kx(1-x) dx = 1$$

$$\text{or, } K \int_0^1 (x-x^2) dx = 1$$

$$K \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = 1$$

$$K \left(\frac{1}{2} - \frac{1}{3} \right) = 1$$

$$K \left(\frac{3-2}{6} \right) = 1$$

$$K = 6$$

Suppose X is a r.v. having p.m.f $P(X)$ given as

X	-3	-1	0	1	2	3	5	8
$P(X)$	0.1	0.2	0.15	0.2	0.1	0.15	0.05	0.05

Compute the probability

(i) X is negative

$$= P(X=-3) + P(X=-1)$$

$$= 0.1 + 0.2$$

$$= 0.30 \quad \text{Q}$$

(ii) X is even

$$= P(X=2) + P(X=8)$$

$$= 0.10 + 0.05$$

$$= 0.15 \quad \text{Q}$$

(iii) X takes a value between 1 and 8 inclusive

$$= P(X=1) + P(X=2) + P(X=3) + P(X=5) + P(X=8)$$

$$= 0.2 + 0.1 + 0.15 + 0.05 + 0.05$$

$$= 0.55 \quad \text{Q}$$

$P(X=3/X \leq 0)$

$$= P(X=3)/P(X \leq 0) = 0.15/(0.1+0.2+0.15) = 0.2/0.45 = 0.44 \quad \text{Q}$$

$P(X \geq 5/X > 0)$

$$= P(X \geq 3)/P(X > 0) = (0.15+0.05+0.05)/(0.2+0.1+0.15+0.05+0.05)$$

$$= 0.25/0.55 = 0.4545 \quad \text{Q}$$

A probability density function is given

$$f(x) = \frac{x+1}{6}, \quad 0 \leq x < 2$$

$$= \frac{x+1}{6}, \quad \text{otherwise}$$

mean E

$$= \int_0^2 x f(x) dx$$

$$= \int_0^2 x \left(\frac{x+1}{6} \right) dx$$

$$= \frac{1}{6} \int_0^2 (x^2 + x) dx$$

$$= \frac{1}{6} \left[\frac{x^3}{3} + \frac{x^2}{2} \right]_0^2$$

$$= \frac{1}{6} \left[\frac{8}{3} + 2 \right]$$

$$= \frac{1}{6} \left[\frac{8+6}{3} \right]$$

$$= \frac{14}{18}$$

$$= 1.11 \quad \text{Q}$$

(ii) $F(1.5)$

$$= \int_0^{1.5} f(x) dx$$

$$= \frac{1}{6} \int_0^{1.5} (x+2) dx$$

$$= \frac{1}{6} \left[\frac{x^2}{2} + 2x \right]_0^{1.5}$$

$$= \frac{1}{6} \left[\frac{2.25}{2} + 3 \right]$$

$$= \frac{(1.125+3)}{6}$$

$$= \frac{4.125}{6}$$

$$= 0.6875$$

(iii) $P(0 < X \leq 1)$

$$= \int_0^1 f(x) dx$$

$$= \frac{1}{6} \int_0^1 (x+2) dx$$

$$= \frac{1}{6} \left[\frac{x^2}{2} + 2x \right]_0^1$$

$$= \frac{1}{6} \left[\frac{1}{2} + 2 \right]$$

$$= \frac{2.5}{6}$$

$$= 0.416$$

(15) A probability curve $f(x) = e^{-x}$, find mean & variance

Soln

$$\text{Mean} = \int_0^\infty x f(x) dx \quad \text{when } x \geq 0$$

$$E(x) = \int_0^\infty x \cdot e^{-x} dx$$

$$= [x e^{-x} - e^{-x}]_0^\infty$$

$$= 0 - 0 + 0 + 1 = 1$$

$$\text{variance } V(x) = \int_0^\infty x^2 f(x) dx - \{E(x)\}^2$$

$$= \int_0^\infty x^2 e^{-x} dx - \{E(x)\}^2$$

$$= [-x^2 e^{-x} + 2x e^{-x} - e^{-x}]_0^\infty - 1^2$$

$$= 0 + 0 + 0 + 0 + 2 \times 1 - 1$$

$$= 2 - 1 = 1 \quad \text{Q}$$

(16) The diameter of an electric cable, say X is assumed to be continuous random variable with $f(x) = 6x(1-x)$, $0 \leq x \leq 1$

(i) check the curve is p.d.f.

$$\text{If curve is p.d.f then } \int_0^1 f(x) dx = \int_0^1 6x(1-x) dx = \int_0^1 (6x - 6x^2) dx$$

$$= 6 \left[\frac{x^2}{2} - \frac{2x^3}{3} \right]_0^1 = 6 \left[\frac{1}{2} - \frac{2}{3} \right] = 6 \times \frac{1}{6} = 1$$

$\therefore$ According to properties of p.d.f $\int f(x) dx = 1$ then it is p.d.f

(i) Mean $(E_x) = \int_0^1 xf(x) dx$
 $= \int_0^1 x \cdot 6x(1-x) dx = 6 \int_0^1 (x^2 - x^3) dx$
 $= 6 \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = 6 \left[\frac{1}{3} - \frac{1}{4} \right] = 6 \times \frac{4-3}{12} = \frac{6}{12} = 0.5$

(ii) If M is median, then
 $\int_0^M f(x) dx = \frac{1}{2}$

$\approx \int_0^M 6x(1-x) dx = \frac{1}{2}$
 $\approx 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^M = \frac{1}{2}$

$\approx 6 \left[\frac{M^2}{2} - \frac{M^3}{3} \right] = \frac{1}{2}$
 $\approx 3M^2 - 2M^3 = \frac{1}{2}$

$\approx 6M^2 - 4M^3 = 1$

$\approx 4M^3 - 6M^2 + 1 = 0$

$\approx 4M^3 - 2M^2 - 4M^2 + 2M - 2M + 1 = 0$

$\approx 4M^2(M-1) - 2M(M-1) - 1(2M-1) = 0$

$\approx (4M-1)(M-1) = 0$

$4M-1=0, 4M^2-2M-1=0$

$M = \frac{1}{4}$ Q

(iii) mode $f'(x) = 0$

$\frac{d}{dx} f(x) = 0$

$\approx \frac{d}{dx} [6x(1-x)] = 0$

$\approx 6 \frac{d}{dx} (x-x^2) = 0$

$\approx 6(1-2x) = 0$

$1-2x=0$

$x = \frac{1}{2}$ Q

(iii) compute $P(\mu - 2\sigma < X < \mu + 2\sigma)$

$P(0.5 - 2 \times 0.1732 < X < 0.5 + 2 \times 0.1732)$

$0.95 P(0.053 < X < 0.95)$

$\int_{0.053}^{0.95} f(x) dx$

$6 \int_{0.053}^{0.95} (x-x^2) dx = 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_{0.053}^{0.95}$

$= 6 \left[0.4512 - 0.2986 - 0.0014 + 0.0009 \right]$

$= 0.993$

(12) A continuous r.v. X follows the probability law

$f(x) = Ax^2, 0 \leq x \leq 1$

So, total probability = 1

$\int_0^1 f(x) dx = 1$

$\approx \int_0^1 Ax^2 dx = 1$

$\approx A \left[\frac{x^3}{3} \right]_0^1 = 1$

$\approx A \frac{1}{3} = 1$

i) $\approx A = 3$

(ii) $\int_{0.2}^{0.5} f(x) dx$

$= \int_{0.2}^{0.5} 3x^2 dx$

$= 3 \left[\frac{x^3}{3} \right]_{0.2}^{0.5}$

$= 3 \left[\frac{0.5^3}{3} - \frac{0.2^3}{3} \right]$

$= 3 \left[0.04167 - 0.00444 \right]$

$= 0.117$ Q

(13) If a random variable X has the density function

$f(x) = \begin{cases} \frac{1}{4}, & -2 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$

then

(i) $P(X < 1)$

$= \int_{-2}^0 f(x) dx + \int_0^1 f(x) dx = \int_{-2}^0 \frac{1}{4} dx + \int_0^1 \frac{1}{4} dx = \frac{1}{4} \left[x \right]_{-2}^0 + \frac{1}{4} \left[x \right]_0^1$

$= \frac{1}{4} \times 2 + \frac{1}{4} = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ Q

We have Variance $(V_x) = \int_{-2}^2 x^2 f(x) dx - \{E(x)\}^2$

$= \int_{-2}^2 x^2 \cdot \frac{1}{4} dx - 0.5^2$

$= \frac{1}{4} \int_{-2}^2 (x^3 - x^4) dx - 0.5^2$

$= \frac{1}{4} \left[\frac{x^4}{4} - \frac{x^5}{5} \right]_{-2}^2 - 0.5^2$

$= \frac{1}{4} \left[\frac{2^4}{4} - \frac{2^5}{5} - \left(\frac{(-2)^4}{4} - \frac{(-2)^5}{5} \right) \right] - 0.5^2$

$= \frac{1}{4} - 0.5^2$

≈ 0.075

$\sigma = \sqrt{0.075} = 0.2736$ Q

(ii) $p(x > 1)$

$$\begin{aligned}
 &= 1 - p(x \leq 1) = 1 - p(-1 \leq x \leq 1) \\
 &= 1 - \int_{-1}^1 f(x) dx = 1 - \int_{-1}^1 \frac{1}{4} dx = 1 - \frac{1}{4} [x]_{-1}^1 = 1 - \frac{1}{4} [1 + 1] \\
 &= 1 - \frac{1}{2} = \frac{1}{2}
 \end{aligned}$$

(iii) $p(2x + 3 > 5)$

$$\begin{aligned}
 &p(1 - p(2x + 3 \leq 5)) \\
 &\text{when } x = -3 \\
 &1 - p(-1 \leq x \leq 5) = 1 - \left[\int_{-1}^0 f(x) dx + \int_0^2 f(x) dx + \int_2^5 f(x) dx \right] \\
 &= 1 - \left[\int_{-1}^0 \frac{1}{4} dx + \int_0^2 \frac{1}{4} dx + 0 \right] \\
 &= 1 - \left[\frac{1}{4} (x)_{-1}^0 + \frac{1}{4} (x)_{0}^2 \right] = 1 - \left[\frac{1}{4} \times 1 + \frac{1}{4} \times 2 \right] \\
 &= 1 - \left[\frac{1}{4} + \frac{1}{2} \right] = 1 - \frac{3}{4} = \frac{1}{4}
 \end{aligned}$$

(iv) A continuous distribution of a variable x in the range $(-3, 3)$ is defined by

$$\begin{aligned}
 f(x) &= \frac{1}{16} (x+3)^2, \text{ for } -3 \leq x < -1 \\
 &= \frac{1}{16} (1-x)^2, \text{ for } -1 \leq x \leq 1 \\
 &= \frac{1}{16} (3-x)^2, \text{ for } 1 \leq x \leq 3
 \end{aligned}$$

we have,

$$\begin{aligned}
 &\int_{-3}^3 f(x) dx = 0 \\
 &= \int_{-3}^{-1} \frac{1}{16} (x+3)^2 dx + \int_{-1}^1 \frac{1}{16} (1-x)^2 dx + \int_1^3 \frac{1}{16} (3-x)^2 dx \\
 &= \left[\frac{1}{16} \left\{ \frac{x^3}{3} + 3x^2 + 9x \right\}_{-3}^{-1} + \frac{1}{16} \left\{ \frac{x^3}{3} - 2x^2 + x \right\}_{-1}^1 + \frac{1}{16} \left\{ \frac{x^3}{3} - 3x^2 + 9x \right\}_1^3 \right] \\
 &= \frac{1}{16} \left[\left\{ \frac{-1}{3} + 9 - 27 + 27 \right\} - \left\{ \frac{-27}{3} + 27 - 27 \right\} \right] + \frac{1}{16} \left[\left\{ \frac{1}{3} - 2 + 1 \right\} - \left\{ \frac{-1}{3} - 2 + 1 \right\} \right] + \frac{1}{16} \left[\left\{ \frac{27}{3} - 27 + 27 \right\} - \left\{ \frac{1}{3} - 9 + 9 \right\} \right] \\
 &= \frac{1}{16} \left[\frac{-1}{3} + 9 - 27 + 27 - \left(\frac{-27}{3} + 27 - 27 \right) + \frac{1}{3} - 2 + 1 - \left(\frac{-1}{3} - 2 + 1 \right) + \frac{27}{3} - 27 + 27 - \left(\frac{1}{3} - 9 + 9 \right) \right] \\
 &= \frac{48}{16} = 1 \quad \text{Proved}
 \end{aligned}$$

Mean (E_x) =

$$\begin{aligned}
 &= \int_{-3}^3 x f(x) dx \\
 &= \frac{1}{16} \left[\int_{-3}^{-1} x(x+3)^2 dx + \int_{-1}^1 x(1-x)^2 dx + \int_1^3 x(3-x)^2 dx \right] \\
 &= \frac{1}{16} \left[\int_{-3}^{-1} (x^3 + 6x^2 + 9x) dx + \int_{-1}^1 (x - 2x^2 + x^3) dx + \int_1^3 (9x - 6x^2 + x^3) dx \right] \\
 &= \frac{1}{16} \left[\left\{ \frac{x^4}{4} + \frac{6x^3}{3} + \frac{9x^2}{2} \right\}_{-3}^{-1} + \left\{ \frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4} \right\}_{-1}^1 + \left\{ \frac{9x^2}{2} - \frac{6x^3}{3} + \frac{x^4}{4} \right\}_1^3 \right] \\
 &= \frac{1}{16} \left[\left\{ \frac{1}{4} - 18 + \frac{27}{2} \right\} - \left\{ \frac{81}{4} - \frac{162}{3} + \frac{81}{2} \right\} + \left\{ \frac{81}{2} - 36 + \frac{81}{4} \right\} - \left\{ \frac{9}{2} - 18 + \frac{9}{4} \right\} \right] \\
 &= 0
 \end{aligned}$$

Variance (V_x) =

$$\begin{aligned}
 &= \int_{-3}^3 x^2 f(x) dx - \{E(x)\}^2 \\
 &= \frac{1}{16} \left[\int_{-3}^{-1} x^2 (x+3)^2 dx + \int_{-1}^1 x^2 (1-x)^2 dx + \int_1^3 x^2 (3-x)^2 dx \right] - 0 \\
 &= \frac{1}{16} \left[\int_{-3}^{-1} (x^4 + 6x^3 + 9x^2) dx + \int_{-1}^1 (x^2 - 2x^3 + x^4) dx + \int_1^3 (9x^2 - 6x^3 + x^4) dx \right] \\
 &= \frac{1}{16} \left[\left\{ \frac{x^5}{5} + \frac{6x^4}{4} + \frac{9x^3}{3} \right\}_{-3}^{-1} + \left\{ \frac{x^3}{3} - \frac{2x^4}{4} + \frac{x^5}{5} \right\}_{-1}^1 + \left\{ \frac{9x^3}{3} - \frac{6x^4}{4} + \frac{x^5}{5} \right\}_1^3 \right] \\
 &= \frac{1}{16} \left[\left\{ \frac{-1}{5} + \frac{6}{5} - \frac{27}{3} \right\} - \left\{ \frac{-243}{5} + \frac{162}{4} - \frac{27}{3} \right\} + \left\{ \frac{27}{3} - \frac{6}{4} + \frac{243}{5} \right\} - \left\{ \frac{9}{3} - \frac{6}{4} + \frac{1}{5} \right\} \right] \\
 &= \frac{-12 + 90 - 180 + 2916 - 7290 + 4860 + 120 - 24 + 120 - 24}{16 \times 16} + \frac{4860 - 7290 + 2916 - 180 + 90 - 12}{16 \times 16} \\
 &= 1 \quad \text{Proved}
 \end{aligned}$$

(v) find K so that the following can serve as the probability of random variable

$$f(x) = \begin{cases} 0, & \text{for } x \leq 0 \\ Kxe^{-4x^2}, & \text{for } x > 0 \end{cases}$$

(i) Determine K

$$\begin{aligned}
 &\text{Total probability} = 1 \\
 &\int_{-\infty}^{\infty} Kxe^{-4x^2} dx = 1 \\
 &\text{or } K \int_{-\infty}^{\infty} x e^{-4x^2} dx = 1 \\
 &K \left[0 - 1 \right] = 1 \\
 &K = 2
 \end{aligned}$$

the Continuous distribution.

Let $f(x) = \frac{1}{2} (1-x^2)$, $0 \leq x \leq 1$. Prove that distribution is symmetric.

Total probability = 1

$$\int_0^1 f(x) dx = 1$$

$$\Rightarrow \int_0^1 \frac{1}{2} (1-x^2) dx = 1$$

$$\Rightarrow \frac{1}{2} \left[x - \frac{x^3}{3} \right]_0^1 = 1$$

$$\Rightarrow \frac{1}{2} \left[\frac{1}{2} - \frac{1}{3} \right] = 1$$

$$\Rightarrow \frac{1}{2} = 1$$

distribution is symmetric then Mean = median = mode.

$$\begin{aligned} \text{Mean} &= \int_0^1 x f(x) dx = \int_0^1 x \cdot \frac{1}{2} (1-x^2) dx = \frac{1}{2} \int_0^1 (x-x^3) dx \\ &= \frac{1}{2} \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1 = \frac{1}{2} \left[\frac{1}{2} - \frac{1}{4} \right] = \frac{1}{2} \left[\frac{1}{4} \right] = \frac{1}{8} \end{aligned}$$

Let $x = 0$, then

for mode, $f'(x) = 0$

$$\int_0^1 f(x) dx = \frac{1}{2}$$

$$\frac{d}{dx} \left\{ \frac{1}{2} (1-x^2) \right\} = 0$$

$$\int_0^1 \frac{1}{2} (1-x^2) dx = \frac{1}{2}$$

$$\text{or } \frac{1}{2} \cdot \frac{d}{dx} (1-x^2) = 0$$

$$\Rightarrow \frac{1}{2} \int_0^1 (1-x^2) dx = \frac{1}{2}$$

$$\text{or } \frac{1}{2} (1-2x) = 0$$

$$\Rightarrow \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{2}$$

$$\Rightarrow 1-2x = 0$$

$$\Rightarrow \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{2}$$

$$\therefore x = \frac{1}{2}$$

$$\Rightarrow \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{2}$$

Mean = median = mode = $\frac{1}{2}$

$$\Rightarrow \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{2}$$

$\therefore$ the distribution is symmetric.

$$\Rightarrow 6M^2 - 4M^3 = 1$$

$$\Rightarrow 4M^3 - 6M^2 + 1 = 0$$

$$\Rightarrow 4M^3 - 6M^2 - 2M + 1 = 0$$

$$\Rightarrow M^2(2M-1) - 2M(2M-1) - 1(2M-1) = 0$$

$$\Rightarrow (2M-1)(2M^2-2M-1) = 0$$

$$\Rightarrow M = \frac{1}{2}$$

(21) A batch of small caliber ammunition is accepted as satisfactory if a sample of 5 shot falls more than 2 feet from the target at a given range. If x , the distance from the target to a given impact point, actually has the density function $f(x) = K \cdot x e^{-x^2}$, $0 < x < 3$ where K is a number which makes the probability density function valid. What is the value of K and what is the probability that the batch will be accepted?

SN: Total probability = 1

$$\int_0^3 K \cdot x e^{-x^2} dx = 1$$

Required probability = $P(0.5 < x < 1.5)$

$$\Rightarrow \int_{0.5}^{1.5} K \cdot x e^{-x^2} dx$$

$$\text{or } K \left[-\frac{1}{2} e^{-x^2} \right]_0^3 = 1$$

$$\Rightarrow \left[-\frac{1}{2} K x e^{-x^2} \right]_{0.5}^{1.5} = 1$$

$$\text{or } -K \left[\frac{1}{2} e^{-9} - \frac{1}{2} e^{-0} \right] = 1$$

$$\Rightarrow -\frac{1}{2} K \left[e^{-9} - 1 \right] = 1$$

$$\Rightarrow -K \left[\frac{1}{2} (e^{-9} - 1) \right] = 1$$

$$\therefore K = \frac{2}{1 - e^{-9}}$$

(22) Find the mean, median and mode for the distribution of $f(x)$.

$$0 < x < \pi, \quad f(x) = \frac{1}{\pi} \sin x$$

$$\text{SN: Mean} = \int_0^\pi x f(x) dx = \int_0^\pi x \sin x dx$$

Discrete

Expectation of Discrete Random variable:-

Let $X = \{x_1, x_2, x_3, \dots, x_n\}$ be a discrete random variable with probabilities $P(x_1), P(x_2), \dots, P(x_n)$ then Mathematical Expectation of X is defined by

$$E(X) = \mu = \sum_{i=1}^n x_i P(x_i) \text{ or } E(X) = \sum x P(x)$$

For example:-

6-faced die is rolled

$$X = S = \{1, 2, 3, 4, 5, 6\}$$

$X = S$	$P(X)$	$X \cdot P(X)$
1	$\frac{1}{6}$	$\frac{1}{6}$
2	$\frac{1}{6}$	$\frac{2}{6}$
3	$\frac{1}{6}$	$\frac{3}{6}$
4	$\frac{1}{6}$	$\frac{4}{6}$
5	$\frac{1}{6}$	$\frac{5}{6}$
6	$\frac{1}{6}$	$\frac{6}{6}$

$$\sum X \cdot P(X) = 21/6 = 3.5$$

Also, Variance (σ^2) of X is defined as

$$V(X) = \sigma^2 = \sum x^2 P(x) - \{E(X)\}^2$$

$$\therefore \sigma = \sqrt{\sum x^2 P(x) - \{E(X)\}^2}$$

Expectation of a continuous random variable

Let $X \in -\infty < X < \infty$ be a continuous random variable with pdf.

Then Expected value of X is defined as

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx, -\infty < x < \infty$$

Variance of X is defined as

$$V(X) = \int_{-\infty}^{\infty} x^2 f(x) dx - \{E(X)\}^2$$

Median & Mode:-

Median is the point which divides the distribution into two

equal parts

If M is median & $a \leq x \leq b$ then

$$\int_a^M f(x) dx = \int_M^b f(x) dx = \frac{1}{2}$$

Mode

mode is the value of x for which $f(x)$ is maximum
for maxima, $f'(x) = 0$

Exercise

(7) what is our mathematical expectation if we stand to win Rs. 100 if a balanced coin comes up head &

Ans.	Expt. (x)	P(x)	$x \cdot P(x)$
Head	100	$\frac{1}{2}$	$100 \cdot \frac{1}{2} = 50$

(8) If a cinema club sells 1000 raffle tickets for a cash prize of Rs. 1000. What is the mathematical expectation of a person who buys the tickets?

Ans.	Expt. (x)	P(x)	$x \cdot P(x)$
buy ticket	1000	$\frac{1}{1000}$	$1000 \cdot \frac{1}{1000} = 1$

(9) what is our mathematical expectation if we win Rs. 50 if a balanced coin comes up heads and lose Rs. 50 if it comes up tails?

Ans.	Expt. (x)	P(x)	$x \cdot P(x)$
H	50	$\frac{1}{2}$	25
T	-50	$\frac{1}{2}$	-25

$$E(X) = 0$$

- (10) The probability that a man fishing at a particular place will catch 1, 2, 3, 4 fish are 0.4, 0.3, 0.2 and 0.1 resp. what is the expected number of fish caught?

Solⁿ No. of caught fish (x) P(x) x P(x)

1	0.4	0.40
2	0.3	0.6
3	0.2	0.6
4	0.1	0.4

$$E(x) = \sum x P(x) = 2$$

- (11) Find the expectation of the number on a dice when thrown two times.

Solⁿ 1st throw 2nd throw

Expt (x)	P(x)	x P(x)	Expt (x)	P(x)	x P(x)
1	1/6	1/6	1	1/6	1/6
2	1/6	2/6	2	1/6	2/6
3	1/6	3/6	3	1/6	3/6
4	1/6	4/6	4	1/6	4/6
5	1/6	5/6	5	1/6	5/6
6	1/6	6/6	6	1/6	6/6

$$E(x)_1 = E(x P(x)) = 2\frac{1}{6}$$

$$E(x)_2 = E(x P(x)) = 2\frac{1}{6}$$

Expectation of two throws $E(x) = E(x_1) + E(x_2)$

$$= 2\frac{1}{6} + 2\frac{1}{6}$$

$$= 7\frac{1}{3}$$

- (12) In a given business venture a man can make a profit of Rs. 100 with probability 0.80 or suffer a loss of Rs. 400 with probability 0.2. Determine his expectation.

Solⁿ Expectation

P(x)	P(x)	x P(x)
P	100	80
L	-400	-80

$$E(x) = 720$$

- (13) A random variable x has the following probability function

Value of x, z	P(x)	x P(x)
-2	0.1	-0.2
-1	$k^2 (0.1)$	$-k^2 (0.1)$
0	0.2	0
1	$2k^2 (0.2)$	$2k^2 (0.2)$
2	0.5	0.6
3	$k^2 (0.1)$	$3k^2 (0.3)$

$$E(x) = k^2 + 0.6$$

$$E(x P(x)) = 0.6 + 4k^2$$

$$= 0.6 + 4 \times 0.1$$

$$= 0.80$$

- (14) A random variable x has the probability distribution

x	P(x)	x P(x)	Variance = $\sum x^2 P(x)$
-1	1/3	-1/3	1/3
0	1/6	0	0
1	1/6	1/6	1/6
2	1/3	2/3	4/3

$$E(x P(x)) = \frac{2}{3} + \frac{1}{6} + \frac{1}{6}$$

$$= \frac{4+1+1}{6} = \frac{6}{6} = 1$$

$$E(x^2 P(x)) = \frac{1}{3} + \frac{1}{6} + \frac{4}{3}$$

$$= \frac{2+1+8}{6} = \frac{11}{6}$$

- (i) Expectation of random variable $E(x) = \sum x P(x) = 0.5$
- (ii) Variance of r.v. $V(x) = E(x^2 P(x)) - [E(x)]^2$

$$= \frac{11}{6} - \left(\frac{1}{2}\right)^2$$

$$= \frac{11}{6} - \frac{1}{4} = \frac{22-3}{12} = \frac{19}{12}$$

- (15) A family has 3 children. Find the probability distribution of these children being boys.

Solⁿ Sample Space for three children is

$S = \{BBB, BBG, BGB, GBB, GBS, GBG, GGB, GGG\}$

$X = \{0, 1, 2, 3\}$ = No. of boys

Expt.	P(x)	x, P(x)
0 (served)	1/3	0
1 (500, 600, 500)	2/3	50
2 (100, 600, 600)	1/3	100
3 (600)	1/3	100

$E(x) = 10/3 = 3\frac{2}{3}$

An importer is offered a shipment of machine tools for £. 1,40,000 and the probabilities that he will be able to sell them for £. 1,80,000, £. 1,70,000 or £. 1,50,000 are 0.32, 0.55 and 0.13. What is the importer's expected gross profit?

Expt.	E(x)	P(x)	x, P(x)
Buy	-1,40,000	1	-1,40,000
Sell	1,80,000	0.32	57,600
"	1,70,000	0.55	93,500
"	1,50,000	0.13	19,500

$E(x) = 30,600$

Paul & Peter enter into a bet according to which A will get £. 200 if it rains on that day, and will lose £. 100 if it doesn't rain. The probability of raining on that day is 0.7. What is the mathematical expectation of A?

Expt.	A(x) - x	P(x)	x, P(x)
Raining	200	0.7	140
Not raining	-100	(1-0.7)=0.3	-30

$E(x) = 110$

A union was negotiating for the probabilities are 0.4, 0.3, 0.2 & 0.1 that the union members will get a £. 150 an hour raise, a £. 100 an hour raise, a £. 50 an hour raise, or a £. 10 an hour raise or no raise at all. What is their expected raise?

Expt.	P(x)	P(x)	x, P(x)
raise	150	0.4	60
"	100	0.3	30
"	50	0.2	10
"	10	0.10	1
no raise	-50	0.2	-10

$E(x) = 100$

60) A probability distribution is given

x	0	1	2	3	4	5
P(x)	0.26	0.25	0.11	0.02	0.25	0.11

① $P(4) = 0.25$

② $P(0 < x < 4) = P(1) + P(2) + P(3) = 0.26 + 0.25 + 0.11 + 0.02 = 0.64$

③ $P(x=4 \cap x=5) = P(x=4) \cdot P(x=5) = 0.25 \times 0.11 = 0.0275$

④ $F(3) = P(x \leq 3) = P(0) + P(1) + P(2) + P(3) = 0.26 + 0.25 + 0.11 + 0.02 = 0.64$

61) In four tosses of a coin, let x be the number of heads. There are 16 possible outcomes with the corresponding value of x. By counting derive the distribution of x and hence calculate expected value of x.

Soln: $x = \{0, 1, 2, 3, 4\}$ Four coins are tossed.
Total no. of cases = $2 \times 2 \times 2 \times 2 = 16$

No. of head	Expt.	P(x)	x, P(x)
0	(TTTT)	1/16	0
1	T T T T, T T T T, T T T T, T T T T	4/16	4/16
2	H T T T, H T T T, H T T T, T T H H, T T H H, T T H H	6/16	12/16
3	H H T T, H H T T, H H T T, T H H H, T H H H, T H H H	4/16	12/16
4	H H H H	1/16	4/16

$E(x) = 2$

Q) A coin is tossed until a head appears. what is the expectation of number of tosses required?

No. of tosses (x)	Experiment	P(x)	x P(x)
1	H	$\frac{1}{2}$	$\frac{1}{2}$
2	TH	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$	$\frac{2}{4}$
3	TTH	$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$	$\frac{3}{8}$
4	TTTH	$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{16}$	$\frac{4}{16}$
∞			

$$E(x) P(x) = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots \infty$$

$$\text{Expectation } E(x) = E(x) P(x) = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots + \infty \quad \text{--- (i)}$$

Multiplying eqⁿ (i) by $\frac{1}{2}$

$$\frac{1}{2} E(x) = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{4}{32} + \dots + \infty \quad \text{--- (ii)}$$

Subtracting eqⁿ (ii) from (i)

$$E(x) = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{32} + \dots + \infty$$

$$\frac{1}{2} E(x) = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{4}{32} + \dots + \infty$$

$$\frac{1}{2} E(x) = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \dots$$

$$\frac{1}{2} E(x) = \text{Geometric series of common ratio } r = \frac{1}{2}$$

$$\text{or } \frac{1}{2} E(x) = \frac{\text{first term}}{1 - \text{common ratio}} \quad [S_{\infty} = \frac{a}{1-r}]$$

$$\approx \frac{1}{2} E(x) = \frac{\frac{1}{2}}{1 - \frac{1}{2}} = \frac{\frac{1}{2}}{\frac{1}{2}}$$

$$E(x) = 2$$

#

Q2) Five urns contain respectively 5 green and 2 white balls, 7 green and 6 white balls, 2 green and 1 white ball, 5 green and 2 white balls, 2 green and 4 white balls. one ball is drawn from each urn. expected number of white balls drawn out.

5G 2W	7G 6W	2G 1W	5G 2W	2G 4W
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for urn I

No. of white (x)	Experiment	P(x)	x P(x)
0	G	$\frac{5}{7}$	0
1	W	$\frac{2}{7}$	$\frac{2}{7}$
		E(x)	$\frac{2}{7}$

$$\text{Urn II } E(x_2) = \frac{4}{13}$$

$$\text{Urn III } E(x_3) = \frac{1}{3}$$

$$\text{Urn IV } E(x_4) = \frac{2}{3}$$

$$\text{Urn V } E(x_5) = \frac{4}{6}$$

$$\text{Expectation of no. of total white ball } E(x) = E(x_1) + E(x_2) + E(x_3) + E(x_4) + E(x_5)$$

$$= \frac{2}{7} + \frac{4}{13} + \frac{1}{3} + \frac{2}{3} + \frac{4}{6}$$

$$= 2.033$$

Q2) Urn A contains 5 cards no. from 1 to 5 and Urn B contains 4 cards no. from 1 to 4. one card is drawn from each of these urns. find the probability function of sum of the number which appears on the cards drawn and mathematical expectation.

for urn A

Urn B

No. of card (x)	P(x)	x P(x)	No. of card (y)	P(y)	y P(y)
1	$\frac{1}{5}$	$\frac{1}{5}$	1	$\frac{1}{4}$	$\frac{1}{4}$
2	$\frac{1}{5}$	$\frac{2}{5}$	2	$\frac{1}{4}$	$\frac{2}{4}$
3	$\frac{1}{5}$	$\frac{3}{5}$	3	$\frac{1}{4}$	$\frac{3}{4}$
4	$\frac{1}{5}$	$\frac{4}{5}$	4	$\frac{1}{4}$	$\frac{4}{4}$
5	$\frac{1}{5}$	$\frac{5}{5}$			

$$E(x_1) = E(x) P(x) = \frac{15}{5}$$

$$\text{Total no. of cards drawn } E(x) = E(x_1) + E(x_2) = \frac{15}{5} + \frac{10}{4} = \frac{17}{2}$$

The expectation of the number on a dice when thrown.

Roll no	x	$P(x)$
1	1	$\frac{1}{6}$
2	2	$\frac{1}{6}$
3	3	$\frac{1}{6}$
4	4	$\frac{1}{6}$
5	5	$\frac{1}{6}$
6	6	$\frac{1}{6}$

$E(X) = \sum x P(x) = 2\frac{1}{2} = 3.5$ Q

if tossed until a tail appears. what is the expectation of the

no. of tosses required?

no. of tosses (x)	prob (P(x))	Expt	$x P(x)$
1	$\frac{1}{2}$	T	$\frac{1}{2}$
2	$\frac{1}{4}$	HT	$\frac{2}{4}$
3	$\frac{1}{8}$	HHT	$\frac{3}{8}$
4	$\frac{1}{16}$	HHHT	$\frac{4}{16}$
...	...	...	...

$E(X) = \sum x P(x) = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots$ Q

Let $E(X) = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots$ --- (i)

Multiplying by $\frac{1}{2}$

$\frac{1}{2} E(X) = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{4}{32} + \dots$ --- (ii)

Subtracting (ii) from (i)

$E(X) = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots$

$\frac{1}{2} E(X) = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{4}{32} + \dots$

$E(X) = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots$

$E(X) =$ Geometrical Series with common ratio $\frac{1}{2}$

We have,

$S = \frac{\text{first no.}}{1 - \text{common ratio}} = \frac{1}{1 - \frac{1}{2}} = \frac{1}{\frac{1}{2}} = 2$

$\therefore \frac{1}{2} E(X) = 1$

$E(X) = 2$ Q

(P2) Suppose X is a r.v. having p.f. P(x) given

x	P(x)	x P(x)	$x^2 P(x)$
-3	0.1	-0.3	0.3
-1	0.2	-0.2	0.2
0	0.15	0	0
1	0.2	0.2	0.2
2	0.10	0.2	0.4
3	0.15	0.45	1.35
5	0.05	0.25	1.25
8	0.05	0.40	3.2

$\sum x P(x) = 1$ $\sum x^2 P(x) = 7.5$

Mean $E(X) = \sum x P(x) = 1$

Variance $V(X) = \sum x^2 P(x) - \{E(X)\}^2$
 $= 7.5 - 1^2 = 6.5$

(P3) Sol. Q. 2

x	P(x)	x P(x)	$x^2 P(x)$
13.5	0.2	2.70	36.45
15.2	0.5	7.60	126.40
19.1	0.3	5.73	109.40

$\sum x P(x) = 16.03$ $\sum x^2 P(x) = 272.25$

Mean = $E(X) = 16.03$
 Variance $V(X) = \sum x^2 P(x) - \{E(X)\}^2$
 $= 272.25 - 16.03^2 = 3.2936$ Q

when, $X: 25x - 8.5$

P(x)	$(25x - 8.5) P(x)$	$(25x - 8.5)^2 P(x)$
0.2	6.80	2168.20
0.5	19.45	7566.50
0.3	14.70	6595.50

$\sum (25x - 8.5) P(x) = 40.95$

$\sum (25x - 8.5)^2 P(x) = 16329.20$

(1) Mean = 40.95, (2) Variance = $E(25x - 8.5)^2 P(x) - \{E(25x - 8.5) P(x)\}^2 = 16329.20 - 40.95^2 = 2496.10$ Q

find the mean and variance of the random variable with probability density function

$$f(x) = xe^{-x}, x \geq 0$$

$$\text{Mean } E(x) = \int_0^{\infty} x^2 e^{-x} dx$$

$$= \left[-x^2 e^{-x} - 2x e^{-x} \right]_0^{\infty}$$

$$= 0 + 0 + 0 + 2e^0$$

$$= 2$$

$$\begin{array}{r} x^2 \rightarrow e^{-x} \\ 2x \rightarrow -e^{-x} \\ 2 \rightarrow -e^{-x} \\ 0 \rightarrow -e^{-x} \end{array}$$

$$\text{Variance } V(x) = \int_0^{\infty} x^2 f(x) dx = \int_0^{\infty} x^2 x e^{-x} dx = \int_0^{\infty} x^3 e^{-x} dx - (E(x))^2$$

$$= \left[-x^3 e^{-x} - 3x^2 e^{-x} - 6x e^{-x} - 6e^{-x} \right]_0^{\infty}$$

$$= 0 + 0 + 0 + 6 - 6$$

$$= 6$$

$$\begin{array}{r} x^3 \rightarrow e^{-x} \\ 3x^2 \rightarrow -e^{-x} \\ 6x \rightarrow -e^{-x} \\ 6 \rightarrow -e^{-x} \end{array}$$

Q. 1. Let x be a continuous r.v. is given as

$f(x) = A x (1-x)^2$ $0 \leq x \leq 1$ elsewhere find $E(x)$ and $V(x)$

$$f(x) = \int_0^1 x f(x) dx$$

we have

$$= \int_0^1 x \cdot A x (1-x)^2 dx$$

$$= 12 \int_0^1 (x^3 - 2x^4 + x^5) dx$$

$$= 12 \left[\frac{x^4}{4} - \frac{2x^5}{5} + \frac{x^6}{6} \right]_0^1$$

$$= 12 \left[\frac{1}{4} - \frac{2}{5} + \frac{1}{6} \right]$$

$$= 12 \times \frac{10-8+4}{60}$$

$$= 12 \times \frac{6}{60} = \frac{2}{5}$$

$$\int_0^1 f(x) dx = 1$$

$$\int_0^1 A x (1-x)^2 dx = 1$$

$$A \int_0^1 (x^3 - 2x^4 + x^5) dx = 1$$

$$A \left[\frac{x^4}{4} - \frac{2x^5}{5} + \frac{x^6}{6} \right]_0^1 = 1$$

$$A \left[\frac{1}{4} - \frac{2}{5} + \frac{1}{6} \right] = 1$$

$$A \left[\frac{10-8+4}{60} \right] = 1$$

$$A \left[\frac{6}{60} \right] = 1$$

$$A = 10$$

$$V(x) = \int_0^1 x^2 f(x) dx - (E(x))^2$$

$$= 12 \int_0^1 x^3 (1-x)^2 dx - \left(\frac{2}{5} \right)^2$$

$$= 12 \int_0^1 (x^3 - 2x^4 + x^5) dx - \frac{4}{25}$$

$$= 12 \left[\frac{x^4}{4} - \frac{2x^5}{5} + \frac{x^6}{6} \right]_0^1 - \frac{4}{25}$$

$$= 12 \left[\frac{1}{4} - \frac{2}{5} + \frac{1}{6} \right] - \frac{4}{25} = \frac{2}{5} - \frac{4}{25} = \frac{6}{25}$$

(B) Same as QN. 21 (same function)

Q. 2. A random variable x has the density $f(x) = a + bx^2$ $0 \leq x \leq 1$ and b is such that mean is $1/3$. Also find the variance of the distribution.

Let total probability = 1

$$\text{Mean } E(x) = \int_0^1 x f(x) dx$$

$$\int_0^1 f(x) dx = 1$$

$$\frac{2}{3} = \int_0^1 x (a + bx^2) dx$$

$$\int_0^1 (a + bx^2) dx = 1$$

$$\frac{2}{3} = \int_0^1 (ax + bx^3) dx$$

$$a \left[x + \frac{bx^3}{3} \right]_0^1 = 1$$

$$\frac{2}{3} = \left[\frac{ax^2}{2} + \frac{bx^4}{4} \right]_0^1$$

$$a + \frac{b}{3} = 1$$

$$\frac{2}{3} = \frac{a}{2} + \frac{b}{4}$$

$$3a + b = 3 \quad \text{--- (1)}$$

$$2a + 3b = 3 \quad \text{--- (2)}$$

Solving eq. (1) & (2)

Now, variance $V(x) = \int_0^1 x^2 f(x) dx - (E(x))^2$

$$3a + b = 3$$

$$= \int_0^1 x^2 (a + bx^2) dx - \left(\frac{2}{3} \right)^2$$

$$3a + b = 3$$

$$= \int_0^1 (ax^2 + bx^4) dx - \frac{4}{9}$$

$$b = 2$$

$$= \left[\frac{ax^3}{3} + \frac{bx^5}{5} \right]_0^1 - \frac{4}{9}$$

$$a = \frac{1}{3}$$

$$= \frac{a}{3} + \frac{b}{5} - \frac{4}{9}$$

$$= \frac{1}{9} + \frac{2}{5} - \frac{4}{9} = \frac{5+12-8}{45} = \frac{9}{45} = \frac{1}{5}$$

$$= \frac{1}{9} + \frac{2}{5} - \frac{4}{9} = \frac{5+12-8}{45} = \frac{9}{45} = \frac{1}{5}$$

Suppose that the number of plants of a particular type which are found in a rectangular region is a geometric area in G.V.

$$P(x) = \frac{a}{x^2} \quad a = 1.2, 5$$

$$= 0 \quad \text{otherwise}$$

Find mean, variance and standard deviation

x	$P(x)$	$x \cdot P(x)$	$x^2 \cdot P(x)$
1	$\frac{1}{1^2}$	$1 \cdot \frac{1}{1^2}$	$1^2 \cdot \frac{1}{1^2}$
2	$\frac{1}{2^2}$	$2 \cdot \frac{1}{2^2}$	$2^2 \cdot \frac{1}{2^2}$
3	$\frac{1}{3^2}$	$3 \cdot \frac{1}{3^2}$	$3^2 \cdot \frac{1}{3^2}$

$$E(x) = \sum x \cdot P(x) = \frac{1}{1} + \frac{2}{4} + \frac{3}{9} = \frac{11}{6}$$

$$E(x^2) = \sum x^2 \cdot P(x) = \frac{1}{1} + \frac{4}{4} + \frac{9}{9} = \frac{11}{3}$$

Total probability = 1

$$E(x) = 1$$

$$1 + \frac{2}{4} + \frac{3}{9} = 1$$

$$\frac{11}{6} + \frac{2}{4} + \frac{3}{9} = 1$$

$$\frac{11}{6} + \frac{2}{4} + \frac{3}{9} = 1$$

$$a = \frac{11}{6}$$

$$\text{Variance (Var)} = E(x^2 \cdot P(x)) - (E(x))^2$$

$$= \frac{11}{3} - \left(\frac{11}{6}\right)^2$$

$$= 1.5778 - 1.3719$$

$$= 0.2057$$

$$\sigma^2 = 0.2057$$

$$\sigma = \sqrt{0.2057} = 0.4535$$

A bag contains 5 white and 3 red balls. Three balls are taken at random and the expected no. of white balls drawn in the three.

x	$P(x)$	$x \cdot P(x)$
1	$\frac{5C_1 \times 3C_2}{8C_3} = 0.2678$	0.2678
2	$\frac{5C_2 \times 3C_1}{8C_3} = 0.5357$	1.0714
3	$\frac{5C_3 \times 3C_0}{8C_3} = 0.1965$	0.5895

$$E(x) = 1.9288$$

(34) Let x be a continuous random variable with pdf.

$$P(x) = ax \quad 0 \leq x \leq 2$$

$$= 0 \quad 2 \leq x \leq 4$$

$$= -x+3a \quad 4 \leq x \leq 6$$

otherwise elsewhere.

(i) Determine constant a .

We have,

$$\text{Total probability} = 1$$

$$\int_0^6 f(x) dx = 1$$

$$\text{or, } \int_0^2 ax dx + \int_2^4 0 dx + \int_4^6 (-x+3a) dx = 1$$

$$\text{or, } \left[\frac{ax^2}{2} \right]_0^2 + \left[-\frac{x^2}{2} + 3ax \right]_4^6 = 1$$

$$\text{or, } 2a + 9a - 18a + 18a + 8a - 12a = 1$$

THEORETICAL DISCRETE DISTRIBUTION

- ① Binomial distribution
- ② Poisson distribution
- ③ Hypergeometric distribution
- ④ Negative binomial distribution
- ⑤ Geometric distribution

→ Binomial distribution

A discrete random variable x is said to have binomial distribution if its pmf is defined as binomial distribution if its pmf is defined as

$$p(x=x) = \begin{cases} C(n, x) p^x q^{n-x} & x = 0, 1, 2, \dots, n \\ 0 & \text{otherwise} \end{cases}$$

where,

n = no. of trial

p = prob. of given event in a single trial

$q = 1 - p$

x = no. of event (we have to find)

e.g. A coin is tossed 3 times what is the probability of obtaining 2 heads?

no. of trial (n) = 3

prob. of head in a single trial (p) = $\frac{1}{2} = 0.50$

$q = 1 - p = 1 - 0.5 = 0.5$

$P(\text{obtaining 2 heads}) = P(x=2)$

we have,

$$\begin{aligned} p(x=2) &= C(n, x) p^x q^{n-x} \\ &= C(3, 2) p^2 q^{3-2} \\ &= C(3, 2) 0.5^2 \times 0.5^1 \\ &= 0.375 \end{aligned}$$

The binomial distribution has following features:-

- (i) no. of trials are finite and fixed
- (ii) Each trial gives only one of two possible outcomes say success or failure (head or tail) etc.
- (iii) All trials are independent
- (iv) $p = 1 - q$
- (v) Mean (μ) = np & Variance = npq

Poisson distribution

A discrete random variable x is said to have poisson distribution if its pmf is defined as

$$p(x=x) = \begin{cases} \frac{e^{-\mu} \mu^x}{x!}, & x = 0, 1, 2, 3, \dots \\ 0, & \text{otherwise} \end{cases}$$

where, μ = mean = np

n = no. of trial

p = probability of success in a single trial

x = required no. of success

Mean & Variance

Mean (μ) = np = average

Variance (σ^2) = np

Following are instances where following distribution may be applied

- 1) No. of bus accidents in a unit time at a particular place
- 2) No. of telephone calls arriving in unit time
- 3) No. of printing mistakes per page to a book
- 4) No. of machines breakdown during one particular day
- 5) prob. of success (p) $\leq 1\%$ = 0.01 etc.

Features of poisson distribution

- 1) No. of trials is indefinitely large ($n \rightarrow \infty$)
- 2) probability of success in a single trial is very small i.e. $p \rightarrow 0$
- 3) Mean (μ) = np

Hypergeometric distribution

A discrete random variable x is said to have hypergeometric distribution if its pmf is defined by

$$p(x=x) = \frac{N_1 C_x \cdot N_2 C_{n-x}}{N C_n}$$

where, N_1 = No. of first type
No. of 2nd "

n = No. of drawn (randomly)

$N = N_1 + N_2$

Required no. of first type drawn = x

" " 2nd " " = $n-x$

Mean & Variance

Mean of Hypergeometric distribution = $\frac{n N_1}{N}$

Variance of hypergeometric distribution = $\frac{n N_1 (N-N_1) (N-n)}{N^2 (N-1)}$

→ (4) Negative Binomial distribution

A discrete random variable x is said to have negative Binomial distribution if its pmf is defined as

$$p(x=x) = \begin{cases} \binom{x+r-1}{r-1} p^r q^x & x = 0, 1, 2, \dots, \infty \\ 0, & \text{otherwise} \end{cases}$$

where

Total no. of trial (n) = $x+r$

No. of success = r (last must be success)

No. of failure = x

p = prob. of success in a single trial, $q = 1-p$

→ (5) Geometric distribution

A discrete random variable x is said to have geometric distribution if its pmf is defined as

$$p(x=x) = \begin{cases} p q^x & x = 0, 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$$

Mean = $\frac{r q}{p}$

Variance = $\frac{r q}{p^2}$

Exercise

- ③ Four coins are tossed simultaneously. What is the probability of getting (i) 2 heads and 2 tails (ii) at least 2 heads and (iii) at least one head.

Sol. no. of trials (n) = 4

Prob. of getting head or tail in single trial (p) = $\frac{1}{2} = 0.5$
 $q = 1 - p = 1 - 0.5 = 0.5$

$P(\text{obtained 2 heads and 2 tails}) = P(X=2)$

We have,
 $P(X=x) = {}^nC_x p^x q^{n-x}$

$$P(X=2) = {}^nC_2 (0.5)^2 (0.5)^{4-2} = 0.375$$

(ii) at least 2 heads

$$P(X \geq 2) = P(X=2) + P(X=3) + P(X=4)$$

$$= {}^4C_2 \times 0.5^2 \times 0.5^{4-2} + {}^4C_3 \times 0.5^3 \times 0.5^{4-3} + {}^4C_4 \times 0.5^4 \times 0.5^{4-4}$$

$$= 0.375 + 0.375 + 0.0625 = 0.8125$$

(iii) at least one head

$$P(X \geq 1) = 1 - P(X=0)$$

$$= 1 - [P(X=0) + P(X=1)]$$

$$= 1 - \left[{}^4C_0 \times 0.5^0 \times 0.5^{4-0} + {}^4C_1 \times 0.5^1 \times 0.5^{4-1} \right]$$

$$= 1 - (0.0625 + 0.25) = 0.9375$$

- ④ 3 dice are thrown 729 times. How many times do we expect at least three dice to show a five or six?

Sol. no. of trials (n) = 6

Prob. of getting 5 or 6 (p) = $\frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

$$q = 1 - p = 1 - \frac{1}{3} = \frac{2}{3}$$

$$P(\text{at least three}) = P(X \geq 3) = 1 - P(X < 3)$$

$$= 1 - \{P(X=0) + P(X=1) + P(X=2)\}$$

$$= 1 - \left\{ {}^6C_0 \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^6 + {}^6C_1 \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^5 + {}^6C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^4 \right\}$$

$$= 1 - 0.6328$$

$$= 0.3671$$

$$\text{the no. of times required} = 0.3671 \times 729 = 267 \text{ times}$$

- ⑤ 12% of the items produced by a machine are defective. What is the probability that out of a random sample of 20 items produced by the machine, 5 are defective?

Sol. no. of trials (n) = 20

Prob. of item produced by machine (p) = $12\% = 0.12$

$$q = 1 - p = 1 - 0.12 = 0.88$$

$$P(\text{obtaining 5 defective}) = {}^nC_x p^x q^{n-x} = {}^{20}C_5 (0.12)^5 (0.88)^{20-5} = 0.0547$$

We have,

$$P(X=x) = {}^nC_x p^x q^{n-x} = {}^{20}C_5 (0.12)^5 (0.88)^{20-5} = 0.0547$$

- ⑥ Suppose the mortality rate for a certain disease is 0.10 and at least 3 will die.

Sol. no. of people (n) = 10

mortality rate (p) = 0.1

$$q = 1 - p = 1 - 0.10 = 0.9$$

$$P(\text{at least 3 will die}) = P(X \geq 3) = 1 - P(X < 3)$$

$$= 1 - \{P(X=0) + P(X=1) + P(X=2)\}$$

$$= 1 - \left\{ {}^{10}C_0 (0.1)^0 (0.9)^{10-0} + {}^{10}C_1 (0.1)^1 (0.9)^{10-1} + {}^{10}C_2 (0.1)^2 (0.9)^{10-2} \right\}$$

$$= 1 - (0.3487 + 0.3771 + 0.1937)$$

$$= 0.0705$$

- ⑦ An insurance salesman sells policies to five men, all of whom are in good health. According to the actuarial table, the probability that a man of the particular age will be alive 30 years hence is $\frac{2}{3}$. What is the probability that in 30 years

(i) all five men

$$(ii) P(\text{at least one man}) = P(X \geq 1) = 1 - P(X=0)$$

no. of trials (n) = 5

probability of alive (p) = $\frac{2}{3}$

$$q = 1 - \frac{2}{3} = \frac{1}{3}$$

$$P(\text{all five men alive}) = P(X=5) = {}^nC_x p^x q^{n-x} = {}^5C_5 \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right)^{5-5} = \frac{32}{243}$$

$$P(X=5) = {}^nC_x p^x q^{n-x} = {}^5C_5 \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right)^{5-5} = \frac{32}{243}$$

$$= 0.1317$$

(iii) at most three men = $P(X \leq 3)$

$$= P(X=0) + P(X=1) + P(X=2) + P(X=3)$$

$$= {}^5C_0 \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right)^5 + {}^5C_1 \left(\frac{2}{3}\right)^1 \left(\frac{1}{3}\right)^4 + {}^5C_2 \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^3 + {}^5C_3 \left(\frac{2}{3}\right)^3 \left(\frac{1}{3}\right)^2$$

$$= 0.0069 + 0.0415 + 0.1317 + 0.0540$$

$$= 0.2331$$

- ⑧ For a binomial distribution consisting of 5 independent trials, the probability of success is 0.4096 and 0.5904 respectively. Find (i) Prob. of getting 2 or 3 success.

So no. of drops $(n) = 5$

$$P(x=1) = 0.4096 \text{ and } P(x=2) = 0.2048$$

For $P(x=1)$

$$P(x=1) = C(5,1) p^1 q^{5-1} \quad (\text{here})$$

$$\text{or, } 0.4096 = 5 p q^4 \quad \text{--- (i)}$$

For $P(x=2)$

$$P(x=2) = C(5,2) p^2 q^{5-2}$$

$$\text{or, } 0.2048 = 10 p^2 q^3 \quad \text{--- (ii)}$$

Dividing eqn (i) by (ii)

$$\frac{0.4096}{0.2048} = \frac{5 p q^4}{10 p^2 q^3}$$

$$\frac{2}{1} = \frac{1}{2} \cdot \frac{q}{p}$$

$$\frac{q}{p} = 4$$

$$q = 4p$$

$$p + q = 1$$

$$p + 4p = 1$$

$$5p = 1$$

$$p = \frac{1}{5}$$

$$q = \frac{4}{5}$$

(13) Between the hours of 2 and 4 pm, the average number of phone calls per minute coming into the switch-board of a company is 2.5. Find the probability that during one particular minute there will be (i) no call at all (ii) exactly 3 calls (iii) at least 5 calls

So, average no. $(\lambda) = 2.5$

(i) we have,

$$P(x=0) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-2.5} 2.5^0}{0!} = 0.0821 \quad R$$

$$(ii) P(x=3) = \frac{e^{-2.5} 2.5^3}{3!} = 0.2158$$

$$(iii) P(x \geq 5) = 1 - P(x < 5) = 1 - \{P(x=0) + P(x=1) + P(x=2) + P(x=3) + P(x=4)\}$$

$$= 1 - \left\{ \frac{e^{-2.5} 2.5^0}{0!} + \frac{e^{-2.5} 2.5^1}{1!} + \frac{e^{-2.5} 2.5^2}{2!} + \frac{e^{-2.5} 2.5^3}{3!} + \frac{e^{-2.5} 2.5^4}{4!} \right\}$$

$$= 1 - (0.0821 + 0.2053 + 0.2565 + 0.133 + 0.1376)$$

$$= 1 - 0.8212$$

$$= 0.1788 \quad B$$

(14) It is known that average no. of that suicide per week in Kashmir. Expect the number of suicide in a month. Find the prob. that there are more suicide in a month (4 week = 1 month)

So, average no. of suicide $\lambda = 1.5/\text{week} = 6/\text{month}$

$$P(5 \text{ or more suicide}) = P(x \geq 5 \text{ or more}) = P(x=5) + P(x > 5)$$

$$= P(x=5) + \{1 - P(x \leq 5)\}$$

$$= \frac{e^{-6} 6^5}{5!} + \left\{ 1 - \left(\frac{e^{-6} 6^0}{0!} + \frac{e^{-6} 6^1}{1!} + \frac{e^{-6} 6^2}{2!} + \frac{e^{-6} 6^3}{3!} + \frac{e^{-6} 6^4}{4!} \right) \right\}$$

$$= 0.1606 + \{1 - (1 + 6 + 18 + 36 + 54)\}$$

$$= 0.1606 + 0.71494$$

$$= 0.8755 \quad B$$

(15) A typist made 2.5 mistakes on an average per page. Find the probability that in a page typed by him (i) there is no mistake (ii) there are two or less mistakes (iii) more than 10 mistakes.

So, average mistakes $(\lambda) = 2.5$

(i) $P(x=0)$ (ii) $P(x=2 \text{ or less}) = P(x=2) + P(x=1) + P(x=0)$

$$P(x=0) = \frac{e^{-2.5} 2.5^0}{0!} = 0.0821 \quad R$$

$$= \frac{e^{-2.5} 2.5^2}{2!} + \frac{e^{-2.5} 2.5^1}{1!} + \frac{e^{-2.5} 2.5^0}{0!}$$

$$= 0.2398 + 0.2988 + 0.1762$$

$$= 0.7148 \quad R$$

(iii) $P(x > 10)$

$$= 1 - P(x \leq 10)$$

$$= 1 - \{P(x=0) + P(x=1) + P(x=2) + P(x=3) + P(x=4) + P(x=5) + P(x=6) + P(x=7) + P(x=8) + P(x=9) + P(x=10)\}$$

$$= 1 - \left\{ \frac{e^{-2.5} 2.5^0}{0!} + \frac{e^{-2.5} 2.5^1}{1!} + \frac{e^{-2.5} 2.5^2}{2!} + \frac{e^{-2.5} 2.5^3}{3!} + \frac{e^{-2.5} 2.5^4}{4!} + \frac{e^{-2.5} 2.5^5}{5!} + \frac{e^{-2.5} 2.5^6}{6!} + \frac{e^{-2.5} 2.5^7}{7!} + \frac{e^{-2.5} 2.5^8}{8!} + \frac{e^{-2.5} 2.5^9}{9!} + \frac{e^{-2.5} 2.5^{10}}{10!} \right\}$$

$$= 1 - 0.999159$$

$$= 0.000841 \quad B$$

18) 2% of electric bulbs manufactured by a certain company are defective, find the prob. that in a sample of 200 bulbs (i) less than 3 bulbs are defective.

sol: prob. of having each bulb defective $(p) = 2\% = 0.02$
 Sample $(n) = 200$
 Mean $(\mu) = np = 200 \times 0.02 = 4$

(i) less than two bulbs defective

$$P(X < 2) = P(X=0) + P(X=1)$$

$$= \frac{e^{-4} \times 4^0}{0!} + \frac{e^{-4} \times 4^1}{1!} = 0.0183 + 0.07326 = 0.0915$$

(ii) more than 2 bulbs defective

$$P(X > 3) = 1 - P(X \leq 3) = 1 - \left\{ \frac{e^{-4} \times 4^0}{0!} + \frac{e^{-4} \times 4^1}{1!} + \frac{e^{-4} \times 4^2}{2!} + \frac{e^{-4} \times 4^3}{3!} \right\}$$

$$= 1 - e^{-4} \{ 1 + 4 + 8 + 10.667 \}$$

$$= 1 - 0.4324$$

$$= 0.5665$$

19) It is known that 5% of books found at a certain library have defective binding. find the probability that 2 of 100 books found by this library will have defective binding using

(i) the formula for the binomial distribution

no. of books $(n) = 100$
 defective probability $(p) = 5\% = 0.05$
 $q = 1 - p = 0.95$

$$P(X=2) = {}^nC_r p^r q^{n-r}$$

$$= {}^{100}C_2 (0.05)^2 (0.95)^{100-2}$$

$$= 0.081$$

(ii) mean $(\mu) = np = 100 \times 0.05 = 5$

$$P(X=2) = \frac{e^{-\mu} \mu^x}{x!} = \frac{e^{-5} 5^2}{2!} = 0.08422$$

(iii) error in percentage $= \left(\frac{0.08422 - 0.081}{0.081} \right) = 0.0092 = 0.322\%$

20) The probability that a person in a family has a fever is 0.01. Searching the thieves by taking them at random what is the prob. that 10 persons are to be examined in order to get 3 thieves.

sol: the probability of being thief $(p) = 0.01$
 no. of person $(n) = 10$
 Mean $(\mu) = np = 0.01 \times 10 = 0.10$

$$P(\text{getting 3 thieves}) = P(X=3) = \frac{e^{-\mu} \mu^x}{x!} = \frac{e^{-0.1} 0.1^3}{3!} = 0.0005$$

21) A well-known baseball player has a lifetime batting average of .30. what is the probability that 40 or fewer times at bat one requires him to achieve 36 hits?

sol: p (average of each game) $= 0.30$
 $P = 0.3$
 $q = 1 - p = 0.7$
 $n = 40$ or lower
 $= 40$ or $50, 30, 33, 36$
 $P(\text{achieving 36 hits}) = P(X=36) = ?$

we have,

$$P(X=x) = \frac{n!}{x!} p^x q^{n-x}$$

$$= 0.7^{40} \left[\frac{40!}{36!} 0.3^{36} + \frac{40!}{37!} 0.3^{37} + \frac{40!}{38!} 0.3^{38} + \frac{40!}{39!} 0.3^{39} \right]$$

$$= 0.7^{40} [740.259 + 246.753 + 63.27 + 11.1 + 1]$$

$$= 0.7^{40} \times 1041.382$$

$$= 0.00281$$

22) An urn contains 25 items in which 15 of them are defective. find the probability of getting at least 5 non-defective items. 12 items are drawn randomly.

sol: Non-defective item $(N_1) = (25 - 15) = 10$
 defective item $(N_2) = 15$
 no. of items drawn $(n) = 12$

$$N = N_1 + N_2 = 10 + 15 = 25$$

$$P(\text{at least 5 non-defective}) = P(X \geq 5) = 1 - P(X \leq 4)$$

$$+ P(n=3) + P(n=4) \}$$

$$= \frac{{}^{10}C_2 \left(\frac{15}{12-2} \right) + {}^{10}C_3 \left(\frac{15}{12-3} \right) + {}^{10}C_4 \left(\frac{15}{12-4} \right)}{{}^{25}C_{12}}$$

$$= \frac{11550 + 135135 + 600600 + 1251350}{5900300}$$

Q. 10. Let containing 17 items, 7 of which are defective; 6 are chosen at random find the prob. of (i) 4 or 5 defective (ii) at least 4 defective (iii) at most 8 defective (iv) 4 defective and 2 non-defective item.

No. of defective item (N_1) = 7
 " non-defective " (N_2) = 12 - 7 = 10
 $N = N_1 + N_2 = 17$
 Sample drawn item (n) = 6

Ans defective
 $P(x=4 \text{ or } 5) = P(x=4) + P(x=5)$

$$= \frac{{}^{N_1}C_x \cdot {}^{N_2}C_{n-x}}{{}^N C_n} + \frac{{}^{N_1}C_{n-x} \cdot {}^{N_2}C_x}{{}^N C_n} = \frac{{}^7 C_4 \left(\frac{10}{17-4} \right) + {}^7 C_5 \left(\frac{10}{17-5} \right)}{{}^{17} C_6}$$

$$= \frac{1575 + 210}{12376}$$

$$= 0.1442 \text{ Q}$$

(At least 4 defective)
 $P(x \geq 4) = 1 - P(x < 4) = 1 - \{ P(x=0) + P(x=1) + P(x=2) + P(x=3) \}$

$$= 1 - \left\{ \frac{{}^7 C_0 \left(\frac{10}{17} \right) + {}^7 C_1 \left(\frac{10}{16} \right) + {}^7 C_2 \left(\frac{10}{15} \right) + {}^7 C_3 \left(\frac{10}{14} \right)}{{}^{17} C_6} \right\}$$

$$= 1 - \frac{210 + 1764 + 4410 + 4200}{12376} = 0.1448 \text{ Q}$$

(At most 3 defective)
 $P(x \leq 3) = P(x=0) + P(x=1) + P(x=2)$

$$= \frac{{}^7 C_0 \left(\frac{10}{17} \right) + {}^7 C_1 \left(\frac{10}{16} \right) + {}^7 C_2 \left(\frac{10}{15} \right)}{{}^{17} C_6} = \frac{210 + 1764 + 4410}{12376} = 0.5558 \text{ Q}$$

(iv) 4 defective and 2 non-defective

$P(x=4) \text{ and } P(x=2)$

$$= \frac{{}^7 C_4 \left(\frac{10}{17} \right)}{{}^{17} C_6} \text{ and } \frac{{}^7 C_2 \left(\frac{10}{17} \right)}{{}^{17} C_6}$$

$$= 0.1273 \text{ and } 0.1273 \text{ Q}$$

(11) In a ward, there are 121 houses in which 65 of them are new. the probability that 7 houses are old out of 25 houses selected randomly

Soln. No. of old house (N_1) = 121 - 65 = 56

" new house (N_2) = 65

$N = N_1 + N_2 = 121$

And drawn item (n) = 25

$P(\text{2 houses are old}) = P(x=7)$

we have,

$$P(x=7) = \frac{{}^{N_1}C_x \cdot {}^{N_2}C_{n-x}}{{}^N C_n} = \frac{{}^{56}C_7 \left(\frac{65}{121-7} \right)}{{}^{121}C_{25}} = 0.0219 \text{ Q}$$

(12) An agriculture cooperative claims that 90% of the watermelons are ripe and ready to eat. And the probability that among 18 watermelons

Soln. $P(\text{Shipped out watermelon}) = 90\%$

$$P = 90\% = 0.90$$

$$Q = 1 - P = 1 - 0.9 = 0.1$$

drawn item (n) = 18

(13) all 18 are ripe and ready to eat

$$P(x=18) = \frac{{}^{18}C_{18} \cdot 0.9^{18} \cdot 0.1^{18-18}}{{}^{18}C_{18}} = 0.1501 \text{ Q}$$

(14) at least 16 are ripe and ready to eat.

$$P(x \geq 16) = 1 - P(x < 16) = 1 - \{ P(x=0) + P(x=1) + P(x=2) + P(x=3) + P(x=4) + \dots + P(x=15) \}$$

$$= 1 - \left\{ \frac{{}^{18}C_0 \cdot 0.9^0 \cdot 0.1^{18} + {}^{18}C_1 \cdot 0.9^1 \cdot 0.1^{17} + {}^{18}C_2 \cdot 0.9^2 \cdot 0.1^{16} + {}^{18}C_3 \cdot 0.9^3 \cdot 0.1^{15} + {}^{18}C_4 \cdot 0.9^4 \cdot 0.1^{14} + {}^{18}C_5 \cdot 0.9^5 \cdot 0.1^{13} + {}^{18}C_6 \cdot 0.9^6 \cdot 0.1^{12} + {}^{18}C_7 \cdot 0.9^7 \cdot 0.1^{11} + {}^{18}C_8 \cdot 0.9^8 \cdot 0.1^{10} + {}^{18}C_9 \cdot 0.9^9 \cdot 0.1^9 + {}^{18}C_{10} \cdot 0.9^{10} \cdot 0.1^8 + {}^{18}C_{11} \cdot 0.9^{11} \cdot 0.1^7 + {}^{18}C_{12} \cdot 0.9^{12} \cdot 0.1^6 + {}^{18}C_{13} \cdot 0.9^{13} \cdot 0.1^5 + {}^{18}C_{14} \cdot 0.9^{14} \cdot 0.1^4 + {}^{18}C_{15} \cdot 0.9^{15} \cdot 0.1^3 + {}^{18}C_{16} \cdot 0.9^{16} \cdot 0.1^2 + {}^{18}C_{17} \cdot 0.9^{17} \cdot 0.1^1 + {}^{18}C_{18} \cdot 0.9^{18} \cdot 0.1^0 \right\}$$

$$= 1 - (1.4 \times 10^{-15} + 1.62 \times 10^{-14} + 1.2792 \times 10^{-13} + 5.245 \times 10^{-12} + 2.00 \times 10^{-11} + 5.057 \times 10^{-10} + 9.565 \times 10^{-9} + 0.0710 + 0.118)$$

$$= 0.7338 \text{ Q}$$

244
 (c) at most 14 are in p. and ready to eat

$$\begin{aligned}
 P(X \leq 14) &= P(X=0) + P(X=1) + P(X=2) + \dots + P(X=14) \\
 &= \binom{18}{0} 0.9^0 0.1^{18} + \binom{18}{1} 0.9^1 0.1^{17} + \binom{18}{2} 0.9^2 0.1^{16} + \dots + \binom{18}{14} 0.9^{14} 0.1^4 \\
 &= 1 \times 10^{-18} + 1.62 \times 10^{-16} + 1.339 \times 10^{-14} + 5.95 \times 10^{-13} + 2 \times 10^{-11} + 5.05 \times 10^{-10} + 9.05 \times 10^{-9} + \\
 &\quad 1.5 \times 10^{-7} + 1.55 \times 10^{-6} + 1.55 \times 10^{-5} + 1.525 \times 10^{-4} + 0.002998 + 0.00324 + 0.02738 + 0.2 \\
 &= 0.982
 \end{aligned}$$

(26) Among the 300 employees of the company, 240 are union members while the other are not. If 8 of the employees are chosen by lot to serve on the committee which administers the pension fund, find the probability that five of them will be union members while the others are not, by using

(a) the formula for the hypergeometric distribution.

No. of union members (N_1) = 240
 " not " (N_2) = 300 - 240 = 60
 $N = N_1 + N_2 = 300$

Chosen employees (n) = 8

$P(\text{five are member \& three are not}) = P(X=5)$

We have,

$$P(X=x) = \frac{\binom{N_1}{x} \binom{N_2}{n-x}}{\binom{N}{n}} = \frac{\binom{240}{5} \binom{60}{3}}{\binom{300}{8}} = 0.1490$$

(b) the formula for the binomial distribution as an approximation.

Mean (μ) = $n \times \frac{N_1}{N} = 8 \times \frac{240}{300} = 6.4$ (from hypergeometric distribution)

here,

mean = $n p$

$p = \frac{\text{mean}}{n} = \frac{6.4}{8} = 0.8$

$q = 1 - p = 0.2$

$P(X=5) = \binom{8}{5} 0.8^5 0.2^{3-5} = 0.1468$

(27) If 6 of 18 new building in a city violate the building code, what is the probability that a building inspector who randomly selects 4 new building, for inspection will catch

(a) how many of the new building that violate the building code.

So, No. of new building who violate building code (N_1) = 6

" " " who not violate " (N_2) = 12

$N = 6 + 12 = 18$

Randomly chosen building (n) = 4

$$P(X=0) = \frac{\binom{N_1}{0} \binom{N_2}{n}}{\binom{N}{n}} = \frac{\binom{6}{0} \binom{12}{4}}{\binom{18}{4}} = 0.1677$$

(b) one of the new building that violate the building code

$$P(X=1) = \frac{\binom{6}{1} \binom{12}{3}}{\binom{18}{4}} = 0.4313$$

(c) Two of the new building that violate the building code

$$P(X=2) = \frac{\binom{6}{2} \binom{12}{2}}{\binom{18}{4}} = 0.3231$$

(d) at least 3 building that violate the building code

$P(X \geq 3) = 1 - P(X \leq 2)$

Find the probability of getting 2 Success in 3 trials of a fair coin.

probability of single trial & coin $(P) = \frac{1}{2}$, $q = 1 - p = \frac{1}{2}$

No. of Success $(r) = 2$

Total no. of trial $(r+x) = 3$

No. of failure $(x) = 3 - 2 = 1$

probability of Success = 1 - probability of failure

$$= 1 - P(x=1)$$

$$= 1 - \left(\frac{r+x-1}{r-1} \right) p^r q^{x-r+1}$$

$$= 1 - \left(\frac{3+1-1}{3-1} \right) 0.5^2 \cdot 0.5^1$$

$$= 1 - \left(\frac{3}{2} \right) 0.25 \times 0.5$$

$$= 1 - \frac{3}{4}$$

$$= \frac{1}{4}$$

If on an average 5 ships out of 10 arrive safely at a port, find the mean and variance of the no. of ship arriving safely at the port in 100 ship.

Average $(\mu) = 5$

$n = 10$

or $n \cdot p$

$$\mu = n \cdot p$$

$$p = \frac{\mu}{n} = \frac{5}{10} = 0.50, q = 1 - p = 1 - 0.5 = 0.5$$

Now, Total no. of ship $(n) = 100$

$$\text{mean}(\mu) = n \cdot p = 100 \times 0.5 = 50$$

$$\text{variance}(\sigma^2) = n \cdot p \cdot q = 100 \times 0.5 \times 0.5 = 25$$

for a binomial distribution, $\mu = 50$ and $\text{variance} = 25$. Correct

or the result

$$\text{mean} = n \cdot p \quad \text{or} \quad \mu = n \cdot p \quad \text{--- (i)}$$

$$\text{variance} = n \cdot p \cdot q \quad \text{or} \quad \sigma^2 = n \cdot p \cdot q \quad \text{--- (ii)}$$

$$p + q = 1$$

$$p = 1 - q = 1 - 0.5 = 0.5 \quad \text{--- correct}$$

Dividing (i) by (ii)

$$\frac{\mu}{\sigma^2} = \frac{n \cdot p}{n \cdot p \cdot q}$$

$$q = \frac{1}{2}$$

(3) If the probability of a defective bolt is $\frac{1}{10}$ find mean, variance & $P(X > 0)$ if number of items = 400.

$P(X > 0)$ if number of items = 400.

So, probability of defective bolt $(p) = \frac{1}{10} = 0.1$, $q = 1 - p = 0.9$

no. of items $(n) = 400$

$$\text{Mean} = n \cdot p = 400 \times 0.1 = 40$$

$$\text{Variance}(\sigma^2) = n \cdot p \cdot q = 400 \times 0.1 \times 0.9 = 36$$

$$P(X > 0) = 1 - P(X = 0)$$

$$= 1 - \left(\frac{400}{0} \right) 0.1^0 \cdot 0.9^{400-0}$$

$$= 1 - 0.9^{400}$$

(34) Over a 10 minute period, counter records an average number

1.3 poisson particles per millisecond. Find a radioactive substance

Determine (a) the probability of one or more particles in

per millisecond and (b) variance

So, Total no. of time $(n) = 10 \text{ minutes} = 600 \text{ milliseconds}$

$$\text{average no. of particles}(\mu) = 1.3$$

we have,

$$n = n \cdot p$$

$$p = \frac{\mu}{n} = \frac{1.3}{600} = 0.00216$$

$$q = 1 - 0.00216 = 0.99784$$

$$(ii) P(X \geq 1) = 1 - P(X = 0)$$

$$= 1 - \left(\frac{600}{0} \right) 0.00216^0 \cdot 0.99784^{600-0}$$

$$= 1 - 0.99784^{600}$$

$$(iii) \text{Variance}(\sigma^2) = n \cdot p \cdot q = 600 \times 0.00216 \times 0.99784 = 1.30$$

(35) At a checkout counter, customers arrive at an average number

2.5 per minute find the probability that at most four will arrive in

given minute.

So, average number $(\mu) = 2.5$

$$P(\text{at most four}) = P(X \leq 4) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4)$$

$$= e^{-2.5} \left(\frac{2.5^0}{0!} + \frac{2.5^1}{1!} + \frac{2.5^2}{2!} + \frac{2.5^3}{3!} + \frac{2.5^4}{4!} \right)$$

$$= e^{-2.5} (1 + 2.5 + 3.125 + 2.604 + 1.6276)$$

$$= 0.991$$

36. Six coin are tossed 64 times. Using the binomial distribution, find the approximate probability of getting 6 heads 1 time.

no. of times $(n) = 64$
 prob. of six coin $(p) = (\frac{1}{2})^6$
 mean $(\mu) = np = 64 \times (\frac{1}{2})^6 = 1$
 $P(\text{getting 6 head at a time}) = P(X=1)$
 $P(X=1) = \frac{e^{-\mu} \mu^x}{x!} = \frac{e^{-1} 1^1}{1!} = e^{-1} = 0.3678$

37. Among the 12 seller collectors on display at a trade show, 9 are flat plate collectors and 3 are concerned collectors. If a person visiting the show randomly selects four of the seller collectors to showcase, what is the probability that 3 of them will be flat plate collectors?

SA No. of flat plate collectors $(N_1) = 9$
 concerned collector $(N_2) = 12 - 9 = 3$
 $N = N_1 + N_2 = 12$

no. of selection of seller collector $(n) = 4$
 $P(3 \text{ flat-plate collector}) = P(X=3)$

we have,
 $P(X=3) = \frac{\binom{N_1}{x} \binom{N_2}{n-x}}{\binom{N}{n}} = \frac{\binom{9}{3} \binom{3}{4-3}}{\binom{12}{4}} = 0.5091$

38. At a checkout counter customers arrive at an average number of 1.5 per minute. find the probability that

SA average no $(\mu) = 1.5$

(i) At most four will arrive in any given minute
 $P(X \leq 4) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4)$

$$= e^{-1.5} \left(\frac{1.5^0}{0!} + \frac{1.5^1}{1!} + \frac{1.5^2}{2!} + \frac{1.5^3}{3!} + \frac{1.5^4}{4!} \right)$$

$$= e^{-1.5} (1 + 1.5 + 1.125 + 0.75 + 0.2344)$$

$$= 0.981$$

(ii) at least 3 will arrive during an interval of minute
 average number per 2 minutes $= 1.5 \times 2 = 3 = (\mu)$

$$P(X \geq 3) = 1 - P(X < 3)$$

$$= 1 - [P(X=0) + P(X=1) + P(X=2)]$$

$$= 1 - e^{-3} \left(\frac{3^0}{0!} + \frac{3^1}{1!} + \frac{3^2}{2!} \right)$$

$$= 1 - e^{-3} (1 + 3 + 4.5)$$

$$= 0.576$$

(iii) At most 5 will arrive during an interval of 6 minutes.
 average number per 6 minutes $(\mu) = 1.5 \times 6 = 9$

$$P(X \leq 5) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

$$= e^{-9} \left(\frac{9^0}{0!} + \frac{9^1}{1!} + \frac{9^2}{2!} + \frac{9^3}{3!} + \frac{9^4}{4!} + \frac{9^5}{5!} \right)$$

$$= e^{-9} (1 + 9 + 40.5 + 121.5 + 273.375 + 492.0225)$$

$$= 0.1156$$

1) Rectangular (Uniform) distribution

(ii) Beta distribution

(iii) Gamma

(iv) Exponential

Normal (Gauss)

1) Rectangular or Uniform distribution

A continuous random variable x is said to have a uniform distribution over an interval (a, b) if its probability density function (pdf) is defined by

$$f(x) = \begin{cases} \frac{1}{b-a} & a \leq x < b \\ 0 & \text{otherwise} \end{cases}$$

Mean & Variance

$$\begin{aligned} \text{Mean} &= \int_a^b x f(x) dx \\ &= \int_a^b x \left(\frac{1}{b-a} \right) dx \\ &= \frac{1}{(b-a)} \int_a^b x dx \\ &= \frac{1}{(b-a)} \left[\frac{x^2}{2} \right]_a^b \\ &= \frac{1}{(b-a)} \left(\frac{b^2 - a^2}{2} \right) \\ &= \frac{b+a}{2} \end{aligned}$$

$$\lambda = \frac{a+b}{2}$$

$$\begin{aligned} \text{Variance } (\sigma^2) &= \int_a^b x^2 f(x) dx - \mu^2 \\ &= \int_a^b x^2 \left(\frac{1}{b-a} \right) dx - \mu^2 \\ &= \frac{1}{(b-a)} \int_a^b x^2 dx - \mu^2 \\ &= \frac{1}{(b-a)} \left[\frac{x^3}{3} \right]_a^b - \mu^2 \\ &= \frac{1}{(b-a)} \left(\frac{b^3 - a^3}{3} \right) - \mu^2 \\ &= \frac{1}{(b-a)} \left(\frac{(b-a)(b^2 + ab + a^2)}{3} \right) - \frac{(a+b)^2}{4} \\ &= \frac{b^2 + ab + a^2}{3} - \frac{(a+b)^2}{4} \\ &= \frac{4b^2 + 4ab + 4a^2 - 3a^2 - 6ab - 3b^2}{12} \\ &= \frac{a^2 - ab + b^2}{12} = \frac{(b-a)^2}{12} \end{aligned}$$

⇒ Gamma distribution

A continuous random variable x is said to have gamma distribution pdf is defined by

$$f(x) = \begin{cases} \frac{1}{\Gamma(\alpha) \beta^\alpha} x^{\alpha-1} e^{-x/\beta} & x > 0 \\ 0 & \text{otherwise} \end{cases} \quad \begin{matrix} \alpha > 0 \\ \beta > 0 \end{matrix}$$

where

$\Gamma(\alpha) = (\alpha-1)!$ if α is a positive integer $\alpha \in \mathbb{R}$ are parameters

Mean & Variance of Gamma distribution

$$\text{Mean } E(x) = \alpha \beta$$

$$\text{Variance } V(x) = \alpha \beta^2$$

particular case of Gamma distribution

if parameter $\alpha = 1$, the gamma distribution is called exponential

$$\text{now, pdf } f(x) = \frac{1}{\beta} e^{-x/\beta}, \quad \begin{matrix} \beta > 0 \\ x > 0 \end{matrix}$$

$$\text{Mean, } E(x) = \beta$$

$$\text{Variance } V(x) = \beta^2$$

⇒ Beta distribution

A continuous random variable x is said to have Beta distribution is defined by

$$f(x) = \begin{cases} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, & \alpha > 0, \beta > 0 \\ 0 & 0 < x < 1 \end{cases} \quad \text{otherwise}$$

Mean & Variance

$$\text{Mean } E(x) = \frac{\alpha}{\alpha+\beta}$$

$$\text{Variance } V(x) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

Normal distribution

Def. A continuous random variable x is said to have normal distribution if its pdf $f(x)$ is defined by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty$$

where,

$$\text{mean} = \mu$$

$$\text{Standard deviation} = \sigma$$

$$\text{Total prob. } \int_{-\infty}^{\infty} f(x) dx = 1$$

CHARACTERISTICS OF NORMAL DISTRIBUTION

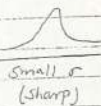
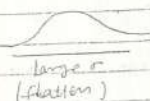
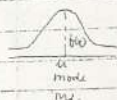
(i) The curve is bell-shaped & symmetrical about the line $x = \mu$

(ii) Mean = Mode = Median

(iii) $x = \mu$ is asymptote to the curve.

(iv) $f(x)$ is max at $x = \mu$ i.e. $[f(x)]_{\text{max}} = \frac{1}{\sigma\sqrt{2\pi}}$

(v) for large value of σ , the curve becomes flatter and out $\rightarrow \infty$ & for small σ it has a sharp peak



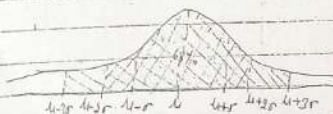
(vi) Mean deviation about mean $\sim \frac{1}{2}\sigma$

(vii) Area property:

$$(a) P(\mu - \sigma < x < \mu + \sigma) = 0.6826$$

$$(b) P(\mu - 2\sigma < x < \mu + 2\sigma) = 0.9544$$

$$(c) P(\mu - 3\sigma < x < \mu + 3\sigma) = 0.9973$$



Standard Normal distribution

In a normal distribution, when mean (μ) = 0 and standard deviation = 1 then the normal distribution is called standard normal distribution

pdf is defined by

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty$$

Cumulative probability distribution of standard normal distribution is defined as

$$\Phi(z) = \int_{-\infty}^z f(z) dz$$

$$f(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{z-\mu}{\sigma}\right)^2}$$

$$\mu = 0, \sigma = 1$$

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-t^2/2} dt$$

Relation between $F(x)$ and $\Phi(z)$

Relation between $F(x)$ & $\Phi(z)$ is defined by

$$(i) F(x) = P(-\infty < x < \infty)$$

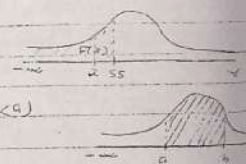
$$(ii) F(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$$

Example

$$(ix) F(55) = P(x \leq 55)$$

$$(iv) F(x > 55) = 1 - P(x \leq 55) = 1 - F(55)$$

$$(v) P(a < x < b) = P(-\infty < x < b) - P(-\infty < x < a) = F(b) - F(a)$$



show that following distribution is rectangular distribution

$$f(x) = \frac{1}{2a}, -a \leq x \leq a$$

$$f(x) = \frac{1}{a-a} = \frac{1}{2a}, -a \leq x \leq a$$

$$\text{Total probability} = \int_{-a}^a f(x) dx = \int_{-a}^a \frac{1}{2a} dx = \frac{1}{2a} [x]_{-a}^a = \frac{1}{2a} (a - (-a)) = \frac{1}{2a} (2a) = 1$$

Since the distribution is rectangular

A random variable X has a uniform distribution over $(-2, 3)$ compute

(i) $P(X=2)$

P.D.F. of X

$$f(x) = \frac{1}{3-(-2)} = \frac{1}{5}$$

$$P(X=2) = f(2) = \frac{1}{5}$$

$P(|X| < 2)$

$$= 1 - P(|X| \geq 2)$$

$$= 1 - P(-2 \leq X \leq 2)$$

$$= 1 - \int_{-2}^2 f(x) dx = 1 - \int_{-2}^2 \frac{1}{5} dx = 1 - \left[\frac{x}{5} \right]_{-2}^2 = 1 - \frac{2}{5} - \left(-\frac{2}{5} \right) = 1 - \frac{4}{5} = \frac{1}{5}$$

$P(|x-2| < 2)$

$$-2 < x-2 < 2$$

$$0 < x < 4$$

$$= P(0 < x < 4) = \int_0^4 f(x) dx = \int_0^4 \frac{1}{5} dx = \left[\frac{x}{5} \right]_0^4 = \frac{4}{5}$$

$$\int_0^4 f(x) dx = \int_0^4 \frac{1}{5} dx = \left[\frac{x}{5} \right]_0^4 = \frac{4}{5}$$

$$= \frac{4}{5} \times 1 = \frac{4}{5}$$

(1) If a random variable has the standard normal distribution, find probability that it will take on a value

(a) less than 1.5

(b) greater than -1.75

(c) between 0.5 to 1.5

So (a) $P(X < 1.5)$ (b) $P(X > -1.75)$ (c) $P(0.5 < X < 1.5)$

$$= \Phi(1.50)$$

$$= 1 - P(X \leq -1.75)$$

$$= \Phi(1.50) - \Phi(0.50)$$

Using table

$$= 1 - \Phi(-1.75)$$

$$= 0.9332 - 0.6915$$

$$= 0.9332$$

$$= 1 - 0.0475$$

$$= 0.8417$$

$$= 0.9509$$

(4) And the value of z if the probability that a random variable has the standard normal distribution will take on the value

(a) less than z is 0.9911

(b) greater than z is 0.1093

$$\Phi(z) = 0.9911$$

$$P(X > z) = 1 - P(X < z)$$

$$\Phi(2.37) = 0.9911 \text{ (using table)}$$

$$0.1093 = 1 - \Phi(z)$$

$$\therefore z = 2.37$$

$$\Phi(z) = 1 - 0.1093 = 0.8907$$

(c) greater than z is 0.6443

$$\Phi(1.23) = 0.8907 \text{ (from table)}$$

$$P(X > z) = 1 - P(X < z)$$

$$\therefore z = 1.23$$

$$0.6443 = 1 - P(X < z)$$

(d) less than z is 0.0212

$$P(X < z) = 1 - 0.6443$$

$$\text{or } P(X < z) = 0.0212$$

$$\Phi(z) = 0.3557$$

$$\text{or } \Phi(z) = 0.0212$$

$$\Phi(-0.37) = 0.3557$$

$$\text{or } \Phi(-2.02) = 0.0212$$

$$\therefore z = -0.37$$

$$\therefore z = -2.02$$

(e) between $-z$ and z is 0.9298

$$P(-z < X < z) = 0.9298$$

$$\Phi(z) - \Phi(-z) = 0.9298$$

$$\text{or } \Phi(z) - [1 - \Phi(z)] = 0.9298$$

$$2\Phi(z) = 1.9298$$

$$\Phi(z) = 0.9649$$

$$\Phi(1.81) = 0.9649$$

$$\therefore z = 1.81$$

5) Given a random variable having the normal distribution with $\mu = 16.2$ and $\sigma^2 = 1.5625$. find the probability that it will take on a value

(i) greater than 16.8

Sol: $\mu = 16.2$, $\sigma^2 = 1.5625$
 $\sigma = 1.25$

$$\begin{aligned} P(X > 16.8) &= 1 - P(X \leq 16.8) \\ &= 1 - F(16.8) \\ &= 1 - \Phi\left(\frac{16.8 - 16.2}{1.25}\right) \\ &= 1 - \Phi(0.48) \\ &= 1 - 0.6844 = 0.3156 \end{aligned}$$

(ii) less than 14.9

$$P(X < 14.9) = \Phi\left(\frac{14.9 - 16.2}{1.25}\right) = \Phi(-1.04) = 0.1492$$

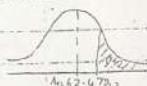
(iii) between 13.6 and 18.8 (iv) between 15.5 and 16.7

$$\begin{aligned} P(13.6 < X < 18.8) &= P(15.5 < X < 16.7) \\ &= F(18.8) - F(13.6) \\ &= \Phi\left(\frac{18.8 - 16.2}{1.25}\right) - \Phi\left(\frac{13.6 - 16.2}{1.25}\right) \\ &= \Phi(2.08) - \Phi(-2.08) \\ &= 0.9812 - 0.0188 \\ &= 0.9624 \end{aligned}$$

$$\begin{aligned} P(15.5 < X < 16.7) &= F(16.7) - F(15.5) \\ &= \Phi\left(\frac{16.7 - 16.2}{1.25}\right) - \Phi\left(\frac{15.5 - 16.2}{1.25}\right) \\ &= \Phi(0.4) - \Phi(-0.56) \\ &= 0.6554 - 0.2878 \\ &= 0.3676 \end{aligned}$$

6) A random variable has a normal distribution with mean $\mu = 62.4$ find its S.d if the probability is 0.2 that it will take on a value greater than 72.20.

Sol: mean (μ) = 62.4
 $\sigma = ?$



$$\begin{aligned} P(X > 72.2) &= 0.20 \\ \text{or } 1 - P(X \leq 72.2) &= 0.20 \\ \text{or } 1 - F(72.2) &= 0.20 \\ \text{or } 1 - 0.2 &= F(72.2) \end{aligned}$$

$$\begin{aligned} \text{or } F(72.2) &= 0.80 \\ \text{or } \Phi\left(\frac{72.2 - 62.4}{\sigma}\right) &= 0.80 \end{aligned}$$

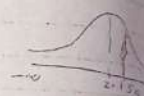
$$\text{or } \Phi\left(\frac{16.8}{\sigma}\right) = \Phi(0.84)$$

$$\text{or } \frac{16.8}{\sigma} = 0.84$$

$$\sigma = \frac{16.8}{0.84} = 20$$

7) find the area under the normal curve for $Z = 1.59$.

$$\text{Sol: } \Phi(1.59) = 0.9382$$



8) find the area to the right of $Z = 0.25$.

$$\begin{aligned} \text{Sol: } P(Z > 0.25) &= 1 - P(Z \leq 0.25) \\ &= 1 - \Phi(0.25) \\ &= 1 - 0.5987 \\ &= 0.4013 \end{aligned}$$



9) if X is uniform distribution with mean 2 and variance $\frac{3}{16}$ find S.d for uniform distribution

$$f(x) = \frac{1}{b-a}, a < x < b$$

$$\text{Mean} = \frac{a+b}{2} = 2$$

$$\text{Variance} = \frac{(b-a)^2}{12}$$

$$\text{or } (4-b)b = 55/16$$

$$a+b = 4$$

$$\frac{(b-a)^2}{12} = \frac{3}{16}$$

$$4b - b^2 = \frac{55}{16}$$

$$\frac{16}{16}$$

$$\frac{(b-a)^2}{12} = \frac{3}{16}$$

$$64b - 16b^2 = 55$$

$$4^2 - 4b = 36/16$$

$$\text{or } 16b^2 - 64b + 55 = 0$$

$$16 - \frac{36}{16} = 4a$$

$$b = \frac{64 \pm \sqrt{64^2 - 4 \times 16 \times 55}}{2 \times 16}$$

$$\frac{16 - 36}{16} = 4a$$

$$= \frac{64 \pm 24}{32}$$

$$\frac{16 - 36}{16} = 4a$$

$$a = 1.25$$

$$\frac{16 - 36}{16} = 4a$$

$$\frac{16 - 36}{16} = 4a$$

$$\frac{16 - 36}{16} = 4a$$

$$f(x) = \frac{1}{b-a} = \frac{1}{4-2} = \frac{1}{2}$$

$$\text{S.d.} = \sqrt{\frac{(b-a)^2}{12}} = \sqrt{\frac{4}{12}} = \frac{1}{\sqrt{3}}$$

$$P(X < 2) = \int_{-\infty}^2 f(x) dx = \int_{-\infty}^2 \frac{1}{2} dx$$

$$= \frac{1}{2} [x]_{-\infty}^2 = \frac{1}{2} [2 - (-\infty)] = \frac{1}{2} [2 - (-\infty)]$$

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$$a = -0.025, b = 0.025$$

$$f(x) = \frac{1}{(b-a)} = \frac{1}{0.025 - (-0.025)} = \frac{1}{0.05}$$

1.7" also as 0.015 (uniform density)

$$P(0.015) = \int_{-0.015}^{0.015} f(x) dx = \int_{-0.015}^{0.015} \frac{1}{0.05} dx = \frac{1}{0.05} [x]_{-0.015}^{0.015} = \frac{1}{0.05} \times 0.03 = 0.6$$

1.8" = 0.012 and 0.012

$$P(0.012) = \int_{-0.012}^{0.012} f(x) dx = \int_{-0.012}^{0.012} \frac{1}{0.05} dx = \frac{1}{0.05} [x]_{-0.012}^{0.012} = \frac{1}{0.05} \times 0.024 = 0.48$$

Gamma distribution

$$\alpha = 2 \text{ and } \beta = 6$$

mean life time of a computer = $\alpha \cdot \beta = 2 \times 6 = 12$

$$P(X > 10) = 1 - P(X \leq 10)$$

pdf of gamma distribution

$$f(x) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}$$

$$\Gamma = (2-1)! = (1)! = 1$$

$$f(x) = \int_0^\infty \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta} dx$$

$$= \frac{1}{36} \int_0^\infty x e^{-x/6} dx$$

$$= \frac{1}{36} \left[-\frac{x^2}{2} e^{-x/6} - \frac{36}{2} e^{-x/6} \right]_0^\infty$$

$$= \frac{1}{36} \left[0 + 0 + \frac{180}{6} e^{-10/6} + \frac{36}{2} e^{-10/6} \right]$$

$$= \left[\frac{-10/6}{36} \left[\frac{180}{6} + \frac{36}{2} \right] \right] = \left[\frac{60+36}{36} \right] \times \frac{e^{-10/6}}{36}$$

$$= \frac{1}{36} \times e^{-10/6} [60+36]$$

$$= 0.5038$$

1.9) 0.45 (uniform density)

$$\alpha = 2, \beta = 3$$

$$P(X < 0.10) = \int_0^{0.10} f(x) dx$$

$$= \int_0^{0.10} x(1-x)^2 dx$$

$$= \int_0^{0.10} (x - 2x^2 + x^3) dx$$

$$= \left[\frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4} \right]_0^{0.10}$$

$$= \left[\frac{0.1^2}{2} - \frac{2}{3} \times 0.1^3 + \frac{0.1^4}{4} \right]$$

$$= 0.00435 \approx 0.0043$$

2) Beta distribution

$$\alpha = 1, \beta = 4$$

$$\Gamma = (4-1)! = (3)! = 6$$

$$\text{mean} = \frac{\alpha}{\alpha+\beta} = \frac{1}{1+4} = 0.2$$

$$P(X > 0.4) = 1 - P(X \leq 0.4)$$

$$= 1 - \int_0^{0.4} f(x) dx$$

$$= 1 - \int_0^{0.4} \frac{1}{\Gamma(\alpha) \Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1} dx$$

$$= 1 - \int_0^{0.4} \frac{1}{1! 3!} x^0 (1-x)^3 dx$$

$$= 1 - \int_0^{0.4} \frac{1}{6} (1-x)^3 dx$$

$$= 1 - \left[-\frac{1}{6} \frac{(1-x)^4}{4} \right]_0^{0.4}$$

$$= 1 - \left[-\frac{1}{24} (1-x)^4 \right]_0^{0.4}$$

$$= 1 - \left[-\frac{1}{24} (1-0.4)^4 + \frac{1}{24} (1-0)^4 \right]$$

$$= 1 - \left[-\frac{1}{24} (0.6)^4 + \frac{1}{24} (1)^4 \right]$$

$$= 1 - \left[-\frac{1}{24} (0.1296) + \frac{1}{24} (1) \right]$$

$$= 1 - \left[-0.0054 + 0.041667 \right]$$

$$= 1 - 0.036267 = 0.963733$$

$$= 0.9637$$

$$f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha) \Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

$$\Gamma = (2-1)! = (1)! = 1$$

$$f(x) = \frac{\Gamma(2+2)}{\Gamma(2) \Gamma(2)} x^{2-1} (1-x)^{2-1} = \frac{6}{1! 1!} x^1 (1-x)^1 = 6x(1-x)$$

$$f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha) \Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

$$= \frac{6}{1! 1!} x^{2-1} (1-x)^{2-1} = 6x(1-x)$$

$$= 6x(1-x)$$

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Beta distribution

$$\alpha = 5 \text{ and } \beta = 2$$

$$P_2 = (\alpha - 1) / (\alpha + \beta) = (5 - 1) / (5 + 2) = 4 / 7 = 0.5714$$

$$\text{Mean}(\mu) = \frac{\alpha}{\alpha + \beta} = \frac{5}{5 + 2} = 0.71428$$

$$P(X \leq 0.5)$$

$$= \int_0^{0.5} f(x) dx$$

$$= 30 \int_0^{0.5} x^4 (1-x)^2 dx$$

$$= 30 \int_0^{0.5} (x^4 - 2x^5 + x^6) dx$$

$$= 30 \left[\frac{x^5}{5} - \frac{2x^6}{6} + \frac{x^7}{7} \right]_0^{0.5} = 30 \left(\frac{0.5^5}{5} - \frac{2 \cdot 0.5^6}{6} + \frac{0.5^7}{7} \right) = (0.00625 - 0.002604) \times 20 = 0.11$$

$$f(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

$$= \frac{\Gamma(7)}{\Gamma(5) \Gamma(2)} x^{4} (1-x)^{1}$$

$$= \frac{7!}{5! 2!} x^4 (1-x)$$

$$= \frac{6!}{5! 2!} x^4 (1-x)$$

$$= 30 x^4 (1-x)$$

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Normal distribution

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}}$$

$$P(X > 200)$$

$$= \int_{200}^{\infty} f(x) dx$$

$$= \int_{200}^{\infty} \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}} dx$$

$$= \frac{1}{\sigma \sqrt{2\pi}} \left[\frac{e^{-\frac{x^2}{2\sigma^2}}}{-\frac{x}{\sigma^2}} \right]_{200}^{\infty} = \frac{1}{\sigma \sqrt{2\pi}} \left[0 + \frac{e^{-\frac{200^2}{2\sigma^2}}}{\frac{200}{\sigma^2}} \right] = e^{-50}$$

$$= 0.25 \left[\frac{e^{-0.25x}}{-0.25} \right]_{200}^{\infty} = 0.25 \left[0 + \frac{e^{-0.25 \times 200}}{0.25} \right] = e^{-50}$$

$$P(X < 10) = \int_0^{10} f(x) dx = \int_0^{10} 0.25 e^{-0.25x} dx = 0.25 \left[\frac{e^{-0.25x}}{-0.25} \right]_0^{10} = 0.25 \left[\frac{e^{-0.25 \times 10}}{-0.25} - \frac{e^{-0.25 \times 0}}{-0.25} \right] = 0.25 \left[\frac{e^{-2.5}}{-0.25} - \frac{1}{-0.25} \right] = 1 - e^{-2.5}$$

$$= 1 - e^{-2.5} = 1 - 0.082 = 0.918$$

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more than 4.8 sec

$$P(X > 4.8) = 1 - P(X \leq 4.8)$$

$$= 1 - F(4.8)$$

$$= 1 - \Phi\left(\frac{4.8 - 4}{0.84}\right)$$

$$= 1 - \Phi(0.95)$$

$$= 1 - 0.8213$$

$$= 0.1787$$

anywhere from 4.7 to 4.82

$$P(4.7 \leq X \leq 4.82)$$

$$= F(4.82) - F(4.7)$$

$$= \Phi\left(\frac{4.82 - 4}{0.84}\right) - \Phi\left(\frac{4.7 - 4}{0.84}\right)$$

$$= \Phi(1.57) - \Phi(0.83)$$

$$= 0.9332 - 0.7968$$

$$= 0.1364$$

SN: $\mu = 15.220$

$$\sigma^2 = 352$$

$$\sigma = 18.221$$

$$(i) P(150 \leq X \leq 1520)$$

$$F(200) - F(100)$$

$$= \Phi\left(\frac{200 - 110}{18.221}\right) - \Phi\left(\frac{100 - 110}{18.221}\right)$$

$$= \Phi(4.94) - \Phi(-0.55)$$

$$= 0.9999 - 0.4898$$

$$= 0.5101$$

$$= 0.5101$$

$$= 0.5101$$

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$$= 0.5101$$

$$(ii) P(X > 300)$$

$$= 1 - P(X \leq 300) = 1 - F(300)$$

$$= 1 - \Phi\left(\frac{300 - 110}{18.221}\right) = 1 - \Phi(10.43)$$

$$= 1 - 1 = 0$$

$$= 1 - 1 = 0$$

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$$(iii) P(X < 100)$$

$$= F(100)$$

$$= \Phi\left(\frac{100 - 110}{18.221}\right) = \Phi(-0.55)$$

$$= 0.4898$$

$$= 0.4898$$

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$$= 0.4898$$

according to Question

$$N \times P(X > 16.25) = 647$$

$$N = \frac{647}{P(X > 16.25)} = \frac{647}{1 - P(X \leq 16.25)} = \frac{647}{1 - \Phi\left(\frac{16.25 - 11}{0.84}\right)} = \frac{647}{1 - \Phi(6.25)} = \frac{647}{1 - 0.9999} = 64700$$

where N = Total No.

$$N = \frac{647}{1 - \Phi\left(\frac{16.25 - 11}{0.84}\right)} = \frac{647}{1 - \Phi(6.25)} = \frac{647}{1 - 0.9999} = 64700$$

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$\mu = 15.72, \sigma^2 = 10.8$
 $\sigma = 3.287$
 $P(X > 60) = P(X > 72)$
 $1 - P(X < 72)$
 $1 - F(72)$
 $1 - \Phi\left(\frac{72 - 15.72}{3.287}\right)$
 $1 - \Phi(1.15)$
 $1 - 0.8746$
 $= 0.1254$
 Total no. of products required = 0.687×1000
 $= 687$



$P(X < 32) = 0.07$
 $F(32) = 0.07$
 $\Phi\left(\frac{32 - \mu}{\sigma}\right) = 0.07$
 $\Phi\left(\frac{32 - \mu}{\sigma}\right) = \Phi(-1.48)$
 $\frac{32 - \mu}{\sigma} = -1.48$
 $32 - \mu = -1.48 \sigma$
 $\mu - 1.48 \sigma = 32$
 $\mu - 1.48 \times 3.287 = 32$
 $\mu - 4.865 = 32$
 $\mu = 36.865$
 Total no. of products required = $36.865 \times 1000 = 36865$

Total no. of products required = 36865
 Total no. of products required = 36865

(23) $\mu = 2, \sigma^2 = 1$
 $\sigma = 1$
 $P(1 < X < 3)$
 $P(-1 < (X-2) < 1)$
 $P(-1 < Z < 1)$
 $P(Z < 1) - P(Z < -1)$
 $\Phi(1) - \Phi(-1)$
 $\Phi(1) - (1 - \Phi(1))$
 $2\Phi(1) - 1$
 $2(0.8413) - 1$
 $= 0.6826$

again $P(X < 63) = 0.89$
 $F(63) = 0.89$
 $\Phi\left(\frac{63 - \mu}{\sigma}\right) = 0.89$
 $\Phi\left(\frac{63 - \mu}{\sigma}\right) = \Phi(1.23)$
 $\frac{63 - \mu}{\sigma} = 1.23$
 $63 - \mu = 1.23 \sigma$
 $\mu = 63 - 1.23 \sigma$
 $\mu = 63 - 1.23 \times 3.287$
 $\mu = 63 - 4.043$
 $\mu = 58.957$

Total no. of products required = $58.957 \times 1000 = 58957$

$F(70) - F(30)$
 $\Phi\left(\frac{70 - \mu}{\sigma}\right) - \Phi\left(\frac{30 - \mu}{\sigma}\right)$
 $\Phi(1) - \Phi(-1)$
 $0.8413 - 0.1587$
 $= 0.6826$

let required value = a
 then $P(X > a) = \frac{90}{900} = 0.1$
 $1 - P(X \leq a) = 0.1$
 $1 - F(a) = 0.1$
 $1 - \Phi\left(\frac{a - \mu}{\sigma}\right) = 0.1$
 $\Phi\left(\frac{a - \mu}{\sigma}\right) = 0.9$
 $\Phi\left(\frac{a - 15.72}{3.287}\right) = 0.9$
 $\frac{a - 15.72}{3.287} = 1.28$
 $a - 15.72 = 1.28 \times 3.287$
 $a - 15.72 = 4.207$
 $a = 19.927$

(24) $\mu = 68.22 \text{ gm}$
 $\sigma^2 = 10.8 \text{ gm}^2$
 $\sigma = 3.286 \text{ gm}$
 to be over 72 gm
 $P(X > 72)$
 $1 - P(X \leq 72)$
 $1 - F(72)$
 $1 - \Phi\left(\frac{72 - 68.22}{3.286}\right)$
 $1 - \Phi(1.15)$
 $1 - 0.8749$
 $= 0.1251$

let 70 and 72 gm
 $P(70 < X < 72)$
 $F(72) - F(70)$
 $\Phi\left(\frac{72 - 68.22}{3.286}\right) - \Phi\left(\frac{70 - 68.22}{3.286}\right)$
 $\Phi(1.15) - \Phi(0.54)$
 $0.8749 - 0.7054$
 $= 0.1695$

Total no. of products over 72 gm
 $= 0.1251 \times 1000 = 125.1 = 125$

(25) $\mu = 50, \sigma = 10$
 $P(X < 90) = 0.13$
 $F(90) = 0.13$
 $\Phi\left(\frac{90 - \mu}{\sigma}\right) = 0.13$
 $\Phi\left(\frac{90 - 50}{10}\right) = 0.13$
 $\Phi(4) = 0.13$
 $4 = \frac{90 - 50}{10}$
 $40 = 90 - 50$
 $90 = 90$

(26) $P(X > 9) = 1 - P(X \leq 9) = 0.14$
 $1 - F(9) = 0.14$
 $1 - \Phi\left(\frac{9 - \mu}{\sigma}\right) = 0.14$
 $\Phi\left(\frac{9 - \mu}{\sigma}\right) = 0.86$
 $\Phi\left(\frac{9 - 50}{10}\right) = 0.86 = \Phi(1.08)$
 $\frac{9 - 50}{10} = 1.08$
 $9 - 50 = 10.8$
 $9 = 60.8$



No. of student = 1000

$$\mu = 42$$

$$\sigma = 24$$

$$P(X > 50)$$

$$= 1 - P(X \leq 50)$$

$$= 1 - F(50)$$

$$= 1 - \Phi\left(\frac{50-42}{24}\right)$$

$$= 1 - \Phi(0.33)$$

$$= 1 - 0.6298$$

$$= 0.3702$$

The no. of student exceeding 50

$$= 0.3702 \times 1000 = 370.2$$

$$P(X < 30) = 0.10$$

$$\Phi\left(\frac{30-42}{24}\right) = 0.10$$

$$\Phi\left(\frac{30-42}{24}\right) = \Phi(-0.5)$$

$$\Phi(-0.5) = 1 - \Phi(0.5)$$

$$1 - \Phi(0.5) = 1 - 0.6915$$

$$= 0.3085$$

$$\mu - 1.28\sigma = 30 \quad \text{--- (1)}$$

$$P(X < 62) = 0.97$$

$$= F(62) = 0.97$$

$$\Phi\left(\frac{62-42}{24}\right) = \Phi(0.83)$$

$$\Phi(0.83) = 0.7967$$

$$62 - 42 = 1.88\sigma$$

$$62 - 42 = 1.88\sigma$$

$$\mu + 1.88\sigma = 62 \quad \text{--- (2)}$$

$$\mu = 77$$

$$\sigma = 9$$

$$P(X > 90) = 0.20$$

$$1 - F(90) = 0.20$$

$$\Phi\left(\frac{90-77}{9}\right) = 0.80 = \Phi(1.44)$$

$$\frac{90-77}{9} = 1.44 \quad \sigma = \frac{13}{1.44} = 9.03$$

$$\frac{90-77}{9} = 1.44$$

$$\frac{90-77}{9} = 1.44$$

$$\frac{90-77}{9} = 1.44$$

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$$\frac{90-77}{9} = 1.44$$

$$P(30 < X < 50)$$

$$= F(50) - F(30)$$

$$= \Phi\left(\frac{50-42}{24}\right) - \Phi\left(\frac{30-42}{24}\right)$$

$$= \Phi(0.33) - \Phi(-0.5)$$

$$= 0.6298 - 0.3085$$

$$= 0.3213$$

The no. of student bet 30 and 50 is

$$0.3213 \times 1000 = 321.3$$

Let the value of score = a

$$P(X > a) = \frac{100}{1000}$$

$$1 - P(X \leq a) = 0.10$$

$$1 - F(a) = 0.10$$

$$1 - \Phi\left(\frac{a-42}{24}\right) = 0.10$$

$$\Phi\left(\frac{a-42}{24}\right) = 0.90$$

$$\Phi\left(\frac{a-42}{24}\right) = \Phi(1.28)$$

$$\frac{a-42}{24} = 1.28$$

$$a - 42 = 1.28 \times 24 = 30.72$$

$$a = 42 + 30.72 = 72.72 \approx 73$$

$$a = 72.72 \approx 73$$

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$$(24) \text{ SD } \mu = 7.5$$

$$S.d. (\sigma) = 5$$

$$(25) P(X > a) = 0.05$$

$$\text{or } 1 - P(X \leq a) = 0.05$$

$$\text{or } 1 - F(a) = 0.05$$

$$\text{or } 1 - \Phi\left(\frac{a-7.5}{5}\right) = 0.05$$

$$\text{or } \Phi\left(\frac{a-7.5}{5}\right) = 0.95$$

$$\text{or } \Phi\left(\frac{a-7.5}{5}\right) = \Phi(1.65)$$

$$\text{or } \frac{a-7.5}{5} = 1.65$$

$$\text{or } a - 7.5 = 8.25$$

$$\text{or } a = 8.25 + 7.5 = 15.75$$

$$(26) \text{ SD } \mu = \text{Normal distribution}$$

$$S.d. (\sigma) = 10$$

$$\text{Mean } (\mu) = 9$$

$$P(X < 30.50) = 0.3264$$

$$\text{or } F(30.50) = 0.3264$$

$$\Phi\left(\frac{30.50-9}{10}\right) = \Phi(2.15)$$

$$\Phi(2.15) = 0.9841$$

$$30.50 - 9 = 21.5\sigma$$

$$30.50 - 9 = 21.5\sigma$$

$$\mu = 85$$

$$(27) \text{ Normal distribution}$$

$$S.d. (\sigma) = 0.01$$

$$\text{Mean } (\mu) = 9$$

$$P(X > 3) = 0.05$$

$$\text{or } 1 - P(X \leq 3) = 0.05$$

$$\text{or } 1 - F(3) = 0.05$$

$$\text{or } 1 - \Phi\left(\frac{3-9}{0.01}\right) = 0.05$$

$$\text{or } \Phi\left(\frac{3-9}{0.01}\right) = 0.95$$

$$\text{or } \Phi\left(\frac{3-9}{0.01}\right) = \Phi(1.65)$$

$$\text{or } \frac{3-9}{0.01} = 1.65$$

$$\text{or } 3 - 9 = 0.0165$$

$$\text{or } 3 - 9 = 0.0165$$

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$$\text{or } 3 - 9 = 0.0165$$

$$(28) P(X < b) = 0.95$$

$$\text{or } F(b) = 0.95$$

$$\text{or } \Phi\left(\frac{b-7.5}{5}\right) = 0.95$$

$$\text{or } \Phi\left(\frac{b-7.5}{5}\right) = \Phi(1.65)$$

$$\text{or } \frac{b-7.5}{5} = 1.65$$

$$\text{or } b - 7.5 = 8.25$$

$$\text{or } b = 8.25 + 7.5 = 15.75$$

$$\text{or } b = 15.75$$

$$\text{or } b = 15.75$$

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$$\text{or } b = 15.75$$

$$(29) \text{ No. of student} = 500$$

$$\text{Mean weight } (\mu) = 151 \text{ lbs}$$

$$S.d. (\sigma) = 15 \text{ lbs}$$

$$P(X > 185) = ?$$

$$P(X > 185) = 1 - P(X \leq 185)$$

$$= 1 - F(185)$$

$$= 1 - \Phi\left(\frac{185-151}{15}\right)$$

$$= 1 - \Phi(2.26)$$

$$= 1 - 0.9881$$

$$= 0.0119$$

$$= 0.0119$$

$$\begin{aligned} \textcircled{b} \quad & P(X < Q) = 0.99 \\ & 2. \quad F(Q) = 0.99 \\ & 3. \quad \frac{Q - 6}{0.02} = \Phi(2.33) \\ & 4. \quad Q - 6 = 0.33 \times 0.02 \end{aligned}$$

Exponential distribution
 $\lambda = 0.5$
 $P(X < 400)$
 $f(x) = \frac{1}{\lambda} e^{-x/\lambda}$
 $= \frac{1}{0.5} e^{-400/0.5}$
 $= 0$

Exponential distribution

$\lambda = 0.005$

At most 10000 calls

$P(0 \leq X \leq 10000)$

$\int_0^{10000} f(x) dx$

$\int_0^{10000} \lambda e^{-\lambda x} dx$

$= 0.005 \cdot \int_0^{10000} e^{-0.005x} dx$

$0.005 \cdot \left[\frac{e^{-0.005x}}{-0.005} \right]_0^{10000}$

$1 - e^{-5}$

$1 - 0.0067 =$

(11) Exponential distribution
 $\lambda = 1/50$, $P(X \leq 70) = ?$
 here,
 $P(X \leq 70) = 1 - P(X > 70)$
 $= 1 - \int_{70}^{\infty} f(x) dx = 1 - \int_{70}^{\infty} (\lambda e^{-\lambda x}) dx$
 $= 1 - \frac{1}{\lambda} \int_{70}^{\infty} e^{-\lambda x} dx = 1 - \frac{1}{\lambda} \left[\frac{e^{-\lambda x}}{-\lambda} \right]_{70}^{\infty}$
 $= 1 - \left[-e^{-\lambda x} \right]_{70}^{\infty} = 1 - (-e^{-\lambda \cdot 70} + e^{-\lambda \cdot \infty}) = 1 - (-e^{-70/50} + 0) = 1 + e^{-70/50} = 1 + e^{-1.4} \approx 1.502$

(ii) Probability from 16,000 to 24,000 miles

$$\begin{aligned}
 P(16000 < X < 24000) &= ? \\
 &= \int_{16,000}^{24,000} f(u) du = \frac{0.005}{16,000} \int_{16,000}^{24,000} u e^{-0.0005u} du \\
 &= 0.0005 \left[\frac{e^{-0.0005u}}{-0.0005} \right]_{16,000}^{24,000} \\
 &= - \left[e^{-0.0005 \times 24,000} - e^{-0.0005 \times 16,000} \right] \\
 &= - [e^{-12} - e^{-8}] \\
 &= e^{-8} - e^{-12} \quad \text{Q.E.D.}
 \end{aligned}$$

(vi) At least 5000 mile

$$\begin{aligned} P(x) &= P(x < 30000) \\ &= 1 - \int_0^{30000} f(x) dx \\ &= 1 - \int_0^{30000} \lambda e^{-\lambda x} dx \\ &= 1 - 0.0005 \int_0^{30000} e^{-0.0005x} dx \\ &= 1 - 0.0005 \left[\frac{e^{-0.0005x}}{-0.0005} \right]_0^{30000} \\ &= 1 + \left[e^{-0.0005 \times 30000} - e^0 \right] \\ &= 1 + e^{-15} - 1 = e^{-15} \end{aligned}$$

(4) Gauss distribution

(ii) $P(X > 9) = 1 - \int_0^9 f(x) dx$

$$f(x) = \frac{1}{\Gamma(x)} x^{x-1} e^{-x/3}$$

$$f(x) = \frac{1}{3^2 \Gamma(2)} x^{2-1} e^{-x/3}$$

$$f(x) = \frac{1}{9x!} x e^{-x/3}$$

$$\therefore P(X > 9) = 1 - \int_0^9 \frac{1}{9} x e^{-x/3} dx$$

$$= 1 - \left[-3x e^{-x/3} - 9 e^{-x/3} \right]$$

$$= 1 - \frac{1}{9} \left[-3 \cdot 9 e^{-3} - 9 e^{-3} + 0 \right]$$

$$= 1 - \left[-3 e^{-3} - e^{-3} + 1 \right]$$

$$= 3e^{-3} + 4e^{-3}$$

$$= 4e^{-3}$$

43) Beta distribution

$$\alpha=1, \beta=4$$

$$\text{mean}(\mu) = \frac{\alpha}{\alpha+\beta} = \frac{1}{1+4} = 0.2$$

$$P(X \geq 0.25)$$

$$f(u) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

$$= \frac{\Gamma(1+4)}{\Gamma(1)\Gamma(4)} \times x^{1-1} \times (1-x)^{4-1}$$

$$= \frac{\Gamma(5)}{\Gamma(1)\Gamma(4)} \times x^0 (1-x)^3$$

$$= \frac{4!}{1! \cdot 3!} (1-x)^3 = 4(1-x)^3$$

$$\therefore \int_{0.25}^1 f(u) du = \int_{0.25}^1 4(1-x)^3 dx$$

$$= 4 \left[\frac{(1-3x+3x^2-x^3)}{0.25} \right]_{0.25}^1$$

$$= 4 \left[\frac{1-3+3-1}{0.25} + \frac{3x^2}{0.25} - \frac{x^3}{0.25} \right]_{0.25}^1$$

$$= 4 \left[\frac{1-3+3-1}{0.25} + \frac{3(1)^2}{0.25} - \frac{(1)^3}{0.25} \right] - 4 \left[\frac{1-3+3-1}{0.25} + \frac{3(0.25)^2}{0.25} - \frac{(0.25)^3}{0.25} \right]$$

$$= 4 \times 0.0291 = 0.3164$$

$$\begin{array}{r} e^{-x/3} \\ 1 \quad - \quad \frac{e^{-x/3}}{1/3} \\ \hline 0 \quad \quad \quad \frac{e^{-x/3}}{1/9} \end{array}$$

No. of Builders = 500

Mean (μ) = 30,000

S.d. (σ) = 10,000

$P(X > 40,000) = ?$

$1 - P(X \leq 40,000)$

$1 - F(40,000)$

$1 - \Phi\left(\frac{40,000 - 30,000}{10,000}\right)$

$1 - \Phi(1)$

$1 - 0.6554$

$= 0.3446$

No. of Builders = 0.3446×500

$= 172.3 \approx 172$

(ii) $P(30,000 < X < 40,000)$

$F(40,000) - F(30,000)$

$\Phi\left(\frac{40,000 - 30,000}{10,000}\right) - \Phi\left(\frac{30,000 - 30,000}{10,000}\right)$

$= \Phi(1) - \Phi(0)$

$= 0.6554 - 0.2743$

$= 0.3811 = 38.11\%$

No. of Builders = 0.3811×500

$P(X > 30,000)$

$1 - P(X \leq 30,000)$

$1 - F(30,000)$

$1 - \Phi\left(\frac{30,000 - 30,000}{10,000}\right)$

$1 - \Phi(0) = 1 - 0.5 = 0.5$

$= 0.5$

$= 0.2557$

JOINT DISTRIBUTION

Joint Probability:

Let $X = (x_1, x_2, x_3, \dots, x_m)$ & $Y = (y_1, y_2, \dots, y_n)$ be two discrete random variables, then the probability of ordered pair (x_i, y_j)

$P(x_i, y_j) \quad i = 1, 2, 3, \dots, m, j = 1, 2, \dots, n$ is called joint probability distribution of X & Y for example

Let $X = \{H, T\}$

$Y = \{1, 2, 3, 4, 5, 6\}$

$X \backslash Y$	1	2	3	4	5	6
H	(H,1)	(H,2)	(H,3)	(H,4)	(H,5)	(H,6)
T	(T,1)	(T,2)	(T,3)	(T,4)	(T,5)	(T,6)

$P(H,1)$

$P(H) \cdot P(1)$

$\frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$

⇒ joint probability mass function (joint pmf)

Let $X = (x_1, x_2, \dots, x_m)$ & $Y = (y_1, y_2, \dots, y_n)$ be two discrete random variables, then joint prob. distribution $P(x_i, y_j) \quad i = 1, 2, \dots, m, j = 1, 2, \dots, n$ is called joint pmf if

(i) $P(x_i, y_j) \geq 0$ for all $(x_i, y_j) \in S$

(ii) $\sum_{i=1}^m \sum_{j=1}^n P(x_i, y_j) = 1$

(iii) probability of any event E is defined by

$$P[(x_i, y_j) \in E] = \sum_{(x_i, y_j) \in E} P(x_i, y_j)$$

⇒ Marginal & Conditional pmf.

Let $P(x_i, y_j) \quad i = 1, 2, \dots, m, j = 1, 2, \dots, n$ be joint pmf of two discrete random variables X & Y then marginal pmf of X is defined by

$$P(X = x_i) = \sum_{j=1}^n P(x_i, y_j)$$

or,

$$P(X) = \sum_j P(x_i, y_j)$$

and joint pdf of X is defined by

$$p(x, y_j) = \sum_{i=1}^m p(x_i, y_j)$$

or, $p(y) = \sum_{i=1}^m p(x_i, y)$

Conditional probability mass function of $X = x_i$ given $Y = y_j$ is given defined by

$$p(x = x_i / Y = y_j) = \frac{p(x = x_i \cap Y = y_j)}{p(Y = y_j)}$$

Joint pdf of random variables X & Y is called joint probability distribution function $f(x, y)$ if

(i) $f(x, y) \geq 0$ for all $x \in X$ or $y \in Y$

(ii) $\iint f(x, y) dx dy = 1$

Let A be event in R given by

$$p[(x, y) \in A] = \iint_A f(x, y) dx dy$$

Marginal & Conditional pdf of X & Y

Let $f(x, y)$ be joint pdf of continuous random variable X and Y . Then

marginal pdf of X is defined by

$$f(x) = \int_{-\infty}^{\infty} f(x, y) dy, -\infty < x < \infty$$

Similarly marginal pdf of Y is defined by

$$f(y) = \int_{-\infty}^{\infty} f(x, y) dx, -\infty < y < \infty$$

Central Limit Theorem

If X_i ($i = 1, 2, \dots, n$) be independent random variables from normal population such that $E(X_i) = \mu_i$ & $V(X_i) = \sigma_i^2$ ($i = 1, 2, \dots, n$) then

it can be proved that under certain general condition, random variable

$T_n = X_1 + X_2 + \dots + X_n$ is asymptotically normal distribution with mean $\mu_{T_n} = \sum_{i=1}^n \mu_i$ & variance $\sum_{i=1}^n \sigma_i^2$

(6) Cr

X \ Y	1	2	3	4	5	6
1	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$
0	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$

$$C_{\text{coin}} = \{1, 0\} = 1/2$$

$$D_{\text{die}} = \{1, 2, 3, 4, 5, 6\} = 1/6$$

(i) $p(X+Y=5) = p(0,5) + p(1,4) = \frac{1}{12} + \frac{1}{12} = \frac{2}{12} = 1/6$

(ii) $p(X+Y=5/X=0) = \frac{p(X+Y=5 \cap X=0)}{p(X=0)} = \frac{p(0,5)}{p(X=0)} = \frac{1/12}{1/6} = 1/2$

(iii) $p(X=Y) = p(1,1) = 1/12$

(7) Given

X \ Y	1	2	3	
1	$0.1/1.6$	$0.2/1.6$	$0.3/1.6$	$p(Y=1) = 0.6/1.6$
2	$0.2/1.6$	$0.3/1.6$	$0.4/1.6$	$p(Y=2) = 0.9/1.6$
3	$0.3/1.6$	$0.4/1.6$	$0.5/1.6$	$p(Y=3) = 1.2/1.6$
	$p(X=1) = 0.6/1.6$	$p(X=2) = 0.9/1.6$	$p(X=3) = 1.2/1.6$	1

(i) Marginal density of X

$$p(x) = p(x=1) + p(x=2) + p(x=3) = 0.6 + 0.9 + 1.2 = 2.7$$

$$p(x=1) = \sum_{j=1}^3 p(x=1, y_j) = p(1,1) + p(1,2) + p(1,3)$$

for $x=1$

$$p(x=1) = p(1,1) + p(1,2) + p(1,3)$$

$$= \frac{0.1}{1.6} + \frac{0.2}{1.6} + \frac{0.3}{1.6} = \frac{0.6}{1.6} = \frac{3}{8}$$

$$p(x=2) = p(2,1) + p(2,2) + p(2,3)$$

$$= \frac{0.2}{1.6} + \frac{0.3}{1.6} + \frac{0.4}{1.6} = \frac{0.9}{1.6} = 0.5625$$

for $x=3$

$$p(x=3) = p(3,1) + p(3,2) + p(3,3)$$

$$= \left(\frac{0.3}{1.6} + \frac{0.4}{1.6} + \frac{0.5}{1.6} \right) = \frac{1.2}{1.6} = 0.75$$

(ii) The conditional probability distribution of X given $Y=1$

$$p(X/Y=1) = \frac{p(X \cap Y=1)}{p(Y=1)}$$

for $x=1$

$$\frac{p(X=1 \cap Y=1)}{p(Y=1)} = \frac{p(1,1)}{p(Y=1)} = \frac{0.1}{1.6} \times \frac{1.6}{0.6} = \frac{1}{6}$$

for $n=2$

$$P(X=2 \cap Y=1) = \frac{P(2,1)}{P(Y=1)} = \frac{0.2}{1.6} \times \frac{1.6}{0.5} = \frac{2}{5} R$$

for $n=3$

$$P(X=3 \cap Y=1) = \frac{P(3,1)}{P(Y=1)} = \frac{0.2}{1.6} \times \frac{1.6}{0.5} = \frac{2}{5} R$$

$$E(X) = \sum_{n=1}^3 n P(X=n)$$

$$= 1 P(X=1) + 2 P(X=2) + 3 P(X=3)$$

$$= \frac{0.6}{1.6} + 2 \times \frac{0.6}{1.6} + 3 \times \frac{0.4}{1.6}$$

$$= \frac{0.6 + 1.2 + 1.2}{1.6} = \frac{3}{2} = 1.5$$

$$E(Y) = \sum_{n=1}^3 n P(Y=n)$$

$$= 1 P(Y=1) + 2 P(Y=2) + 3 P(Y=3)$$

$$= 1 \times \frac{0.5}{1.6} + 2 \times \frac{0.6}{1.6} + 3 \times \frac{0.5}{1.6}$$

$$= \frac{0.5 + 1.2 + 1.5}{1.6} = \frac{3.2}{1.6} = 2$$

$$V(X) = \sum_{n=1}^3 n^2 P(X=n) - \{E(X)\}^2$$

$$= 1^2 P(X=1) + 2^2 P(X=2) + 3^2 P(X=3) - 1.5^2$$

$$= \frac{0.6}{1.6} + 4 \times \frac{0.6}{1.6} + 9 \times \frac{0.4}{1.6} - \frac{2.25}{1.6} = \frac{0.6 + 2.4 + 3.6 - 2.25}{1.6} = \frac{4.35}{1.6}$$

Covariance let X and Y

$$Cov(X, Y) = E(XY) - E(X) \cdot E(Y)$$

$$E(XY) = \sum_{n=1}^3 \sum_{m=1}^3 xy P(n, m)$$

$$= 1 \times 1 P(1,1) + 2 \times 1 P(2,1) + 3 \times 1 P(3,1) + 1 \times 2 P(1,2) +$$

$$+ 1 \times 3 P(1,3) + 2 \times 2 P(2,2) + 2 \times 3 P(2,3) + 3 \times 2 P(3,2) + 3 \times 3 P(3,3)$$

$$= 1 \times \frac{0.1}{1.6} + 2 \times \frac{0.2}{1.6} + 3 \times \frac{0.2}{1.6} + 1 \times \frac{0.2}{1.6} + 1 \times \frac{0.3}{1.6} + 2 \times \frac{0.2}{1.6} + 2 \times \frac{0.3}{1.6} + 3 \times \frac{0.2}{1.6} + 3 \times \frac{0.3}{1.6} + 3 \times \frac{0.4}{1.6}$$

$$E(XY) = \frac{0.1}{1.6} + \frac{0.2}{1.6} + \frac{0.2}{1.6} + \frac{0.2}{1.6} + \frac{0.3}{1.6} + \frac{0.2}{1.6} + \frac{0.3}{1.6} + \frac{0.3}{1.6} + \frac{0.3}{1.6} + \frac{0.4}{1.6} = \frac{0.1 + 0.2 + 0.2 + 0.2 + 0.3 + 0.2 + 0.3 + 0.3 + 0.4}{1.6} = \frac{2.7}{1.6} = 1.6875$$

$$Cov(X, Y) = E(XY) - E(X) \cdot E(Y) = 1.6875 - 1.5 \times 2 = -0.1875$$

$$(iv) E(Y|X) = \sum_{m=1}^3 \sum_{n=1}^3 P(n, m)$$

$$= P(1,1) + P(1,2) + P(1,3)$$

$$= \frac{0.1}{1.6} + \frac{0.2}{1.6} + \frac{0.2}{1.6} = \frac{0.5}{1.6} = 0.3125$$

$$F(2,2) = \sum_{n=1}^2 \sum_{m=2}^3 P(n, m)$$

$$= P(1,2) + P(2,2) + P(3,2)$$

$$= \frac{0.1}{1.6} + \frac{0.2}{1.6} + \frac{0.2}{1.6} = \frac{0.5}{1.6} = 0.3125$$

$$F(3,3) = \sum_{n=1}^3 \sum_{m=1}^3 P(n, m)$$

$$= P(1,1) + P(1,2) + P(1,3) + P(2,1) + P(2,2) + P(2,3) + P(3,1) + P(3,2) + P(3,3)$$

$$= \frac{0.1}{1.6} + \frac{0.2}{1.6} + \frac{0.2}{1.6} + \frac{0.2}{1.6} + \frac{0.2}{1.6} + \frac{0.3}{1.6} + \frac{0.2}{1.6} + \frac{0.2}{1.6} + \frac{0.4}{1.6}$$

$$= \frac{0.1 + 0.2 + 0.2 + 0.2 + 0.3 + 0.2 + 0.2 + 0.4}{1.6} = \frac{2.8}{1.6} = 1.75$$

$X \backslash Y$	0	1	2
0	0.15	0.1	0.2
1	0.65	0.3	0.2

marginal distribution of X

$$P(X=0) = \sum_{n=0}^2 P(n, y)$$

$$= 0.15 + 0.65 = 0.8$$

$$= P(n, 0) + P(n, 1)$$

$$at n=0$$

$$P(n=0) = P(0,0) + P(0,1) = 0.15 + 0.1 = 0.25$$

$$at n=1$$

$$P(n=1) = P(1,0) + P(1,1) = 0.1 + 0.2 = 0.3$$

$$at n=2$$

$$P(n=2) = P(2,0) + P(2,1) = 0.2 + 0.2 = 0.4$$

$$Similarly,$$

$$\text{marginal distribution of } Y$$

$$P(Y=0) = \sum_{n=0}^2 P(n, 0)$$

$$P(Y=0) = 0.15 + 0.65 = 0.8$$

$$P(Y=1) = 0.1 + 0.3 = 0.4$$

$$P(Y=2) = 0.2 + 0.2 = 0.4$$

we have,

$$P(1,1) = 0.3$$

$$P(X=1) = 0.4, P(Y=1) = 0.4$$

$$P(n=1) \cdot P(Y=1) = 0.4 \times 0.4 = 0.16$$

$$also P(1,1) = P(X=1 \cap Y=1)$$

$$\text{also } P(X=2 \cap Y=1) = 0.2$$

$$P(X=2) = 0.4, P(Y=1) = 0.4$$

$$P(X=2) \cdot P(Y=1) = 0.4 \times 0.4 = 0.16$$

$$P(X=2) \cdot P(Y=1) = 0.4 \times 0.4 = 0.16$$

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$$P(X=2) \cdot P(Y=1) = 0.4 \times 0.4 = 0.16$$

X \ Y	0	1
0	0.28	0.37
1	0.22	0.13

Find marginal distribution of X

$$P(X=x) = \sum_{y=0}^1 P(x,y)$$

$$= P(x,0) + P(x,1)$$

For $x=0$

$$P(x=0) = P(0,0) + P(0,1) = 0.28 + 0.22 = 0.50$$

For $x=1$

$$P(x=1) = P(1,0) + P(1,1) = 0.37 + 0.13 = 0.50$$

X	0	1
0	0.50	0.37
1	0.22	0.13

Find the marginal distribution of Y

$$P(Y=y) = \sum_{x=0}^1 P(x,y)$$

$$= P(0,y) + P(1,y)$$

For $y=0$

$$P(y=0) = P(0,0) + P(1,0) = 0.28 + 0.37 = 0.65$$

For $y=1$

$$P(y=1) = P(0,1) + P(1,1) = 0.22 + 0.13 = 0.35$$

X \ Y	0	1
0	1/5	4/15
1	2/5	1/5
2	1/5	2/5

Find the value of A

$$\text{Total probability} \sum_{x=0}^2 \sum_{y=0}^1 P(x,y) = 1$$

$$P(0,0) + P(0,1) + P(1,0) + P(1,1) + P(2,0) + P(2,1) + P(2,2) + P(0,2) = 1$$

$$\frac{1}{5} + \frac{2}{5} + \frac{1}{5} + \frac{2}{5} + \frac{1}{5} + \frac{2}{5} + \frac{2}{5} + \frac{2}{5} = 1$$

$$\frac{11}{5} + A = 1 \Rightarrow A = 1 - \frac{11}{5} = \frac{5-11}{5} = -\frac{6}{5}$$

(ii) If $f(0,0)=0$, $f(1,0)=\frac{1}{4}$ and $f(2,0)=\frac{3}{4}$. Find $f_{X/Y}(x-1/y=0)$

$$f_{X/Y}(x-1/y=0) = \frac{P(X=1 \cap Y=0)}{P(Y=0)} = \frac{P(1,0)}{P(Y=0)} = \frac{1/15}{4/15} = \frac{1}{4}$$

$$(ii) f_{X/Y}(x-1/y=0) = \frac{P(X=-1 \cap Y=0)}{P(Y=0)} = \frac{P(-1,0)}{P(Y=0)} = \frac{1/15}{4/15} = \frac{1}{4}$$

$$(12) f(x,y) = \frac{1}{24}(2x+y)$$

(i) X \ Y	0	1	2
0	0	2/24	4/24
1	1/24	3/24	5/24
2	2/24	4/24	6/24

$$(ii) P(X/k=x) = \frac{P(Y \cap X=x)}{P(X=x)} = \frac{P(Y,2)}{P(X=2)}$$

$$P(X=0)$$

(13) Marginal pdf of X

$$f(x) = \int_{-\infty}^{\infty} f(x,y) dy$$

$$= \int_0^{\infty} \frac{2(1+y)^{-4}}{2(1+y)^4} dy = \frac{2}{2(1+y)^4} \int_0^{\infty} (1+y)^{-4} dy$$

$$= \frac{2}{2(1+y)^4} \left[\int_0^{\infty} (1+y)^{-4} dy + \int_0^{\infty} \frac{(y+1)^{-3}}{(1+y)^4} dy \right]$$

$$= \frac{2}{2(1+y)^4} \left[\int_0^{\infty} (1+y)^{-4} dy + \int_0^{\infty} (1+y)^{-3} dy - \int_0^{\infty} (1+y)^{-4} dy \right]$$

$$= \frac{2}{2(1+y)^4} \left[\int_0^{\infty} (1+y)^{-3} dy + \int_0^{\infty} (1+y)^{-2} dy - \int_0^{\infty} (1+y)^{-3} dy \right]$$

$$= \frac{2}{2(1+y)^4} \left[\int_0^{\infty} (1+y)^{-2} dy + \int_0^{\infty} (1+y)^{-1} dy - \int_0^{\infty} (1+y)^{-2} dy \right] = \frac{2}{2(1+y)^4} \left[\frac{(1+y)^{-1}}{-1} + \frac{(1+y)^0}{0} - \frac{(1+y)^{-1}}{-1} \right]$$

$$\frac{0}{2(1+y)} \left[\frac{-2(1+y) + 5 + 3}{6} \right]$$

$$= \frac{-2}{2(1+y)} \times \left(\frac{-2 - 2x + 5 + 3}{6} \right)$$

$$= \frac{-2}{4} \frac{3-2x}{(1+y)^4}$$

Marginal pdf of y

$$f(y) = \int_0^1 f(x,y) dx$$

$$= \int_0^1 \frac{3(1+x+y)}{2(1+y)^4} dx = \frac{3}{4} \frac{3-2y}{(1+y)^4}$$

Conditional distribution of y for x=2

$$p(y/x=2) = \frac{f(2,y)}{f(2)} = \frac{\frac{3(1+2+y)}{2(1+y)^4} \times \frac{2}{3}}{\frac{3(1+2+y)}{2(1+y)^4} \times \frac{2}{3}} = \frac{6(1+y)}{(2-2y)(1+y)^4}$$

(15) $f(x,y) = e^{-(x+y)}, x \geq 0, y \geq 0$

So marginal pdf of x

$$f(x) = \int_0^\infty f(x,y) dy = \int_0^\infty e^{-(x+y)} dy = \left[\frac{e^{-(x+y)}}{-1} \right]_0^\infty = 0 + e^{-x}$$

Marginal pdf of y

$$f(y) = \int_0^\infty f(x,y) dx = \int_0^\infty e^{-(x+y)} dx = \left[\frac{e^{-(x+y)}}{-1} \right]_0^\infty = 0 + e^{-y}$$

nm

$$f(x) \cdot f(y) = e^{-x} \cdot e^{-y} = e^{-(x+y)} = f(x,y)$$

∴ x and y are independent

(16)

	0	1	
0	$\frac{2}{9}$	$\frac{1}{9}$	$P(X=0) = \frac{3}{9}$
1	$\frac{1}{9}$	$\frac{5}{9}$	$P(Y=1) = \frac{6}{9}$
	$P(X=0) = \frac{3}{9}$	$P(X=1) = \frac{6}{9}$	

Marginal distribution of x

$$P(X=x) = \sum_{y=0}^1 P(x,y)$$

$$P = P(X=0) + P(X=1)$$

at $x=0$

$$P(X=0) = P(0,0) + P(0,1)$$

$$= \frac{2}{9} + \frac{1}{9} = \frac{3}{9}$$

at $x=1$

$$P(X=1) = P(1,0) + P(1,1)$$

$$= \frac{1}{9} + \frac{5}{9} = \frac{6}{9}$$

Marginal distribution of y

$$P(Y=y) = \sum_{x=0}^1 P(x,y)$$

$$= P(0,y) + P(1,y)$$

at $y=0$

$$P(Y=0) = P(0,0) + P(1,0)$$

$$= \frac{2}{9} + \frac{1}{9} = \frac{3}{9}$$

at $y=1$

$$P(Y=1) = P(0,1) + P(1,1)$$

$$= \left(\frac{1}{9} + \frac{5}{9} \right) = \frac{6}{9}$$

nmw

$$P(X=0 | Y=1) = \frac{1}{6}$$

$$P(X=0) = \frac{3}{9}, P(Y=1) = \frac{6}{9}$$

$$P(X=0) \cdot P(Y=1) = \frac{1}{9} \times \frac{6}{9} = \frac{2}{9} \neq \frac{1}{6}$$

also,

$$P(X=1 | Y=1) = \frac{5}{6}$$

$$P(X=1) = \frac{6}{9}, P(Y=1) = \frac{6}{9}$$

$$P(X=1) \cdot P(Y=1) = \frac{5}{9} \times \frac{6}{9} = \frac{10}{9} \neq \frac{5}{6}$$

∴ x & y are dependent

(17) $f(x,y) = (1-e^{-x})(1-e^{-y})$

(i) Marginal pdf of x

$$f(x) = \int_0^\infty f(x,y) dy = \int_0^\infty (1-e^{-x})(1-e^{-y}) dy = \int_0^\infty (1-e^{-x} - e^{-x}e^{-y}) dy$$

$$= \left[y - e^{-x}y + e^{-x} - e^{-x}e^{-y} \right]_0^\infty = 0 + 0 + 0 - 1 + e^{-x} = e^{-x} - 1$$

y=0

$$\text{un-normalized pdf of } Y \\ (11) \Rightarrow \int_{-\infty}^{\infty} f(x) dx = \int_0^{\infty} (1-e^{-x})(1-e^{-x}) dx = (e^2 - 1)$$

original pdf of Y

$$f_Y(y) = \int_0^y 10e^{-x} dx = \int_0^y (1 - e^{-x}) dx = (e^{-y} - 1)$$

$\therefore X$ & Y are independent.

$\therefore X$ & Y are independent.

$$= \int_0^1 \int_0^2 f(x,y) dx dy = \int_0^2 \int_0^1 (1-e^{-x})(1-e^{-y}) dy dx$$

$$= \int_0^1 \int_0^2 f(x,y) dx dy = \int_0^2 \int_0^1 (1-e^{-x})(1-e^{-y}) dy dx$$

$$\int_0^1 \int_0^2 (1 - e^{-x} + e^{-y} + e^{-x-y}) dx dy = \int_0^2 [x - e^{-x} + 2e^{-y} - e^{-x-y}]_0^2 dy$$

$$\int (2 + e^{-y} - 7e^{-y} - e^{-3}e^{-y} - 1 - e^{-1} + e^{-y} + e^{-1}e^{-y}) dy$$

$$\{2y + e^{-1} + 3e^{-2} + e^{-3}e^{-2} - e^{-1}y - e^{-2} - e^{-1}e^{-2}\}_1^2$$

$$12 + 2e^{-1/2} + 3e^{-2} + e^{-5} - 2e^{-1} - e^{-1/2} - e^{-1/3} - e^{-1} - 3e^{-1} - e^{-1} + e^{-1} + e^{-1/2}$$

$$1 + 3e^{-2} + e^{-5} - 4e^{-1} - e^{-4}$$

$$\sin(\pi/2) = 2, \quad 0 < x < 1, \quad 0 < y < x$$

original density $A \cdot x$

$$f(x) = \int_0^x \log y \, dy = -2x$$

original drawing of 4

$$f(x) = \int_0^1 \ln(xy) \, dy = \int_0^1 2 \, dy = 2$$

conditional density function $P(Y/Z=z) = \frac{P(Y \cap X=z)}{P(X=z)} = \frac{P(Y=z)}{P(Y=1)} = \frac{1}{2} = \frac{1}{4} \times 2$

$$P(X/Y=3) = \frac{P(X \cap Y=3)}{P(Y=3)} = \frac{1/9}{1/3} = \frac{2}{2} = 1$$

(19) Joint pdf of x & y is

$$\rho(x, y) = \frac{x+y}{2!}, \quad x = \{1, 2\}, y = \{1, 2\}$$

$X \backslash Y$	1	2	3	
1	$\frac{2}{9}$	$\frac{3}{9}$	$\frac{4}{9}$	$P(Y=1) = \frac{9}{9}$
2	$\frac{3}{9}$	$\frac{4}{9}$	$\frac{5}{9}$	$P(Y=2) = \frac{10}{9}$
	$P(X=1) = \frac{9}{9}$	$P(X=2) = \frac{10}{9}$	$P(X=3) = \frac{11}{9}$	

Marginal pdf of $x = \sum_{j=1}^2 P(x, y)$

$$P(X=1) = P(X_1=1) + P(X_2=1)$$

at $x=1$,
 $P(x=1) = P(1,1) + P(1,2) = \frac{2}{21} + \frac{3}{21} = \frac{5}{21}$

$$\text{At } r=2, \quad f(r=2) = P(2,1) + P(2,2) = \frac{3}{21} + \frac{4}{21} = \frac{7}{21}$$

$$\text{for } n=3, \quad P(n=3) = P(3,1) + P(3,2) = \frac{4}{21} + \frac{5}{21} = \frac{9}{21}$$

$$\text{Marginal p.d. of } Y = \sum_{x=1}^3 P(x, y)$$

$$P(Y=j) = P(1,j) + P(2,j) + P(3,j)$$

$$P(Y=1) = P(1,1) + P(2,1) + P(3,1) = \frac{1}{21} + \frac{1}{21} + \frac{1}{21} = \frac{3}{21}$$

$$\begin{aligned} \alpha + \beta = 2, \quad p(x=2) &= p(1,2) + p(2,2) + p(3,2) \\ &= \frac{3}{21} + \frac{4}{21} + \frac{5}{21} \\ &= \frac{12}{21} \quad \underline{B} \end{aligned}$$

(20) $f(x, y) = A(x+y)$, $0 < x < 2$, $0 < y < 1$

(i) determine A

marginal pdf & total probability $= \iint f(x,y) dx dy = 1$

$$\int_0^1 \int_0^1 A(x,y) dx dy = 1$$

$$\frac{1}{3} \int_0^1 \left[\frac{x^2}{2} + xy \right]_0^2 dy = 0.1$$

$$\text{or, } A \int_0^1 (2+2y) dy = 1$$

$$\text{or } A \left[2y + \frac{2y^2}{2} \right]_0^1 = 1$$

$$\text{or } A(2+1) = 1$$

$$A = \frac{1}{3}$$

Find Marginal pdf of x and y

Marginal pdf of X

$$f(x) = \int_0^1 f(x,y) dy = \int_0^1 A(x+y) dy = \frac{1}{3} \int_0^1 (x+y) dy$$

$$= \frac{1}{3} \left[xy + \frac{y^2}{2} \right]_0^1 = \frac{1}{3} \left[x + \frac{1}{2} \right] = \frac{x+1}{6}$$

Marginal pdf of Y

$$f(y) = \int_0^2 A(x+y) dx = \frac{1}{3} \int_0^2 \left[\frac{x^2}{2} + xy \right]_0^2 = \frac{2+2y}{3}$$

$$\therefore \frac{x}{y=y} = \frac{P(X=x \cap Y=y)}{P(Y=y)} = \frac{f(x,y)}{f(y)} = \frac{\frac{1}{3}(x+y)}{\frac{2+2y}{3}} = \frac{x+y}{2+2y}$$

$$2) f(x,y) = e^{-(x+y)} \quad x \geq 0, y \geq 0$$

$$(i) P(X < 1) = \int_0^1 \int_0^\infty f(x,y) dx dy = \int_0^1 \int_0^\infty e^{-(x+y)} dx dy = \int_0^1 \left[-e^{-(x+y)} \right]_0^\infty dy$$

$$= - \int_0^1 (e^{-(1+y)} - e^{-y}) dy = - \left[\frac{e^{-(1+y)}}{-1} - \frac{e^{-y}}{-1} \right]_0^1$$

$$= - \left[-e^{-1-y} + e^{-1} - e^{-1} + e^0 \right] = - \left[e^{-1} - 1 \right] = 1 - e^{-1}$$

(ii) Marginal pdf of X

$$f(x) = \int_0^\infty f(x,y) dy = \int_0^\infty e^{-(x+y)} dy = \left[\frac{e^{-(x+y)}}{-1} \right]_0^\infty$$

$$= - \left[e^{-\infty} - e^{-x} \right] = e^{-x}$$

Marginal pdf of Y

$$f(y) = \int_0^\infty f(x,y) dx = \int_0^\infty e^{-(x+y)} dx = \left[\frac{e^{-(x+y)}}{-1} \right]_0^\infty$$

$$= - \left[e^{-\infty} - e^{-y} \right] = e^{-y}$$

$$(iii) f(x,y) = e^{-x} \cdot e^{-y} = e^{-(x+y)} = f(x)f(y)$$

$\therefore X$ & Y are independent.

$$(22) f(x,y) = 4xy e^{-(x^2+y^2)} \quad x > 0, y > 0$$

(i) Marginal pdf of X

$$f(x) = \int_0^\infty f(x,y) dy = \int_0^\infty 4xy e^{-(x^2+y^2)} dy = -2x \left[e^{-(x^2+y^2)} \right]_0^\infty$$

$$= -2x \left[e^{-\infty} - e^{-x^2} \right] = 2x e^{-x^2}$$

Marginal pdf of Y

$$f(y) = \int_0^\infty f(x,y) dx = \int_0^\infty 4xy e^{-(x^2+y^2)} dx = -2y \left[e^{-(x^2+y^2)} \right]_0^\infty$$

$$= -2y \left[e^{-\infty} - e^{-y^2} \right] = 2y e^{-y^2}$$

hence

$$f(x)f(y) = 2x e^{-x^2} \cdot 2y e^{-y^2} = 4xy e^{-(x^2+y^2)} = f(x,y)$$

$\therefore X$ & Y are independent.

(ii) find the conditional density of X given $Y=y$

$$P(X/Y=y) = \frac{P(X \cap Y)}{P(Y=y)} = \frac{f(x,y)}{f(y)} = \frac{4xy e^{-(x^2+y^2)}}{2y e^{-y^2}} = 2x e^{-x^2}$$

(23) Find Marginal density of X

$$f(x) = \int_0^\infty f(x,y) dy = \int_0^\infty \frac{1}{2} y^2 dy = \frac{1}{2} \left[\frac{y^3}{3} \right]_0^\infty = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

Marginal density of Y

$$f(y) = \int_0^\infty f(x,y) dx = \int_0^\infty \frac{1}{2} y^2 dx = \frac{1}{2} y^2 \int_0^\infty 1 dx = \frac{1}{2} y^2 \times \infty = \infty$$

hence

$$f(x)f(y) = \frac{1}{6} \times \infty = \frac{1}{6} \times \infty = f(x,y)$$

$\therefore X$ & Y are independent.

$$R(x) = 5 \text{ ohm / each resistor}$$

$$R^2(x) = 0.2 \text{ ohm}^2 / \text{Each resistor}$$

for 40 resistors

$$\text{Mean } E(x) \text{ or } \mu_1 = 40 \times 5 = 200 \text{ ohms}$$

$$\text{variance } E(x^2) \text{ or } \sigma^2 = 0.2 \times 40 = 8 \text{ ohm}^2$$

$$\text{S.d. } \sigma = 2$$

$$P(X > 98 \text{ ohms}) \approx$$

$$P(X > 98) = 1 - P(X \leq 98)$$

$$= 1 - F(98)$$

$$= 1 - \Phi\left(\frac{98 - 200}{2}\right) \quad \text{Since } F(x) = \Phi\left(\frac{x - \mu}{\sigma}\right)$$

$$= 1 - \Phi(-51)$$

$$= 1 - 0.00001557$$

$$= 0.99998443$$

$$\text{Load limit } (T_0) = 1000 \text{ lbs.}$$

$$\text{Load factor } (C_1) = 1.0$$

$$\text{mean } (C_2) = 175 \text{ lbs.}$$

$$\text{S.d. } (C_3) = 20 \text{ lbs.}$$

$$P(C_1 \leq 1.0) = P\left[\frac{T_0 - E(T_0)}{\sqrt{V(T_0)}} \leq \frac{1000 - E(T_0)}{\sqrt{V(T_0)}}\right]$$

$$= P\left[Z \leq \frac{1000 - 725}{\sqrt{70^2}}\right]$$

$$= P\left[Z \leq \frac{1000 - 100 \times 175}{\sqrt{100 \times 20^2}}\right]$$

$$= P[Z \leq 1.67]$$

$$= 0.5 - P(1.67 < Z < 0)$$

$$= 0.5 - 0.4525 \quad [\text{using table, 8}]$$

$$= 0.0475$$



THEORY OF ESTIMATION

Parameter and Statistics

Any population constant is called a parameter. for example population mean, population variance etc.

Any constant taken from sample is called statistics. for example sample mean, sample s.d. etc.

Estimate

The sample mean denoted by $\bar{x}$ is also known as estimate of population mean μ .

The methods of judgement about a population parameter is called statistical estimation. There are two types of statistical estimation.

(i) point estimation

(ii) Interval estimation

(i) point estimation

It is a single value is calculated from the sample as an estimate of an unknown population parameter. this process is known as point estimation.

(ii) Interval estimation

Jerzy Neymann first illustrated the concept of interval estimation which include an unknown population parameter (μ) with certain probability.

The procedure of finding the interval is called interval estimation and the interval where C_1 & C_2 are lower & upper interval. C_1 & C_2 are called border (i.e. $P(C_1 < \mu < C_2) = 1 - \alpha$).

⇒ Confidence interval for the mean for large sample

Let $x_1, x_2, \dots, x_n$ be random normal distribution with known & known population s.d. (σ) $C_1, C_2 = (\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}})$ where

using (1-α) prob
and Conf
Sample
1) unbiased
2) consistency
3) efficiency
4) sufficiency

$$\mu \in (\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}})$$

gt population size (N) is given

$$(c_1, c_2) = (\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}})$$

Confidence Interval for the mean (small sample, $n < 30$)

Confidence Interval

$$(c_1, c_2) = (\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}})$$

where, $\bar{x}$ = Sample mean

n = " size

s = " s.d.

degree of freedom = $n-1$

max error

$$z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$$

$$z_{\alpha/2} \sqrt{\frac{PQ}{n}}$$

Confidence Interval for population proportion ($n \geq 30$)

Sample size = n

Number of 1st term = x , no. of 2nd term = $n-x$

proportion of first term ($\hat{p}$) = $\frac{x}{n}$

proportion of second term ($\hat{q}$) = $1-\hat{p}$

Confidence interval

$$(c_1, c_2) = \hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

$$= (\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}, \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}})$$

Criteria of good estimator

A good estimator is one which is as close to the true value of the parameter as possible. The criteria of a good estimator for evaluating

the statistic are

1. Bias

for different of means

Confidence Interval for large sample

$$(c_1, c_2) = (\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

for small sample

$$(c_1, c_2) = (\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, n_1+n_2-2} s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$s = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}}$$

18

Unbiasedness:-

An estimator is unbiased if average of sample statistics of same size is equal to the parameter

Consistency:-

A statistic is said to be consistent estimator of the population parameter if as sample size increases the value of the sample statistic very close to the value of the population parameter.

Efficiency:-

An estimator with lesser variability is said to be more efficient

Sufficiency:-

A statistic is said to be a sufficient estimator of the population parameter if it uses all the information about population parameter contained in the sample.

EXERCISE-13

1. Here,

$$\text{Population mean } (\mu) = \frac{1+2+3+6+8}{5} = \frac{20}{5} = 4$$

Sample of size 3 can be selected from population size 5 in 5C3 = 10 ways

Alt way

Sample	Observation	Sample mean ($\bar{x}$)	$P(\bar{x})$	$\bar{x} \cdot P(\bar{x})$
1	1, 2, 3	6/3	1/10	6/10
2	1, 2, 6	3/3	1/10	3/10
3	1, 2, 8	11/3	1/10	11/10
4	1, 3, 6	10/3	1/10	10/10
5	1, 3, 8	12/3	1/10	12/10
6	1, 6, 8	15/3	1/10	15/10
7	2, 3, 6	11/3	1/10	11/10
8	2, 3, 8	13/3	1/10	13/10
9	3, 6, 8	17/3	1/10	17/10
10	3, 6, 8	17/3	1/10	17/10

Hence Average of Sample mean $= E \bar{x} P(\bar{x})$
 $= 4$

Hence, the sample mean of size 3 is unbiased estimator of the population mean.

population mean $(\bar{x}) = \frac{2+4+5+6+7+8}{6} = 3\frac{2}{3} = 16/3$

sample of size 2 can be selected from population size 6 in ${}^6P_2 = 15$ diff ways

Sample	Observation	Sample mean $(\bar{x})$	$P(\bar{x})$	$\bar{x} P(\bar{x})$
1	3, 4	6/2	1/15	6/30
2	3, 5	7/2	1/15	7/30
3	3, 6	8/2	1/15	8/30
4	3, 7	9/2	1/15	9/30
5	3, 8	10/2	1/15	10/30
6	4, 5	9/2	1/15	9/30
7	4, 6	10/2	1/15	10/30
8	4, 7	11/2	1/15	11/30
9	4, 8	12/2	1/15	12/30
10	5, 6	11/2	1/15	11/30
11	5, 7	12/2	1/15	12/30
12	5, 8	13/2	1/15	13/30
13	6, 7	13/2	1/15	13/30
14	6, 8	14/2	1/15	14/30
15	7, 8	15/2	1/15	15/30

$E \bar{x} P(\bar{x}) = 140/30 = 14/3$

Average of Sample mean $E \bar{x} P(\bar{x}) = 14/3$

Hence the sample mean of size 2 is unbiased estimator of population mean.

③ ^{iv} population mean $(\bar{x}) = \frac{3+6+9+15+27}{5} = \frac{60}{5} = 12$

Sample of size 3 can be selected from population size 5 in 5P_3 ways

Sample	Observation	Sample mean $\bar{x}$	$P(\bar{x})$	$\bar{x} P(\bar{x})$
1	3, 6, 9	18/3	1/10	18/10
2	3, 6, 15	24/3	1/10	24/10
3	3, 9, 27	39/3	1/10	39/10
4	3, 9, 15	27/3	1/10	27/10
5	3, 15, 27	45/3	1/10	45/10
6	3, 15, 27	45/3	1/10	45/10
7	6, 9, 15	30/3	1/10	30/10
8	6, 9, 27	42/3	1/10	42/10
9	6, 15, 27	48/3	1/10	48/10
10	9, 15, 27	51/3	1/10	51/10

Average of Sample mean $E \bar{x} P(\bar{x}) = 12$

Hence the sample mean of size 3 is unbiased estimator of the population mean.

④ ^{iv} population mean $(\bar{x}) = \frac{7+9+2}{3} = \frac{18}{3} = 6$

Sample of size 2 can be selected from the population size 3 in ${}^3P_2 = 3$ ways

Sample	Observation	Sample mean $\bar{x}$	$P(\bar{x})$	$\bar{x} P(\bar{x})$
1	7, 9	11/2	1/3	11/6
2	7, 2	9/2	1/3	9/6
3	9, 2	6/2	1/3	6/6

$E \bar{x} P(\bar{x}) = 26/6 = 13/3$

Average of Sample mean $E \bar{x} P(\bar{x}) = 13/3$

Hence the sample mean of size 2 is unbiased estimator of the population mean.

Population mean ($\bar{x}$) = $\frac{12+9+7+11+8}{5} = 47/5$

Sample of size 3 can be selected from population size 5 in $5C_3 = 10$ different ways

Sample	Observation	Sample mean ($\bar{x}$)	$P(\bar{x})$	$\bar{x}P(\bar{x})$
1	12, 9, 7	28/3	1/10	28/30
2	12, 9, 11	32/3	1/10	32/30
3	12, 9, 8	29/3	1/10	29/30
4	12, 7, 11	30/3	1/10	30/30
5	12, 7, 8	27/3	1/10	27/30
6	12, 11, 8	31/3	1/10	31/30
7	9, 7, 11	27/3	1/10	27/30
8	9, 7, 8	24/3	1/10	24/30
9	9, 11, 8	28/3	1/10	28/30
10	7, 11, 8	26/3	1/10	26/30

$\sum \bar{x}P(\bar{x}) = \frac{282}{30} = 9.4$

Average of sample mean $E\bar{x}P(\bar{x}) = 9.4/5$

Hence the sample mean of size 3 is unbiased estimator of the popⁿ mean.

Sample of size (n) = 10

Population S.D. (σ) = 5.10

Sample mean ($\bar{x}$) = 21.6

$1-\alpha = 95\%$

$\alpha = 0.05$

We have,

Confidence Interval (C_1, C_2) = $\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
 $= 21.6 \pm Z_{0.05} \frac{5.10}{\sqrt{10}} = 21.6 \pm Z_{0.05} \times 0.51$
 $= 21.6 \pm 1.96 \times 0.51 = 21.6 \pm 0.9996$
 $= (20.6, 22.6)$

(11) 50%

$1-\alpha = 99\%$

Sample size (n) = 9

$\alpha = 0.01$

Max error (E) = 0.45 min

Pop S.D. (σ) = 1.4 min

We have,

Max Error = $Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

or, $0.45 = Z_{0.01} \frac{1.4}{\sqrt{9}}$

or, $0.45 \times \sqrt{9} = 2.58 \times 1.4$

$n = \left(\frac{2.58 \times 1.4}{0.45} \right)^2 = 208.74 \approx 209$

(12) Max Error (E) = 9

Sample of size $n = 64$

$1-\alpha = 0.9$

$\alpha = 0.10$

Popⁿ Variance (σ^2) = 2.56

$\sigma = 1.6$

We have,

Max Error = $Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = Z_{0.10} \frac{1.6}{\sqrt{64}} = 1.645 \times \frac{1.6}{8} = 0.329$

(13) Popⁿ S.D. (σ) = 0.75

(14) $1-\alpha = 95\% \Rightarrow \alpha = 5\% = 0.05$

Size of sample (n) = 50

Sample mean ($\bar{x}$) = 4.85

We have,

Confidence interval (C_1, C_2) = $\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 4.85 \pm Z_{0.05} \frac{0.75}{\sqrt{50}}$
 $= 4.85 \pm 0.1078$
 $= (4.642, 5.0578)$

(15) $1-\alpha = 98\% \Rightarrow \alpha = 2\% = 0.02$

Sample size (n) = 40

Sample mean ($\bar{x}$) = 4.56

We have,

$(C_1, C_2) = \bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 4.56 \pm Z_{0.02} \frac{0.75}{\sqrt{40}} = 4.56 \pm 0.2471$
 $= (4.312, 4.8071)$

$C_1 - C_2 = 0.4$
 Sample size (n) = 9
 $1 - \alpha = 95\%$
 $\alpha = 5\% = 0.05$
 S.d (σ) = 0.75

we have,
 $C_1 - C_2 = \bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} - \bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
 $C_1 - C_2 = 2 z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
 $0.4 = 2 z_{0.025} \frac{0.75}{\sqrt{n}}$
 $n = \left(\frac{2 \times 1.96 \times 0.75}{0.4} \right)^2 = 54.0221755 \approx 55$

Sample size (n) = 650
 S.d (σ) = 20
 Mean ($\bar{x}$) = 30
 $1 - \alpha = 95\%$
 $\alpha = 5\% = 0.05$

we have,
 $(C_1, C_2) = \bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 30 \pm z_{0.025} \frac{20}{\sqrt{650}}$
 $= 30 \pm 1.96 \times \frac{20}{\sqrt{650}}$
 $(C_1, C_2) = (28.152, 31.8479)$

Sample size (n) = 100
 Population (N) = 500
 Mean ($\bar{x}$) = 148.2 hr
 S.d (σ) = 24.3 hr
 $1 - \alpha = 90\%$
 $\alpha = 10\% = 0.1$

$(C_1, C_2) = \left(\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right) \sqrt{\frac{N-n}{N-1}}$
 $= 148.2 \pm z_{0.05} \frac{24.3}{\sqrt{100}} \sqrt{\frac{500-100}{500-1}}$
 $= (148.2 \pm 1.96 \times 2.2295) \quad (\text{using table page 391})$
 $= (148.2 \pm 4.3698)$
 $= 143.83, 152.569$

Sample size (n) = 10

$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{40 + 70 + 110 + 101 + 37 + 88 + 95 + 98 + 107 + 100}{10}$
 $= 97.20$

x_i	δx	$(x - \bar{x})^2$
1	70	739.84
2	170	519.84
3	110	163.84
4	101	11.44
5	23	201.64
6	88	84.64
7	95	4.84
8	91	0.64
9	107	96.04
10	100	7.84
		$\Sigma (x - \bar{x})^2 = 1533.60$

Sample standard deviation (s) = $\sqrt{\frac{\Sigma (x - \bar{x})^2}{n-1}}$
 $= \sqrt{\frac{1533.60}{10-1}} = \sqrt{170.4} = 13.05$
 we have,
 $(C_1, C_2) = \bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$
 $= 97.2 \pm t_{0.05, 9} \times \frac{13.05}{\sqrt{10}}$
 $= 97.2 \pm 2.212 \times 4.1215$
 $(C_1, C_2) = (96.99, 107.42)$

(18) $\bar{x} = \frac{2.2 + 1.8 + 3.1 + 2.0 + 2.4 + 2.1 + 2.5 + 1.2}{8} = 2.1$

x	$(x - \bar{x})^2$
2.2	0.01
1.8	0.09
3.1	1.00
2.0	0.01
2.4	0.09
2.1	0.01
2.5	0
1.2	0.81
$\Sigma (x - \bar{x})^2 = 2.02$	

Sample standard deviation (s) = $\sqrt{\frac{\Sigma (x - \bar{x})^2}{n-1}}$
 $= \sqrt{\frac{2.02}{8-1}} = 0.537$
 $1 - \alpha = 95\%$
 $\alpha = 5\% = 0.05$
 we have,
 $(C_1, C_2) = \bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$
 $= 2.1 \pm t_{0.025, 7} \times \frac{0.537}{\sqrt{8}}$
 $= 2.1 \pm 2.365 \times 0.1887$
 $= (2.1 \pm 0.449)$
 $(C_1, C_2) = (1.65, 2.55)$

(19) No. of sample (n) = 14

mean ($\bar{x}$) = 17.85
 S.d (σ) = 1.255
 $1 - \alpha = 95\% \Rightarrow \alpha = 5\%$

we have,
 $(C_1, C_2) = \bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$
 $= 17.85 \pm t_{0.025, 13} \times \frac{1.255}{\sqrt{14}}$
 $= 17.85 \pm 2.16 \times 0.335$
 $= (16.72, 18.98)$

192

(22) Size size $(n) = 81$

$$1 - \alpha = 95\% \Rightarrow \alpha = 5\% = 0.05$$

$$\text{proportion } (\hat{p}) = 1/81, \hat{q} = 1 - 1/81 = 80/81$$

we have,

$$(C_1, C_2) = \hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}} = \frac{1}{81} \pm Z_{0.05} \sqrt{\frac{1}{81} \times \frac{80}{81} \times \frac{1}{81}} = (0.0123 \pm 1.96 \times 0.024)$$

which is not exist.

(23) Sample size $(n) = 500$

No. of bad pineapples = 65

$$\text{No. of good " } (x) = 500 - 65 = 435$$

$$\text{proportion of good " } (\hat{p}) = \frac{435}{500} = 0.87$$

$$\hat{q} = 1 - 0.87 = 0.13$$

$$1 - \alpha = 95\%$$

$$\alpha = 5\% = 0.05$$

Confidence Interval

$$(C_1, C_2) = \hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

$$= 0.87 \pm Z_{0.05} \sqrt{\frac{0.87 \times 0.13}{500}}$$

$$= 0.87 \pm 1.96 \times 0.015 = 0.87 \pm 0.0294$$

$$\therefore (C_1, C_2) = (0.8406, 0.8994)$$

(24) no. of dice thrown $(n) = 9000$

obtaining 3 or 4 no. $(x) = 3240$

$$\text{proportion } (\hat{p}) = \frac{x}{n} = \frac{3240}{9000} = 0.36, \hat{q} = 1 - 0.36 = 0.64$$

$$\alpha = 5\%$$

we have,

$$(C_1, C_2) = \hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}} = 0.36 \pm Z_{0.05} \sqrt{\frac{0.36 \times 0.64}{9000}}$$

$$= 0.36 \pm 1.96 \sqrt{0.000256}$$

$$= 0.36 \pm 0.00992$$

$$\therefore (0.35, 0.36992)$$

25) 90%.

$$1 - \alpha = 90\% \Rightarrow \alpha = 10\% = 0.10$$

Total no. of sample (n) = 50 item.

no. of defective item (x) = 18

$$\text{Proportion of defective item } (\hat{p}) = \frac{x}{n} = \frac{18}{50} = 0.36$$

$$\hat{q} = 1 - \hat{p} = 1 - 0.36 = 0.64$$

we have,

$$\begin{aligned} (e_1, e_2) &= \hat{p} \pm Z_{\alpha} \sqrt{\frac{\hat{p}\hat{q}}{n}} = 0.36 \pm Z_{0.10} \sqrt{\frac{0.36 \times 0.64}{50}} \\ &= 0.36 \pm 1.645 \times 0.0678 \\ &= 0.36 \pm 0.11166 \\ &= (0.25, 0.47) \end{aligned}$$

26) 91%.

x	$(x - \bar{x})^2$	x	$(x - \bar{x})^2$
23.2	229.86	44.50	34.09
51.5	51.29	35.70	8.77
34.00	21.73	33.50	26.64
46.70	64.62	39.3	0.41
45.60	48.15	22.00	272.59
32.00	37.96	51.2	157.22
41.40	7.50	$\Sigma = 695.9$	$\Sigma (x - \bar{x})^2 = 1220.526$
37.20	2.13	$\bar{x} = 38.6611$	
42.50	14.74		
46.90	67.88		
51.5	164.84		
38.40	5.11		

$$\begin{aligned} \text{sample s.d. (s)} &= \sqrt{\frac{\Sigma (x - \bar{x})^2}{n-1}} \\ &= \sqrt{\frac{1220.526}{47}} \\ &= 8.473 \end{aligned}$$

$$\alpha = 0.02 = 0.02$$

We have,

$$\begin{aligned} \text{Confidence Interval } (C_1, C_2) &= \bar{x} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \\ &= 38.6611 - t_{0.02, 19} \times \frac{8.473}{\sqrt{20}} \\ &= 38.6611 - 2.567 \times \frac{8.473}{\sqrt{20}} \\ &= (33.5944, 43.7277) \end{aligned}$$

(2) n_1, n_2	$(x_1 - \bar{x}_1)^2$	$(x_2 - \bar{x}_2)^2$	$(x_1 - \bar{x}_1)^2$	$(x_2 - \bar{x}_2)^2$
107	413.3	12.1	25	
148	427.25	11.5	19.6	
123	18.25	11.2	2.89	
165	1419.02	15.1	4.84	
102	641.4	13.7	16	
119	69.38	12.9	0	
$\Sigma x_1 = 764$	$\Sigma (x_1 - \bar{x}_1)^2 = 2989.32$	$\Sigma x_2 = 74$	$\Sigma (x_2 - \bar{x}_2)^2 = 10.40$	
$n_1 = 6$	$s.d. = \sqrt{\frac{\Sigma (x_1 - \bar{x}_1)^2}{n_1 - 1}}$	$n_2 = 6$	$s.d. = \sqrt{\frac{\Sigma (x_2 - \bar{x}_2)^2}{n_2 - 1}}$	
$\bar{x}_1 = 127.33$	$= \sqrt{\frac{2989.32}{5}}$	$\bar{x}_2 = 12.3$	$= \sqrt{\frac{10.40}{5}}$	
	$= 24.45$		$= 1.4213$	

$$s = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{6 \times 24.45^2 + 6 \times 1.4213^2}{6 + 6 - 2}} = \sqrt{\frac{4798.87}{10}} = 21.906$$

We have,

$$\begin{aligned} (C_1, C_2) &= |\bar{x}_1 - \bar{x}_2| \pm t_{\alpha/2, n-1} s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = |127.33 - 12.3| \pm t_{0.02, 10} \times 21.906 \sqrt{\frac{1}{6} + \frac{1}{6}} \\ &= 115.03 \pm 2.228 \times 21.906 \times 0.577 \\ &= (-2.649, 23.63) \end{aligned}$$

(29) n	Type A	Type B
Sample size (n)	58	27
mean ($\bar{x}$)	70.70	76.13
s.d. (s)	1.8	2.42

$$\begin{aligned} \text{we have,} \\ \text{Confidence interval } (C_1, C_2) &= (\bar{x}_1 \pm Z_{\alpha/2} \frac{s_1}{\sqrt{n_1}}) - (\bar{x}_2 \pm t_{\alpha/2, n_2} \frac{s_2}{\sqrt{n_2}}) \\ &= (70.7 \pm 2.05 \times \frac{1.8}{\sqrt{58}}) - (76.13 \pm 2.05 \times \frac{2.42}{\sqrt{27}}) \\ &= (70.7 \pm 0.596 \times \frac{1.8}{\sqrt{58}}) - (76.13 \pm 0.953 \times \frac{2.42}{\sqrt{27}}) \\ &= (70.7 \pm 0.463) - (76.13 \pm 0.953) \\ &= (70.7 - 0.463, 70.7 + 0.463) - (76.13 - 0.953, 76.13 + 0.953) \\ &= (-4.936, -5.924) \end{aligned}$$

$$\begin{aligned} (30) \text{ Sample size } (n) &= 400 \\ \text{Proportion } (p) &= 0.12, \hat{p} = 1 - 0.12 = 0.88 \\ 1 - \alpha &= 98\% \Rightarrow \alpha = 2\% \end{aligned}$$

$$\begin{aligned} \text{we have,} \\ (C_1, C_2) &= \hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}} = 0.12 \pm 2.05 \sqrt{\frac{0.12 \times 0.88}{400}} \\ &= 0.12 \pm 2.05 \times 0.0362 \\ &= (0.12 \pm 0.0742) \\ &= (0.0458, 0.1942) \end{aligned}$$

$$\begin{aligned} (31) \text{ Sample size } (n) &= 400 \\ \text{proportion } (p) &= 1 - 0.12 = 0.88, \hat{p} = 1 - 0.88 = 0.12 \\ 1 - \alpha &= 98\% \Rightarrow \alpha = 2\% \end{aligned}$$

$$\begin{aligned} \text{we have,} \\ (C_1, C_2) &= \hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}} = 0.88 \pm 2.05 \sqrt{\frac{0.12 \times 0.88}{400}} \\ &= 0.88 \pm 0.0374 \\ &= (0.8426, 0.9174) \end{aligned}$$

Month	Surface (Cal/cm ² /day) ($\bar{x}_1$)	$(x_1 - \bar{x}_1)^2$	Surface (Cal/cm ² /day) ($\bar{x}_2$)	$(x_2 - \bar{x}_2)^2$
Jan	363	9145.06	582	5765.01
Feb	472	2495.52	492	58.69
Mar	518	4192.02	556	5268.55
Apr	521	4590.01	549	4301.18
May	621	3159.51	479	19.50
Jun	582	17889.02	422	3771.99
Jul	365	7788.02	315	28364.15
Aug	412	1701.52	414	4818.14
Sep	469	248.02	505	465.84
Oct	468	217.02	552	4703.68
Nov	371	6765.02	492	73.69
Dec	320	15190.02	502	556.17

$$\sum x_1 = 5439 \quad \sum (x_1 - \bar{x}_1)^2 = 100748.22 \quad \sum x_2 = 5801 \quad \sum (x_2 - \bar{x}_2)^2 = 55191.87$$

$$n_1 = 12$$

$$n_2 = 12$$

$$\bar{x}_1 = \frac{5439}{12} = 453.25$$

$$\bar{x}_2 = \frac{5801}{12} = 483.416$$

$$S_1 = \sqrt{\frac{\sum (x_1 - \bar{x}_1)^2}{n_1 - 1}}$$

$$= \sqrt{\frac{100748.22}{12 - 1}}$$

$$= 95.70$$

$$S_2 = \sqrt{\frac{\sum (x_2 - \bar{x}_2)^2}{n_2 - 1}}$$

$$= \sqrt{\frac{55191.87}{12 - 1}}$$

$$= 70.84$$

$$S = \sqrt{\frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2 - 2}}$$

$$= \sqrt{\frac{12 \times 95.70^2 + 12 \times 70.84^2}{12 + 12 - 2}}$$

$$= 87.936$$

we have,

$$(C_1, C_2) = |\bar{x}_1 - \bar{x}_2| \pm t_{0.05, 22} \times 87.936 \sqrt{\frac{1}{12} + \frac{1}{12}}$$

$$= |453.25 - 483.416| \pm 2.074 \times 87.936 \times 0.408$$

$$= 30.166 \pm 75.94574.41$$

$$= (-44.29, 104.57) \quad \underline{\underline{B}}$$

Notes:-	Country A	Country B	SI
Mean height (in inches)	$\bar{x}_1 = 67.42$	$\bar{x}_2 = 67.35$	
S.d. in inches	$\sigma_1 = 2.58$	$\sigma_2 = 2.10$	
Sample size	$n_1 = 1000$	$n_2 = 1200$	
$1 - \alpha = 98\%$			
$\Rightarrow \alpha = 2\% = 0.02$			

we have,

$$(C_1, C_2) = |\bar{x}_1 - \bar{x}_2| \pm Z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$= |(67.42 - 67.35)| \pm 2.002 \sqrt{\frac{3.572}{1000} + \frac{2.502}{1200}}$$

$$= |(67.42 - 67.35)| \pm 2.33 \times 0.1059$$

$$= (0.17 \pm 0.2587)$$

$$(C_1, C_2) = (-0.0887, 0.4287) \quad \underline{\underline{B}}$$

TEST OF HYPOTHESIS

Hypothesis are assumption or guesses about the population parameter with the help of sample result.

Such assumption may or may not be true.

→ Test of Significance (Test of hypothesis)

There are two types of hypothesis.

(i) Null hypothesis

(ii) Alternative hypothesis.

(i) Null hypothesis

The first procedure in test of significance is the setting up null hypothesis.

A hypothesis of no difference betⁿ population mean & specified mean or no difference betⁿ two population means is called null hypothesis. It is denoted by H_0 .

(e.g. $H_0: \mu = \mu_0$ (Specified mean)

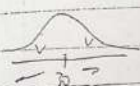
$H_0: \mu_1 = \mu_2$

(ii) Alternative hypothesis:

A hypothesis of difference betⁿ population mean and specified mean or difference betⁿ two population mean is called alternative hypothesis. It is denoted by H_1 .

Single or Double Sample

- $H_1: \mu \neq \mu_0$ (Specified) - two tailed.
- $H_1: \mu > \mu_0$ (right tailed)
- $H_1: \mu < \mu_0$ (left tailed)



(i) Two tailed

$\mu \neq \mu_0, \mu > \mu_0$ or $\mu < \mu_0$

(ii) one tailed

- greater than, - Exceed, more than, Superior, Superior -
- less than

PU

→ Types of error in Test of hypothesis

The decision to accept or reject null hypothesis may be wrong or right. There are four possible decisions which are

	Reject H_0	Accept H_0
H_0 true	wrong (Type I)	Right
H_0 false	Right	wrong (Type II)

There are two types of error

(i) Type I error (ii) Type II error

(i) Type I error

The wrong decision or error of rejecting H_0 when it is true is called type I error. The probability of type I error

(ii) is denoted by α . (i.e. $P(\text{type I error}) = \alpha$

(ii) Type II error

The error of accepting H_0 when it is false is called type-II error.

→ Critical region

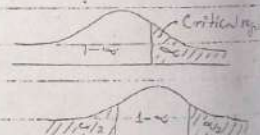
The area covered by a

type-I error is called

Critical region & the probability of type I error α is called level of significance.

Level of significance

In practical point of view, level of significance α is generally taken as 5% and 1%.



⇒ procedure of testing of hypothesis

- (i) Set up null & alternative hypothesis.
- (ii) for H_1 , decide whether it is one tailed or two tailed
- (iii) Decide appropriate level of significance depending upon the reliability of estimates.
- (iv) Compute appropriate test statistics (calculated value)
- (v) Write tabulated value at given α (critical value)
- (vi) If $|t_{cal}| \leq Z_{tabulated}$ at α , then null hypothesis (H_0) is accepted.
- (vii) If $|t_{cal}| > Z_{tabulated}$ at α , H_0 is rejected.
- (viii) write conclusion.

⇒ formula

(i) Large single sample ($n \geq 30$)

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \quad \text{where, } \bar{x} = \text{Sample mean, } n = \text{Sample size}$$

$\mu_0 = \text{Specified mean, } \sigma = \text{S.d. (population) (specified)}$

(ii) Small single sample ($n < 30$)

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \quad \text{where, } \bar{x} = \text{Sample mean}$$

$\mu_0 = \text{Specified mean}$
 $s = \text{Sample S.d., } n = \text{Sample size}$
 degree of freedom = $n-1$

(iii) proportion (population) ($n \geq 30$)

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}}$$

where,
 $\hat{p} = \text{proportion of Success in a sample}$
 $p_0 = \text{Specified } \text{pop.} \text{ proportion of Success}$
 $q_0 = 1 - p_0$
 $n = \text{Sample size}$

(iv) Test of significance for difference of means of two sample (large)

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

where,

1st type

2nd type

sample size = n_1

sample size = n_2

" mean = $\bar{x}_1$

" mean = $\bar{x}_2$

Pop. S.d. = σ_1

pop. S.d. = σ_2

(v) Test of significance for difference of means of two sample (small)

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{where } s = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}}$$

(vi) Test of significance for difference of proportions 1st sample

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p} \hat{q} (\frac{1}{n_1} + \frac{1}{n_2})}} \quad \text{where, } \hat{p}_1 = \text{proportion of Success 1st sample}$$

$\hat{p}_2 = \text{proportion of Success 2nd sample}$

$$\hat{p} = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2}, \quad \hat{q} = 1 - \hat{p}$$

(vii) Paired T-test

paired T-test is used when

- (i) The sample sizes are equal i.e. ($n_1 = n_2$)
- (ii) The pair of observation (x_i, y_i), $i = 1, 2, 3, \dots, n$ are independent of each other. the problem is to test that there is significant difference or not.

gt is defined as

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

where,

$$\bar{d} = \frac{\sum d_i}{n} \quad d_i = x_i - y_i$$

&

$$s_d = \sqrt{\frac{\sum d_i^2 - n \bar{d}^2}{n-1}}$$

② Solⁿ: Sample Size (n) = 400

no. of top quality (X) = 50

$$\text{proportion of top quality } (\hat{p}) = \frac{x}{n} = \frac{50}{400} = \frac{1}{8} = 0.125$$

Null Hypothesis (H_0): $P = P_0 (0.2)$

$$P_0 = 1 - p_0 = 1 - 0.2 = 0.8$$

Alternate " (H_1) : $P \neq P_0$ (two tailed)

we have,

$$Z = \frac{\hat{p} - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} = \frac{0.125 - 0.20}{\sqrt{\frac{0.2 \times 0.8}{400}}} = -3.75$$

$$|Z_{\text{calc}}| = 3.75$$

Z_{tabled} at 4.1% Two tailed = 2.58

Since $|Z_{\text{calc}}| > Z_{\text{table}}$ at 1% hence Null Hypothesis H_0 is rejected

i.e. H_1 is accepted

Conclusion: 20% was wrong at 1% level of significance

③ Solⁿ: Size of sample (n) = 3000

no. of males (x) = 1632

$$\text{proportion of males } (\hat{p}) = \frac{x}{n} = \frac{1632}{3000} = 0.544$$

Null Hypothesis (H_0): $P = P_0 (1/2)$

$$P_0 = 1 - P_0 = 1 - \frac{1}{2} = \frac{1}{2}$$

Alternate " (H_1) : $P \neq P_0$ (Two tail)

we have,

$$Z = \frac{\hat{p} - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} = \frac{0.544 - 0.50}{\sqrt{\frac{0.5 \times 0.5}{3000}}} = 4.82$$

$$|Z_{\text{calculate}}| = 4.82$$

Z_{tabled} at $\alpha = 5\%$ Two tailed = 1.96

Since $|z|_{cal} > z_{tabulated}$ at 5% $\therefore$ null hypothesis is rejected i.e. H_1 is accepted.
 Conclusion - No. of males & females in the city are not equal.

- ④ Size of Sample (n) = 200 Piece
 Lost luggage (x) = 17
 proportion ($\hat{p}$) = $\frac{x}{n} = \frac{17}{200} = 0.085$
 Null Hypothesis (H_0): $P = P_0 (0.06)$, $Q_0 = 1 - 0.06 = 0.94$
 alternate " (H_1): $P \neq P_0$ (one tailed)

we have,

$$z = \frac{\hat{p} - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} = \frac{0.085 - 0.06}{\sqrt{\frac{0.06 \times 0.94}{200}}} = 1.488$$

$|z|_{cal} = 1.488$
 tabulated at $\alpha = 0.05$, one tailed = 1.645
 Since $|z|_{cal} < z_{tabulated}$ at $\alpha = 0.05$, therefore Null hypothesis is accepted.
 Conclusion - The claim of airline that 6% of all luggage is never found is right.

- ⑤ Size of Sample (n) = 60
 no. of out of order transistor (x) = 7
 proportion ($\hat{p}$) = $\frac{x}{n} = \frac{7}{60} = 0.1166$
 Null Hypothesis (H_0): $P = P_0 (0.10)$, $Q_0 = 1 - 0.10 = 0.90$
 alternate " (H_1): $P \neq P_0$ (two tailed)
 we have,

$$z = \frac{\hat{p} - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} = \frac{0.1166 - 0.10}{\sqrt{\frac{0.10 \times 0.9}{60}}} = 0.43$$

$|z|_{cal} = 0.43$
 z_{tab} at $\alpha = 0.01$ (two tailed) = 2.58
 Since $|z|_{cal} < z_{tab}$ at 0.01 $\therefore$ null hypothesis is accepted.
 Conclusion - manufacturer claim is correct.

- ⑥ Sample size (n) = 120
 no. of birth control pills (x) = 20
 proportion ($\hat{p}$) = $\frac{x}{n} = \frac{20}{120} = \frac{1}{6} = 0.1666$
 Null Hypothesis (H_0): $P = P_0 (0.25)$, $Q_0 = 1 - 0.25 = 0.75$
 alternate " (H_1): $P < P_0$ (one tailed)
 we have,

$$z = \frac{\hat{p} - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} = \frac{0.1666 - 0.25}{\sqrt{\frac{0.25 \times 0.75}{120}}} = -2.11$$

$|z|_{calculated} = 2.11$
 tabulated at $\alpha = 5\%$, one tailed = 1.645
 Since $|z|_{cal} > z_{tabulated}$ at $\alpha = 5\%$, hence null hypothesis H_0 is rejected i.e. H_1 is accepted.
 Conclusion - social worker believe is correct.

- ⑦ City I
 Sample size (n_1) = 1000
 the own colour TV (x_1) = 318
 proportion ($\hat{p}_1$) = $\frac{x_1}{n_1} = \frac{318}{1000} = 0.318$
 City II
 $n_2 = 1200$
 $x_2 = 450$
 $\hat{p}_2 = \frac{x_2}{n_2} = \frac{450}{1200} = 0.375$

$$P = \frac{x_1 + x_2}{n_1 + n_2} = \frac{318 + 450}{1000 + 1200} = 0.349, \hat{Q} = 0.651$$

Null Hypothesis (H_0) $\hat{p}_1 = \hat{p}_2$

Alternate " (H_1) : $\hat{p}_1 \neq \hat{p}_2$ (Two-tailed)

we have,

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{0.515 - 0.375}{\sqrt{0.379 \times 0.621 \left(\frac{1}{1000} + \frac{1}{1000}\right)}} = -2.793$$

$$|Z|_{cal} = 2.793$$

$Z_{tabulated}$ at $\alpha = 1\%$ two-tailed = 2.58

Since $|Z|_{cal} > Z_{tabulated}$ $\therefore$ null hypothesis (H_0) is rejected. i.e. Alternate hypothesis H_1 is accepted.

Conclusion :- There is significance betⁿ color TV.

(10) Size of men (n_1) = 400

Size of women (n_2) = 600

favor of proposal (x_1) = 200

favor of proposal (x_2) = 325

$$\hat{p}_1 = \frac{200}{400} = 0.5$$

$$\hat{p}_2 = \frac{325}{600} = 0.5416$$

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{200 + 325}{400 + 600} = \frac{525}{1000} = 0.525, \hat{q} = 0.475$$

Null Hypothesis (H_0) = $\hat{p}_1 = \hat{p}_2$

Alternate " (H_1) = $\hat{p}_1 \neq \hat{p}_2$ (Two-tailed)

we have,

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{0.5 - 0.5416}{\sqrt{0.525 \times 0.475 \left(\frac{1}{400} + \frac{1}{600}\right)}} = -1.29$$

$$|Z|_{cal} = 1.29$$

$Z_{tabulated}$ at $\alpha = 5\%$ two-tailed = 1.96

Since $|Z|_{cal} < Z_{tab}$ $\therefore$ null hypothesis is accepted

Conclusion :- men & women are favorable for proposal.

(11) sample size (n_1) = 500

$n_2 = 100$

Imported output outside (x_1) = 16

$n_2 = 3$

$$\text{proportion } (\hat{p}_1) = \frac{16}{500} = 0.032$$

$$\hat{p}_2 = \frac{3}{100} = 0.03$$

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{16 + 3}{500 + 100} = 0.0316, \hat{q} = 0.96834$$

Null Hypothesis (H_0) = $\hat{p}_1 = \hat{p}_2$

Alternate " (H_1) = $\hat{p}_1 > \hat{p}_2$ (One-tailed)

we have,

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{0.032 - 0.03}{\sqrt{0.0316 \times 0.96834 \left(\frac{1}{500} + \frac{1}{100}\right)}} = 0.1043$$

$$|Z|_{cal} = 0.1043$$

$Z_{tabulated}$ at $\alpha = 5\%$ one-tailed = 1.645

Since $|Z|_{cal} < Z_{tabulated}$ $\therefore$ null hypothesis (H_0) is accepted

Conclusion :- we may conclude that the machine has not improved significantly.

(12) Sample size (n) = 64

mean breaking strength ($\bar{x}$) = 54 lbs

Standard deviation (σ) = 10 lbs

Specified mean (μ_0) = 50 $\alpha = 5\% = 0.05$ (assuming)

Null hypothesis (H_0) = $\mu = \mu_0 (50)$

Alternate " (H_1) = $\mu > \mu_0$, one-tailed

we have

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{54 - 50}{10/\sqrt{64}} = 3.20$$

$$|Z|_{cal} = 3.2$$

$Z_{tabulated}$ at $\alpha = 5\%$ one-tailed = 1.645

Since $|z|_{cal} > z_{tabulated}$ $\therefore$ null hypothesis is rejected
 i.e. Alternative hypothesis (H_1) is accepted.
 Conclusion: the claim that the average breaking strength
 of glass rod > 50 is right.

(23) Sample size (n) = 6400

mean of English men ($\bar{x}_1$) = 67.85 mean of Australian ($\bar{x}_2$) = 68.55
 S.d. (σ_1) = 3.56 S.d. (σ_2) = 2.52

Null hypothesis (H_0): $\mu = \mu$

Alternate " (H_1): $\mu > \mu$ (one tailed)

We have

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{67.85 - 68.55}{\sqrt{\frac{3.56^2}{6400} + \frac{2.52^2}{6400}}} = -15.589$$

$$|z|_{cal} = 15.589$$

$z_{tabulated}$ at $\alpha = 1\%$, one tailed = 2.33

Since $|z|_{cal} > z_{tabulated}$ at 1% $\therefore$ null hypothesis
 (H_0) is rejected i.e. Alternative hypothesis (H_1) is accepted

Conclusion: Data indicate that Australian are on
 the average taller than English men is right.

(24) Σx	x	$(x - \bar{x})^2$	x	$(x - \bar{x})^2$
70	70	917.06	95	3.161
120	120	539.26	98	1.493
110	110	179.82	107	103.439
83	83	189.83	100	10.381
88	88	77.05		
				$\Sigma(x - \bar{x})^2 = 3817.59$

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{96.728}{9} = 10.7478 \quad S.d. (s) = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n-1}}$$

Sample size (n) = 9, Specified mean (μ_0) = 100 = 15.023

Null hypothesis (H_0) = $\mu = \mu_0$

Alternate hypothesis (H_1) = $\mu \neq \mu_0$ two tailed

We have

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{96.728 - 100}{15.023/\sqrt{9}} = -0.64$$

$$|t|_{cal} = 0.64$$

$t_{tabulated}$ at $\alpha = 5\%$ & degree of freedom $n-1 = 8$, is 2.306

Since $|t|_{cal} < t_{tabulated}$ $\therefore$ null hypothesis is accepted

Conclusion: These data do not support the population is also

$$(C.I.)_{\bar{x}} = \bar{x} \pm t_{\alpha, n-1} \frac{s}{\sqrt{n}} = 96.728 \pm t_{0.05, 8} \times \frac{15.023}{\sqrt{9}} \\ = 96.728 \pm 2.306 \times \frac{15.023}{3} \\ = (85.19, 108.36)$$

(25) $\Sigma (x - \bar{x})^2$

$$70 \quad 16 \quad S.d. (s) = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n-1}} = \sqrt{\frac{90}{10-1}} = \sqrt{10} = 3.162$$

Sample size (n) = 10

Specified mean (μ_0) = 64

Null hypothesis (H_0) = $\mu = \mu_0$

Alternate " (H_1) = $\mu > \mu_0$ (one tailed)

We have

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{66 - 64}{3.162/\sqrt{10}} = 2$$

$$|t|_{cal} = 2$$

$t_{tabulated}$ at 5% & degree of freedom 9, one tailed =

Since $|t|_{cal} > t_{tabulated}$ $\therefore H_0$ is rejected i.e. H_1

Conclusion: Average height is greater than 64 inch

(12) $\alpha = 5\%$ Sample size $(n) = 11$
 Mean $(\bar{x}) = 14.2 \text{ cm}$
 S.d. $(s) = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = \sqrt{\frac{219.6}{11-1}} = 4.686 \text{ cm}$
 Specified mean $(\mu_0) = 15 \text{ cm}$
 null hypothesis $(H_0): \mu = \mu_0$
 Alternate $(H_1): \mu \neq \mu_0$ two tailed
 we have,

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{14.2 - 15}{4.686/\sqrt{11}} = -0.566$$

 $|t|_{\text{cal}} = 0.566$
 t_{tabled} at $\alpha = 5\%$, degree of freedom 10,
 two tailed is 2.228
 Since $|t|_{\text{cal}} < t_{\text{tabled}}$, H_0 is accepted.
 Conclusion: - Old View matches the experimental data.

we have,

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{1234 - 1026}{40872 \sqrt{\frac{1}{8} + \frac{1}{2}}} = 9.392$$

 $|t|_{\text{cal}} = 9.392$
 t_{tabled} at $\alpha = 0.01$, $n_1 + n_2 - 2 = 13$ (one tailed) = 2.65
 Since $|t|_{\text{cal}} > t_{\text{tabled}}$, null hypothesis (H_0) is rejected i.e. H_1 is accepted.
 Conclusion: - Hence type I sample is superior to type II.

(41) χ^2

Prior training (x_i)	84	78	36	37	59	69	83	96	90	65
after training (y_i)	90	58	56	49	62	81	84	86	89	74
$d_i = (x_i - y_i)$	-6	-10	-20	-12	-8	-12	-1	10	1	-10
d_i^2	36	100	400	144	64	144	1	100	1	100

Null hypothesis $(H_0) = \mu_x = \mu_y$ (Training is not effective)
 Alternate hypothesis $(H_1) = \mu_x < \mu_y$ (Training is effective) one tailed

we have,

$$\bar{d} = \frac{\sum d_i}{n} = \frac{-63}{10} = -6.3$$

$$s.d. = \sqrt{\frac{\sum d_i^2 - n\bar{d}^2}{n-1}} = \sqrt{\frac{1125 - 10 \times (-6.3)^2}{10-1}} = 8.924$$

$$t = \frac{\bar{d}}{s.d./\sqrt{n}} = \frac{-6.3 \times \sqrt{10}}{8.924} = -2.291$$

$|t|_{\text{calculated}} = 2.291$
 t_{tabled} at $\alpha = 0.05$, degree of freedom, 9 = 1.833
 Since $|t|_{\text{cal}} > t_{\text{tabled}}$, H_0 is rejected i.e. H_1 is accepted.
 Conclusion: - Hence training is effective.

(32) χ^2

Type I	Type 2
Sample no. $(n_1) = 8$	$n_2 = 7$
Sample mean $(\bar{x}_1) = 1234 \text{ hr}$	$\bar{x}_2 = 1086 \text{ hr}$
Sample S.d. $(s_1) = 36 \text{ hr}$	$s_2 = 49 \text{ hr}$

Null hypothesis $(H_0): \mu_1 = \mu_2$
 Alternate $(H_1): \mu_1 > \mu_2$ (one tailed)

$$S = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{8 \times 36^2 + 7 \times 49^2}{8+7-2}} = 40.732$$

Regression

The literal meaning of regression is stepping back towards the average or is to go back to original condition.

Defⁿ

Regression analysis is a mathematical measure of the average relationship betⁿ two or more variables in terms of the original units of the data.

There are two types of variables:-

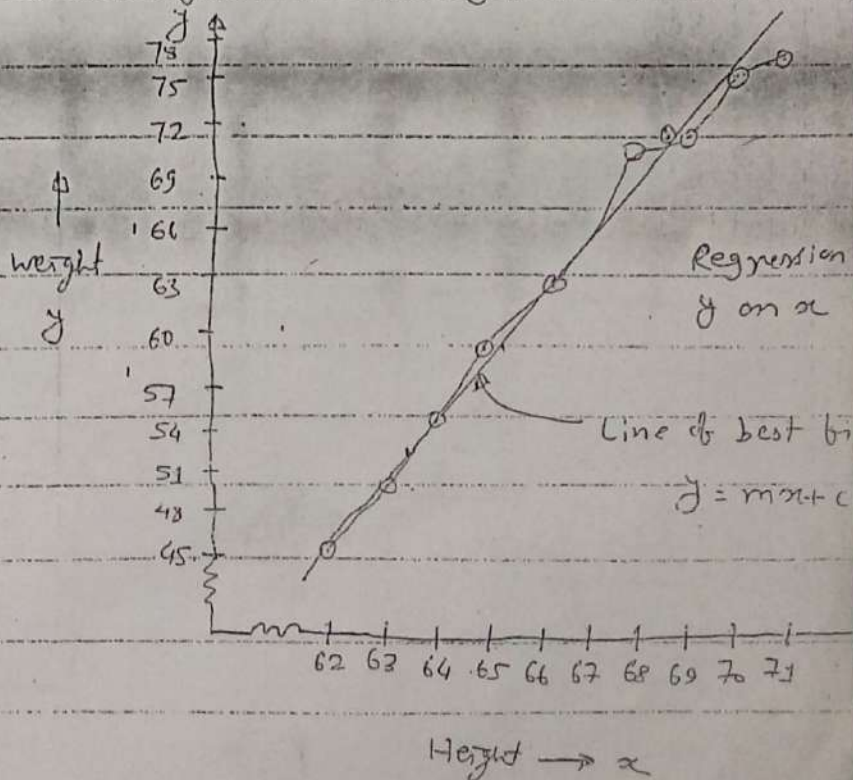
(i) Independent variable (x)

(ii) Dependent variable (y)

Scatter diagram

Height in inch (x)	62	63	64	65	67	68	69	70	71
Weight in kg (y)	45	50	55	59	62	70	72	75	76

where, dependent variable (y) & independent variable (x) are plotted in the diagram, the diagram is called scatter diagram.



⇒ The method of least square

The mathematical equation for a regression of y on x is of the form

$$y = mx + c. \text{ It is also called line of best fit.}$$

where,

$$\text{slope}(m) = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

$$y\text{-intercept } (c) = \bar{y} - m\bar{x}$$

⇒ correlation

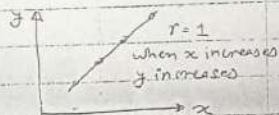
The correlation betⁿ two variables x and y is known as simple correlation. This means, the change in one variable is accompanied by the proportionate change in the other variable.

→ Determination of correlation by graphical method
Coefficient of correlation

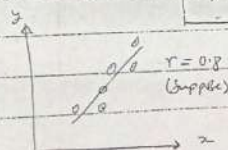
The coefficient of correlation is denoted by r and defined by

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} \quad -1 \leq r \leq 1$$

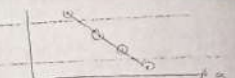
(i) Perfect positive correlation



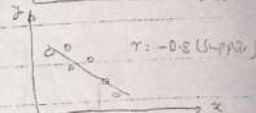
(ii) Positive correlation



(iii) perfect negative correlation



(iv) Negative Correlation



(v) No relationship betⁿ x & y



→ $0.5 < r \rightarrow$ High
 $|r| \leq 0.5 \rightarrow$ Low

⇒ Probable error (P.E.)

An old measure of ascertaining the reliability of the value of correlation coefficient (r) is used by probable error

probable error is defined by

$$\begin{aligned} P.E. &= 0.6745 \times \text{standard error} \\ &= 0.6745 \times \frac{1-r^2}{\sqrt{n}} \end{aligned}$$

⇒ Rank correlation

The rank correlation coefficient of two variables is denoted by R & defined by

$$R = 1 - \frac{6 \sum d^2}{n(n^2-1)} \quad \text{for distinct observation}$$

$$1 - \frac{6 \left[\sum d^2 + \frac{m_1(m_1^2-1)}{12} + \frac{m_2(m_2^2-1)}{12} + \frac{m_3(m_3^2-1)}{12} + \dots \right]}{n(n^2-1)} \quad \text{(for repeated observations)}$$

Line of regression

A line of regression is the straight line which gives the best fit. Regression equation of y on x

(a) $y - \bar{y} = b_{yx}(x - \bar{x})$ where slope, $b_{yx} = r \frac{\sigma_y}{\sigma_x}$

(b) Regression equation of x on y

$$x - \bar{x} = b_{xy}(y - \bar{y}) \quad \text{where } b_{xy} = r \frac{\sigma_x}{\sigma_y}$$

here,

$$b_{yx} \times b_{xy} = r \frac{\sigma_y}{\sigma_x} \times r \frac{\sigma_x}{\sigma_y} = r^2$$

$$\therefore \boxed{b_{yx} \times b_{xy} = r^2}$$

Exercise -15

x	y	(x- $\bar{x}$)	(y- $\bar{y}$)	(x- $\bar{x}$)(y- $\bar{y}$)	(x- $\bar{x}$) ²
1	2.4	-3	-1.6	4.8	9
2	3	-2	-1	2	4
3	3.6	-1	-0.4	0.4	1
4	4	0	0	0	0
6	5	2	1	2	4
8	6	4	2	8	16
$\Sigma x = 24$	$\Sigma y = 24$			$\Sigma (x-\bar{x})(y-\bar{y}) = 17.2$	$\Sigma (x-\bar{x})^2 = 34$
$n = 6$	$n = 6$				
$\bar{x} = \frac{24}{6} = 4$	$\bar{y} = \frac{24}{6} = 4$				

we have line of best fit (regression line of y on x) is
 $y = mx + c$ --- (1)

$$m = \frac{\Sigma (x-\bar{x})(y-\bar{y})}{\Sigma (x-\bar{x})^2} = \frac{17.2}{34} = 0.506$$

$$c = \bar{y} - m\bar{x}$$

$$= 4 - 0.506 \times 4 = 1.976$$

putting the value of m & c in eqn (1)

$$y = 0.506x + 1.976$$

x	y	(x- $\bar{x}$)	(y- $\bar{y}$)	(x- $\bar{x}$)(y- $\bar{y}$)	(x- $\bar{x}$) ²
0	12	-12.5	-8	100.0	156.25
5	15	-7.5	-5	37.5	56.25
10	12	-2.5	-3	7.5	6.25
15	22	2.5	2	5.0	6.25
20	24	7.5	4	30.0	56.25
25	30	12.5	10	125	156.25
$\Sigma x = 75$	$\Sigma y = 120$			$\Sigma (x-\bar{x})(y-\bar{y}) = 305$	$\Sigma (x-\bar{x})^2 = 437.5$

$$\begin{aligned} \Sigma x &= 75 & \Sigma y &= 120 \\ n &= 6 & n &= 6 \\ \bar{x} &= \frac{75}{6} = 12.5 & \bar{y} &= \frac{120}{6} = 20 \end{aligned}$$

we have line of best fit (regression line of y on x) is

$$y = mx + c \text{ --- (1)}$$

$$\text{where } m = \frac{\Sigma (x-\bar{x})(y-\bar{y})}{\Sigma (x-\bar{x})^2} = \frac{305}{437.5} = 0.6972$$

$$c = \bar{y} - m\bar{x}$$

$$= 20 - 0.6972 \times 12.5$$

$$= 11.28$$

putting the value of m & c in eqn (1)

$$y = 0.697x + 11.28$$

θ	v	$(\theta-\bar{\theta})$	$(v-\bar{v})$	$(\theta-\bar{\theta})(v-\bar{v})$	$(\theta-\bar{\theta})^2$
0	1.8	-7.5	0.425	-3.1875	56.25
5	1.45	-2.5	0.025	-0.29125	6.25
10	1.18	2.5	-0.175	-0.4375	6.25
15	1.0	7.5	-0.575	-2.68125	56.25
				$\Sigma (\theta-\bar{\theta})(v-\bar{v}) = -6.825$	$\Sigma (\theta-\bar{\theta})^2 = 125$

$$\Sigma \theta = 30 \quad \Sigma v = 5.43$$

$$n = 4 \quad n = 4$$

$$\bar{\theta} = 7.5 \quad \bar{v} = 1.3575$$

we have line of best fit (regression line v on θ) is

$$v = a + b\theta \text{ --- (1)}$$

$$\text{where } b = \frac{\Sigma (\theta-\bar{\theta})(v-\bar{v})}{\Sigma (\theta-\bar{\theta})^2} = \frac{-6.825}{125} = -0.0546, \quad a = \bar{v} - b\bar{\theta}$$

$$= 1.3575 - (-0.0546 \times 7.5)$$

$$= 1.758$$

$$v = 1.758 - 0.0546\theta$$

Types of Theoretical Discrete Distributionsi- Binomial Distribution:

$$P(r) = {}^n C_r \cdot p^r q^{n-r}$$

$$E(x) = n \cdot p$$

$$V(x) = n \cdot p \cdot q$$

- % given.
- rate.
- Event p known

ii- Poisson's Distribution:

This is applied when:

- n is greater than 30.
- p is very small.
- $n \cdot p = m$ is infinite.

$$P(r) = \frac{e^{-m} \cdot m^r}{r!}$$

- Average given = m.
- n is greater & p is given then, $m = np$.

$\therefore m$ is called average of P.D.

iii- Hypergeometric Distribution:

From N items out of which N_1 is 1st kind and N_2 are 2nd kind. then probability of r items of the 1st kind:

$$P(r) = \frac{{}^{N_1} C_r \cdot {}^{N_2} C_{n-r}}{{}^N C_n}$$

iv- Negative Binomial distribution:

If x are failure during the r is success in x+r trials.

$$P(x) = {}^{x+r-1} C_{r-1} \cdot p^r \cdot q^x$$

$$(*) \text{ No. of class} = 1 + 3.22 \log_{10} N$$

$$\text{size of class} = \frac{L-S}{K}$$

Theoretical Continuous Distribution :

i- Rectangular or Uniform Distribution :

$$\begin{aligned}
 - f(x) &= \frac{1}{b-a}, \quad a < x < b. \\
 - E(x) &= \frac{a+b}{2} \\
 - V(x) &= \frac{(b-a)^2}{12}
 \end{aligned}
 \quad \left. \begin{array}{l} \text{- Interval.} \\ \text{- Every.} \end{array} \right\}$$

ii- Gamma Distribution :

$$\begin{aligned}
 - f(x) &= \frac{1}{\beta^2 \Gamma(2)} \cdot x^{2-1} \cdot e^{-x/\beta} \\
 \Gamma(2) &= \int_0^{\infty} x^{2-1} \cdot e^{-x} \cdot dx \\
 - E(x) &= 2 \cdot \beta \\
 - V(x) &= 2 \cdot \beta^2
 \end{aligned}$$

iii- Beta Distribution :

$$\begin{aligned}
 - f(x) &= \frac{1}{\beta(\alpha \cdot \beta)} x^{\alpha-1} \cdot (1-x)^{\beta-1} \\
 - B(\alpha \cdot \beta) &= \int_0^1 x^{\alpha-1} \cdot (1-x)^{\beta-1} dx \\
 V(x) &= \frac{\alpha \beta}{(\alpha+\beta)^2 (\alpha+\beta+1)}, \quad E(x) = \frac{\alpha}{\alpha+\beta}
 \end{aligned}$$

iv- Normal distribution :

$$Z = \frac{x - \mu}{\sigma} = z_1$$

where μ = mean

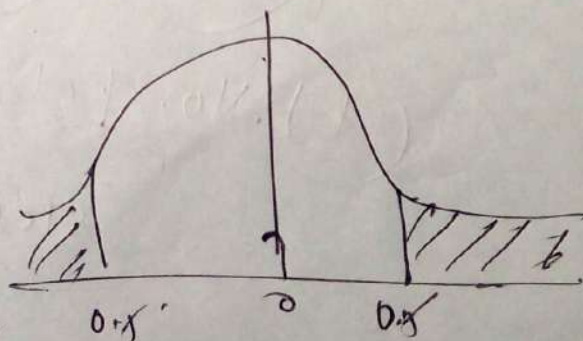
σ = S.D.

Find, μ & σ is important

Let $z = z_1$ &

z_1 = less than 0.5 = -ve sign.

z_1 = more than 0.5 = +ve sign.



Sampling and Estimations:

i- Estimating Population mean (μ) for large sample ($n > 30$):

$$- \mu = \bar{x} \pm S.E(\bar{x}) \cdot Z_{2/2}$$

$$- S.E(\bar{x}) = \frac{\sigma}{\sqrt{n}} \quad - \text{for population size not given.}$$

$$- S.E(\bar{x}) = \frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}} \quad - \text{for population size given.}$$

ii- Estimating population mean (μ) for large sample ($n > 30$):

- P is population proportion.

$$- P = p \pm S.E(p) \cdot Z_{2/2}$$

$$p = \frac{x}{n}$$

$$q = 1 - p$$

$$- S.E(p) = \sqrt{\frac{pq}{n}} \quad - N \text{ is given.}$$

$$- S.E(p) = \sqrt{\frac{pq}{n}} \cdot \sqrt{\frac{N-n}{N-1}} \quad - N \text{ is given.}$$

iii- Confidence interval for population mean for sample $n (\leq 30)$.

Let, $\bar{x}$ = sample mean, n = sample size.

S = sample deviation

$$\mu = \bar{x} \pm S.E(\bar{x}) \cdot t_{2/2, n-1}$$

$$- S.E = \frac{S}{\sqrt{n-1}} \quad - \text{sample S.D is given}$$

$$- S.E(\bar{x}) = \frac{S}{\sqrt{n}} \quad - \text{sample S.D (S) is not given}$$

$$S^2 = \frac{\sum (x - \bar{x})^2}{n-1}$$

iv- Determine (n), $n = ?$

for

$$n = \frac{\sigma^2}{e^2} \cdot Z_{2/2}^2$$

where, $e = \mu - \bar{x}$

$$n = \frac{pq}{e^2} \cdot Z_{2/2}^2$$

where, $e = p - \bar{p}$

chapter-14.

Test of Hypothesis.

i. H-test of single mean for large sample ($n > 30$).

- Find $z_{\alpha/2}$ from significant level.

- Find $z = \frac{\bar{x} - \mu}{S.E(\bar{x})}$

where $S.E(\bar{x}) = \frac{\sigma}{\sqrt{n}}$

- Compare z & $z_{\alpha/2}$ and get result.

ii. H-test of single proportion for large sample ($n > 30$).

- Find $z_{\alpha/2}$ from significant level.

- Find $z = \frac{p - P}{S.E(p)}$

where, $S.E(\bar{x}) = \sqrt{\frac{pq}{n}}$

- Compare z & $z_{\alpha/2}$ and get result.

iii. H-test betⁿ two mean large samples ($n > 30$).

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

iv. H-test betⁿ two proportion large sample ($n > 30$).

$$z = \frac{p_1 - p_2}{S.E(p_1 - p_2)}$$

$$S.E(p_1 - p_2) = \sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$\hat{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

v- H-test of single mean for small sample ($n \leq 30$)

$$t = \frac{\bar{x} - \mu}{SE(\bar{x})}$$

where $SE(\bar{x}) = \frac{s}{\sqrt{n-1}}$ when s is given.

$$SE(\bar{x}) = \frac{s}{\sqrt{n}} \text{ when } s \text{ is not given}$$

$$\text{then, } s^2 = \frac{1}{n-1} \sum (x - \bar{x})^2, \quad \bar{x} = \frac{\sum x}{n}$$

vi- H-test betn two mean for small sample ($n \leq 30$)

$$t = \frac{(\bar{x}_1 - \bar{x}_2)}{SE(\bar{x})} \sim t_{n_1+n_2-2}$$

$$\text{where } SE(\bar{x}) = s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$s = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

test for different of two mean:-

Before add (2)	After add (4)	$d = x - y$	$(d - \bar{d})$	$(d - \bar{d})^2$	d^2

$$\bar{d} = \frac{\sum d}{n}$$

$$s^2 = \frac{\sum (d - \bar{d})^2}{n-1}, \quad t = \frac{\bar{d}}{\frac{s}{\sqrt{n}}}$$

$$t_{\alpha, n-1} = 5\% = 0.05 = 1.896$$

Compare

Chapter-15. Regression and Correlation.

i - Rank Correlation Coefficient:

$$R = 1 - \frac{6 \sum d^2}{n(n^2-1)}$$

$$d = R_1 - R_2$$

Coeff. of Correlation
 $r = \frac{n \sum dx \cdot dy - \sum dx \sum dy}{\sqrt{n \sum dx^2 - (\sum dx)^2} \sqrt{n \sum dy^2 - (\sum dy)^2}}$

$$P.E. = 0.6745 \sqrt{\frac{1-r}{n}}$$

ii. Rank Correlation for Repeated Ranks:

$$R = 1 - \frac{6 \left[\sum d^2 + \frac{\sum m_i(m_i^2-1)}{12} \right]}{n(n^2-1)}$$

2. Regression Analysis:

$$y = a + bx \quad \text{--- (i)}$$

$$\text{where, } a = \frac{\sum y}{n} - b \frac{\sum x}{n}$$

$$b = \frac{n \sum xy - \sum x \cdot \sum y}{n \sum x^2 - (\sum x)^2}$$

3. Standard error of estimate:

$$S_e = \sqrt{\frac{\sum (y - y_e)^2}{(n-2)}}$$

y = Observed value of y .

y_e = Estimated value of y .

4. Coefficient of determination (r^2):

$$r^2 = 1 - \frac{(\sum (y - y_e)^2)}{\sum (y - \bar{y})^2}$$

Chapter - 15

Simple Linear Regression and Correlation:

Correlation:

Correlation analysis is a statistical tool which study the relationship between two or more than two variables. The relation between two variable is called simple correlation. If the relation between more than two variable is called multiple correlation. Correlation can be +ve & -ve. If the increase in the value of one variable result in a corresponding increase in the other variable then it is known as positive correlation. For eg. Income & expenditure of the family, interest rate & deposit in a bank, age of husband & wife.

On the other hand, if the increase in one variable result in a corresponding decrease in the other variables then it is known as negative correlation. For eg. Price & demand of a commodity, Pressure and Volume etc.

Correlation and Regression Analysis :

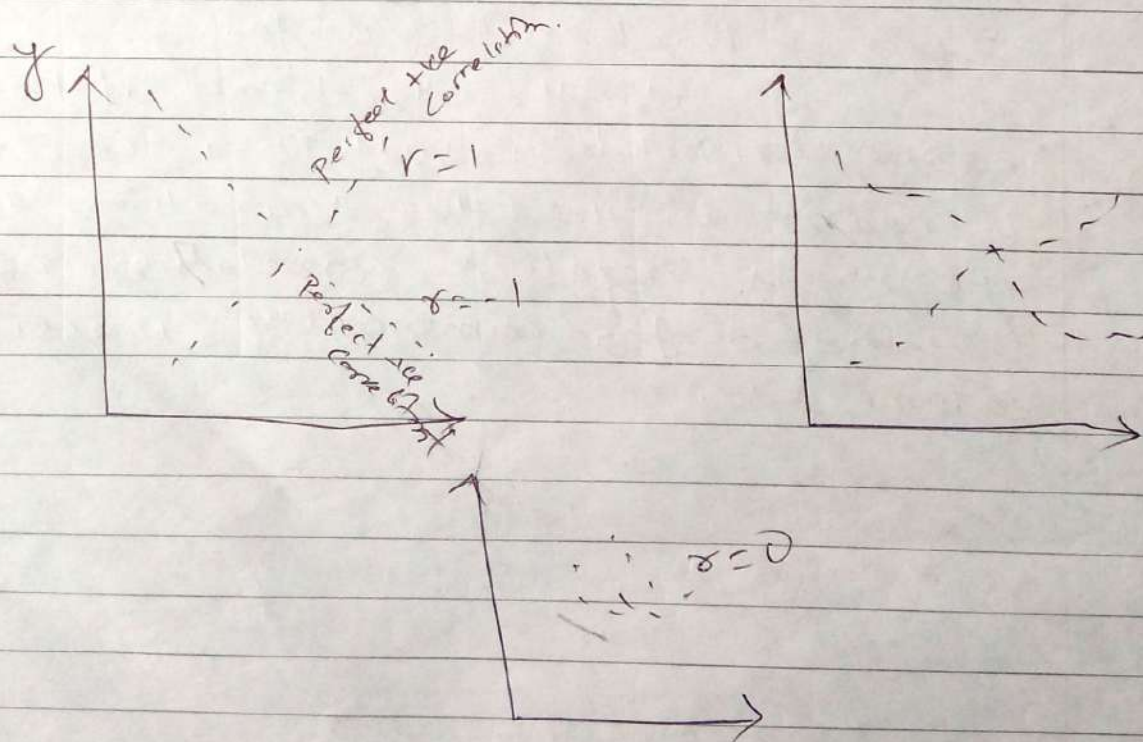
Methods of studying Correlations.

- i - Scatter diagram.
- ii - Karl Pearson's correlation Coefficient.
- and
iii - Spearman's rank Correlation.

i - Scatter diagram :

It is the simplest method of studying correlation between two variable by diagrammatic method.

Independent variable 'x' is plotted along 'x' axis & dependent variable 'y' is plotted along 'y' axis. The pairs of variables from the given data are shown in the form of dots on 'xy' plane. The totality of these dots determines the type of co-relation between two variables.



If all the points lie in a straight line from lower left to upper right direction then it is known as perfect positive correlation.

If the points are not in a straight line but there is an increasing trend then it is called high degree of true correlation. Similarly if all the points lie in a straight line from upper left to lower right direction then it is called perfect negative correlation.

If the plotted points do not show any trend either increasing or decreasing then there is no relationship between the two variables.

Q.P. - Karl Pearson's correlation coefficient $\frac{0}{6}$

$$r = \frac{\sum [(x - \bar{x}) \cdot (y - \bar{y})]}{\sqrt{\sum (x - \bar{x})^2} \cdot \sqrt{\sum (y - \bar{y})^2}}$$

$$= \frac{n \sum x \cdot y - \sum x \cdot \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \cdot \sqrt{n \sum y^2 - (\sum y)^2}}$$

$$= \frac{n \sum (dx \cdot dy) - \sum dx \cdot \sum dy}{\sqrt{n \sum dx^2 - (\sum dx)^2} \cdot \sqrt{n \sum dy^2 - (\sum dy)^2}}$$

$$= \frac{n \sum (dx \cdot dy) - \sum dx \cdot \sum dy}{\sqrt{n \sum dx^2 - (\sum dx)^2} \cdot \sqrt{n \sum dy^2 - (\sum dy)^2}}$$

$$= \frac{n \sum (dx \cdot dy) - \sum dx \cdot \sum dy}{\sqrt{n \sum dx^2 - (\sum dx)^2} \cdot \sqrt{n \sum dy^2 - (\sum dy)^2}}$$

where, $dx = x - A$

$dy = y - B$

Properties of Correlation Coefficient :-

- i- $-1 \leq r \leq 1$.
- ii- If $r = 0$, then there is no correlation between x & y .
- iii- If $r = 1$, then, ~~the~~ there is perfect ^{pos} correlation.
- iv- If $r = -1$, then there is perfect -ve correlation.

Q.1

Calculate the Correlation Coefficient from the following data :-

1- Marks in math (x) :-	67	68	65	68	72	72	69	71
Marks in statistics (y) :-	65	66	67	67	68	69	70	72

Solⁿ:-

x	y	$dx = x - A$	$dy = y - B$	dx^2	dy^2	$dx \cdot dy$
67	65	-1	-2	1	4	2
68	66	0	-1	0	1	0
65	67 = B	-3	0	9	0	0
68	67	0	0	0	0	0
72	68	4	1	16	1	4
72	69	4	2	16	4	8
69	70	1	3	1	9	3
71	72	3	5	9	25	15
Total		$\Sigma dx = 8$	$\Sigma dy = 8$	52	44	32

$$n = 8$$

then,

$$r = \frac{n \Sigma (dx \cdot dy)}{\Sigma dx \cdot \Sigma dy}$$

$$\frac{\sqrt{n \Sigma dx^2 - (\Sigma dx)^2} \cdot \sqrt{n \Sigma dy^2 - (\Sigma dy)^2}}{n}$$

$$= \frac{8 \times 32 - 8 \cdot 8}{\sqrt{8 \times 52 - 8^2} \times \sqrt{8 \times 44 - 8^2}}$$

$$= 0.6$$

Interpretation :

Moderate positive.

Probable Error %

probable error is a major of testing for testing the reliability of the sample correlation coefficient (r). It is given by

$$P.E. = 0.6745 \left(\frac{1-r^2}{\sqrt{n}} \right).$$

- If $r < P.E.$, then correlation is not at all significant.
- If $r > 6(P.E.)$, then r is significant.
- In other conditions nothing can be concluded.

Note :

Confidence interval for population correlation coefficient (ρ).

$$S = r \pm P.E.$$

(9.0.1)

$$P.E. = 0.6745 \left(\frac{1-(0.6)^2}{\sqrt{8}} \right) \\ = 0.15$$

Here, $6(P.E.) = 6 \times 0.15 = 0.9$.

$\therefore$ No conclusion can be done regarding the significant of sample correlation coefficient.

important.

- 1-a- Histogram, polygon, frequency curve, O-give diagram.
- Percentile, Decile, Quartile, mean.
b- More consistency, better scores (x) and (y) mode.
- Combine mean and variance and corrected mean.

- 2-a- Rules of probability.
b- Bayes's theorem & Numerical.

- 3-a- Continuous and discrete random variable.
b- theoretical discrete random variable distribution

- 4-a- Theoretical continuous random variable distribution.
b- joint distribution.

- 5-a- Theory of estimation.
b-

- 6-a-
b- Correlation and regression.

- 7- short Note.

Night:

Repeat:

- 2-a-b, } dis marks
5-a-b, }
6-a-b. }

important.

- 1-a- Histogram, polygon, frequency curve, O-give diagram.
 - Percentile, Decile, Quartile, mean,.
- b- More consistency, better scores, (x) and (y) mode.
 - Combine mean and variance and corrected mean.
- 2-a- Rules of probability.
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- 3-a- Continuous and discrete random variable.
 - b- theoretical discrete random variable distribution
- 4-a- Theoretical continuous random variable distribution.
 - b- joint distribution.
- 5-a- Theory of estimation.
 - b-
- 6-a-
 - b- Correlation and regression.
- 7 - short Note.

Night.:

Repeat:

- | | |
|--------|-------------|
| 2-a-b, | } dis marks |
| 5-a-b, | |
| 6-a-b. | |

chapter - 10 :

- i- Binomial Distribution $\therefore P(x) = nC_x p^x q^{1-x}$
ii- Poisson Distribution $\therefore$

$$P(x) = \frac{e^{-m} \cdot m^x}{x!}$$

iii- but question in per unit m (mean per unit) $\therefore$

Hypergeometric Distribution $\therefore P(x) = \frac{n! C_x \cdot N_2 C_{n-x}}{N! C_n}$

iv- Negative Binomial Distribution $\therefore P(x) = \frac{n! C_{x-1} p^n q}{x!}$

chapter - 8 :

Random Variable and Probability distribution

$\rightarrow$ Discrete random Variable $P(E) = \sum_{i=1}^n P(x_i)$

$\rightarrow$ Continuous

$$P(E) = \int_a^b f(x) dx$$

chapter - 9 :

Mathematical Expectation $\therefore$

$\rightarrow$ For Discrete random Variable $\therefore$

$$E(x) = \sum_{i=1}^n x_i P_i, E(x^2) = \sum_{i=1}^n x_i^2 \cdot P_i$$

$$V(x) = E(x^2) - [E(x)]^2$$

$\rightarrow$ For Continuous Random Variable $\therefore$

chapter - 11 :

Page 24